

LCFC Confidential

HS460&HS560&HS760(NM-XXXX) MB Schematics Document

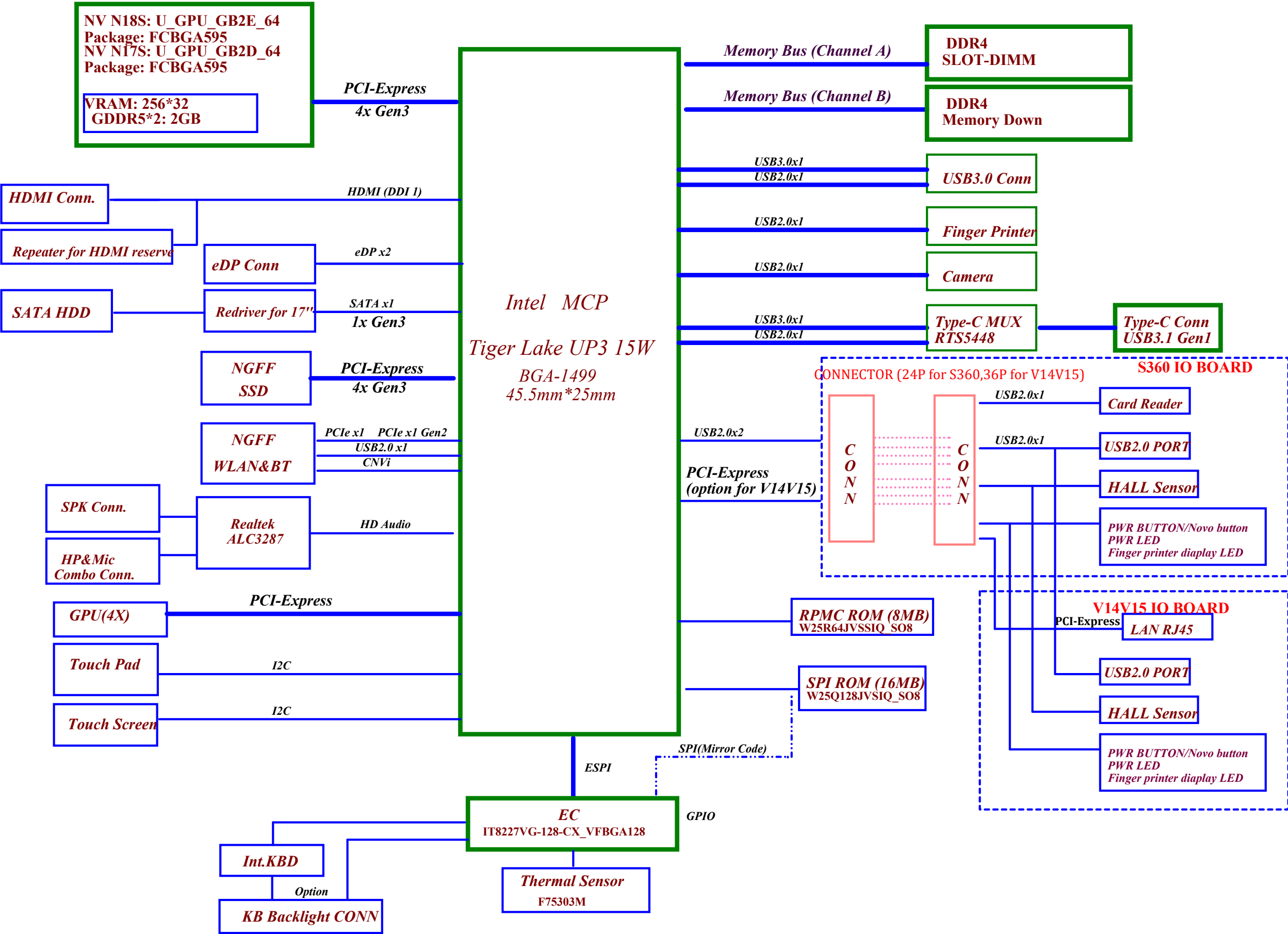
Tiger Lake UP3 with DDR4 + Slot-DIMM

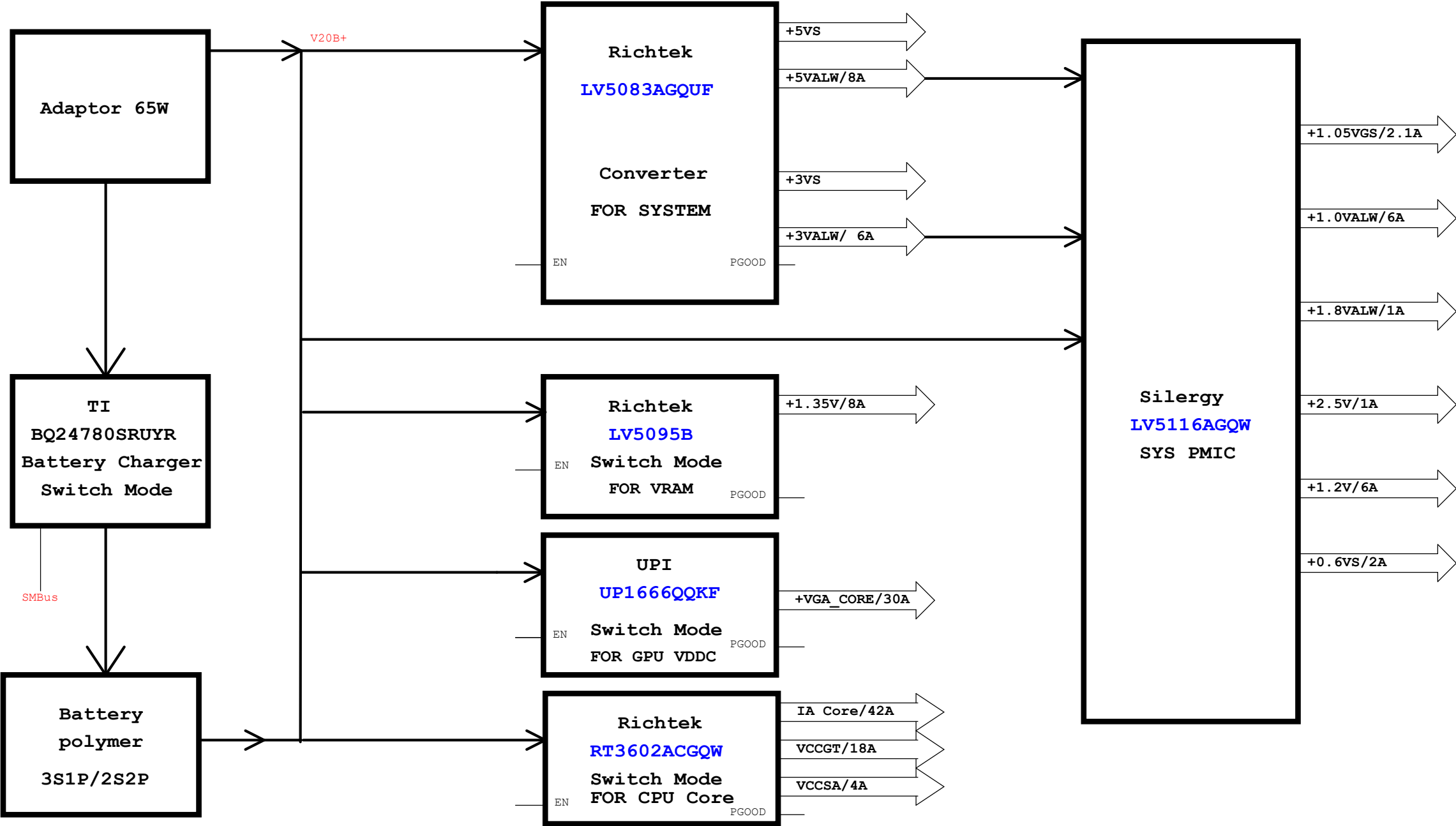
2020-06

REV: 0.1

Security Classification	LCFC Highly Confidential Information			Title	S360-TGL	LCFC
Issued Date	2012/07/01	Deciphered Date	2014/07/01			
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Cover Page





Voltage Rails (0 --> Means ON , X --> Means OFF)

Power Plane State	V20B+	+3VALW +5VALW +3VALW_PCH +1.8VALW +1.05VALW	+1.2V +2.5V_DDR +VCCST	+5VS +3VS +VCCIO +VCCSTG +VCCSA +VCC_GT +CPU_CORE +0.6VS
S0	0	0	0	0
S3	0	0	0	X
S3 Battery only	0	0	0	X
S5 S4 AC Only	0	0	X	X
S5 S4 Battery only	0	X	X	X
S5 S4 AC & Battery don't exist	X	X	X	X

STATE \ SIGNAL		SLP_S3#	SLP_S4#	SLP_S5#	+VALW	+V	+VS	Clock
Full ON		HIGH	HIGH	HIGH	ON	ON	ON	ON
S3 (Suspend to RAM)		LOW	HIGH	HIGH	ON	ON	OFF	OFF
S4 (Suspend to Disk)		LOW	LOW	LOW	ON	OFF	OFF	OFF
S5 (Soft OFF)		LOW	LOW	LOW	ON	OFF	OFF	OFF

HSIO PORT	Function
USB3.0	1 USB3.0 Conn
	2 USB3.0 Conn
	3 NC
	4 NC
	5 NC
	6 NC
USB2.0	1 USB3.0 Conn
	2 NC
	3 USB3.0 Conn
	4 USB2.0 conn
	5 Card reader
	6 Touch Screen
	7 Camera
	8 NC
	9 NC
	10 Bluetooth
PCIE	5~8 X4 DGPU
	9 WLAN
	10 NC
	11 SATA HDD
	12 NC
	13~16 X4 PCIE/SATA SSD

SMBUS Control Table

	SOURCE	BATT	Charger	DGPU	IT8586E	Memory Down	PCH	PMIC	SODIMM	Thermal Sensor	WLAN WiMAX
EC_SMB_CLK1 EC_SMB_DA1	IT8586E +3VL_EC	V	V	X	V +3VL_EC	X	X	X	X	X	X
EC_SMB_CLK2 EC_SMB_DA2	IT8586E +3VS	X	X	V +3VG_AON	V +3VS	X	V +3VALW_PCH	X	X	V	X
EC_SMB_CLK3 EC_SMB_DA3	IT8586E +3VL_EC	X	X	X	V +3VL_EC	X	X	V	X	X	X
PCH_SMB_CLK PCH_SMB_DATA	PCH +3VALW_PCH	X	X	X	X	X	V +3VALW_PCH	X	V +3VS	X	V +3VS

EC SMBus1 address		EC SMBus2 address		EC SMBus3 address		PCH SM Bus address	
Device	Address	Device	Address	Device	Address	Device	Address
Smart Battery	need to update	Thermal Sensor(NCT7718W)	1001_100xb	PMIC	need to update	DDR4 SODIMM	need to update
Charger	0001 0010 b	PCH	need to update			Wlan	Reserved
		DGPU	need to update				

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				C		Notes For VGA		0.1	
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Virtual Symbol

PCB	ZZZ1 PCB 21M NM-D472 REV1 M/B DA800018210 HS460_DIS@ ZZZ2 PCB 21M NM-D472 REV1 M/B DA800018210 HS560_DIS@ ZZZ3 PCB 21M NM-D473 REV1 M/B DA800018310 HS760_DIS@ ZZZ4 PCB 21R NM-D475 REV1 M/B DA800018510 HV460_DIS@ ZZZ5 PCB 21R NM-D475 REV1 M/B DA800018510 HV560_DIS@ ZZZ6 PCB 21M NM-D471 REV1 M/B DA800018110 HS460_UMA@ ZZZ31 PCB 21M NM-D471 REV1 M/B DA800018110 HS560_UMA@ ZZZ32 PCB 21M NM-D471 REV1 M/B DA800018110 HS760_UMA@ ZZZ33 PCB 21R NM-D474 REV1 M/B DA800018410 HV460_UMA@ ZZZ34 PCB 21R NM-D474 REV1 M/B DA800018410 HV560_UMA@
CPU	UC1 S IC FH8069004531502 QV8G B1 3G BGA 1448 SA0000AY230 UC1 S IC FH8069004530601 QVBD B1 2.4G BGA SA0000AY230 UC1 S IC FH8069004529905 QVBA B1 2.8G BGA SA0000AY230 UC1 S IC FH8069004531901 QVBS B1 1.8G BGA 1448 SA0000BAC20 UC1 S IC FH8069004531802 QVBK B1 2G BGA 1449 SA0000B4920 UC1 S IC FH8069004531602 QVBH B1 3G BGA 1449 SA0000AY340 UC1 S IC FH8069004531301 QVBE B1 2.4G BGA SA0000AY240
GPU	UG1 S IC N175-G5-A1 BGA 595P GPU MP SA0000AEQ20 UG1 S IC N185-G5-A1 BGA 603P GPU MP SA0000AQJ40 PR6816 1180W 22.2K +/-1% 0402 SD00001N200 OPT17@ PR6816 1180W 22.2K +/-1% 0402 SD00001N200 OPT18@
DRAM MD	D16Gb MIC 3200 MT40A1G16KD-062E-E SA0000ADK00 D8Gb MIC 3200 MT40A512M16TB-062E-J SA0000R510 D8Gb SAM 3200 KAA8G165WC-BCWE SA0000ACB10 D16Gb HYX 3200 H5ANAG6CMR-XNC SA0000AC520 D8Gb HYX 3200 H5AN8G6NDJR-XNC SA0000AT510 D16Gb SAM 3200 KAAAG165WA-BCWE SA0000AC220 S IC D4 512MX16/3200 SDQC8GW16XCWE9N1T FBGA 96P SA0000BC360
VRAM	S IC D5 256M32/2000 K4G80325FC-HC2 SA00007Z720 S IC D5 256M32 H5GC8H24AJR-R2C SA00009FV00 S IC D5 256M32 MT51J256M32HF-80-B SA00007Z930
X76	
DRAM SODIM	
Re-Driver	UK1 S IC P3EQX8741STZDEX TQFN SATA REDRIVER SA000009400 SATA_6741@

Virtual Symbol_Power

15W	25W
PR1045 16.9K_0402_1% SD003416926J 15W@ PR1044 24.3K_0402_1% SD000011300 15W@ PR1043 11.8K_0201_1% SD003416926J 15W@ PR1039 1180W 22.2K +/-1%_0402 SD003416926J 15W@	PR1045 12.1K_0402_1% SD003412126J 25W@ PR1044 21.5K_0402_1% SD003421526J 25W@ PR1043 11.8K_0201_1% SD003416926J 25W@ PR1039 1200W 14.3K_1%_0201 SD003416926J 25W@
PR1008 1/20W_200K_1%_0402 SD003416926J 15W@ PR1021 1/8K_0402_1% SD003417836T 15W@ PR1011 178K_0402_1% SD003417836T 15W@	PR1008 1/20W_83.1K_1%_0402 SD003416926J 25W@ PR1021 22.6K_0402_1% SD003416926J 25W@ PR1011 19.6K_0402_1% SD003416926J 25W@
PR1024 1/20W_8.45K_1%_0201 SD000013JBT 15W@	PR1024 1/20W_14K_1%_0201 SD00001V000 25W@

PU2000 RT_OPT@
RT8816CGQW WQFN 20P PWM
SA0000B3F00

PU2000 UPL_OPT@
UP1666QKQF WQFN 20P CONTROLLER
SA00008G000

GPU Power Solution

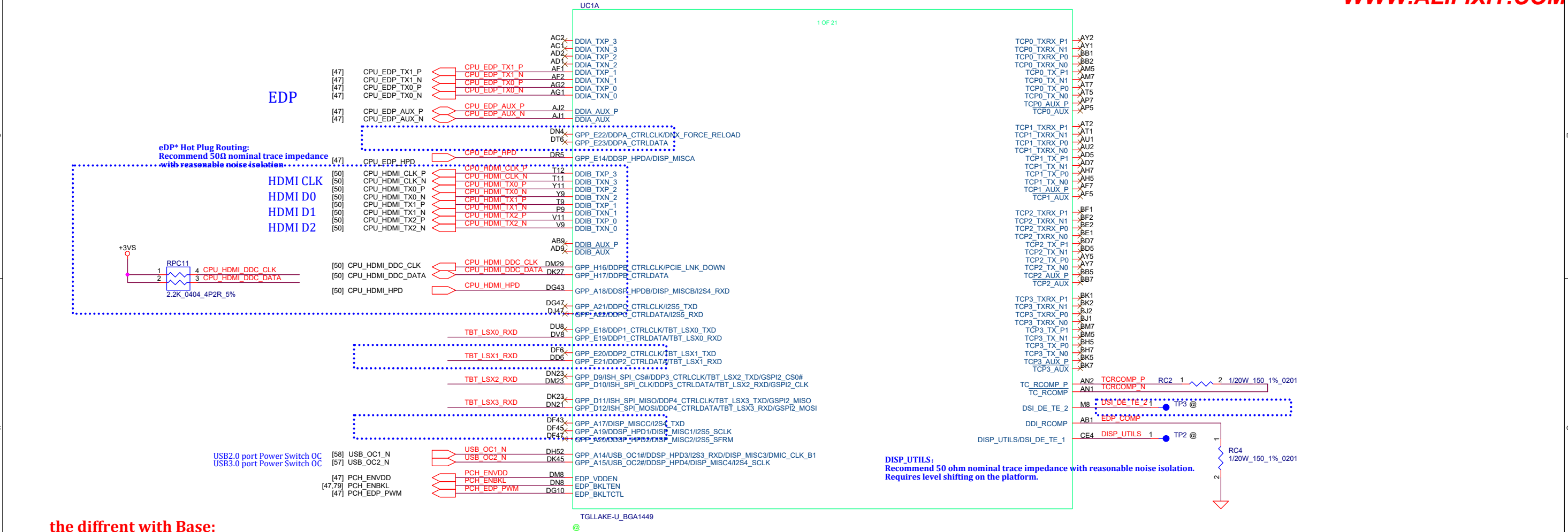
HDMI Royalty

ZZZ38 HDMI@
HDMI PN
RC00000040J

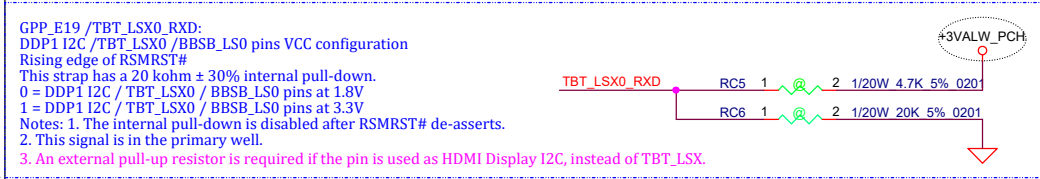
D8G_MIC_S@ ZZZ6 MIC D8G S360 X764BF1200P	D8G_SAM_S@ ZZZ7 SAM D8G S360 X764BF1200Q	D8G_HYX_S@ ZZZ8 HYX D8G S360 X764BF1200M	D8G_MIC_V@ ZZZ23 MIC D8G V14V15 X764BE1200K	D4G_SMT_V@ ZZZ36 SMT D4G V14V15 X764BE1200L	D8G_SAM_V@ ZZZ20 SAM D8G V14V15 X764BE1200M	D8G_HYX_V@ ZZZ22 HYX D8G V14V15 X764BE1200R
D4G_SAM_S@ ZZZ9 SAM D4G S360 X764BF1200K SSAMV2GN17@	D4G_MIC_S@ ZZZ10 MIC D4G S360 X764BF1200N	D4G_HYX_S@ ZZZ11 HYX D4G S360 X764BF1200R	D4G_SMT_S@ ZZZ35 SMT D4G S360 X764BF1200L	D4G_MIC_V@ ZZZ19 MIC D4G V14V15 X764BE1200Q	D4G_HYX_V@ ZZZ18 HYX D4G V14V15 X764BE1200P	
ZZZ12 SAM V2G N17 S360 X764BF1200J	ZZZ13 MIC V2G N18 S360 X764BF1200G	ZZZ14 MIC V2G N17 S360 X764BF1200E	ZZZ24 SAM V2G N17 V14V15 X764BE1200J	ZZZ25 MIC V2G N17 V14V15 X764BE1200E	ZZZ26 HYX V2G N17 V14V15 X764BE1200G	
SHYNY2GN18@ ZZZ15 HYN V2G N18 S360 X764BF1200H	SHYNY2GN17@ ZZZ16 HYN V2G N17 S360 X764BF1200F	SSAMV2GN18@ ZZZ17 SAM V2G N18 S360 X764BF1200D	VHYXV2GN18@ ZZZ27 HYX V2G N18 V14V15 X764BE1200F	VMICV2GN18@ ZZZ28 MIC V2G N18 V14V15 X764BE1200H	VSAMV2GN18@ ZZZ29 SAM V2G N18 V14V15 X764BE1200D	

BOM Structure

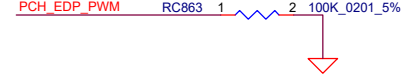
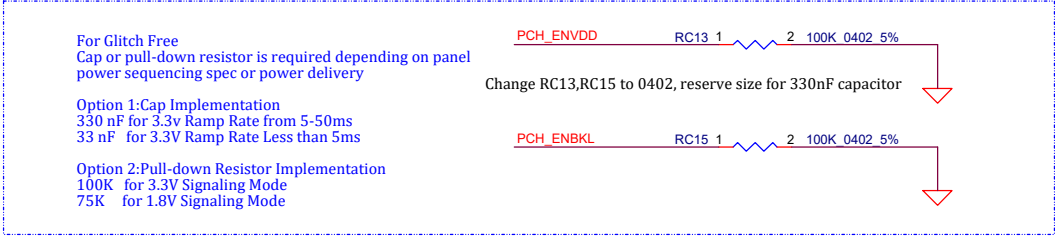
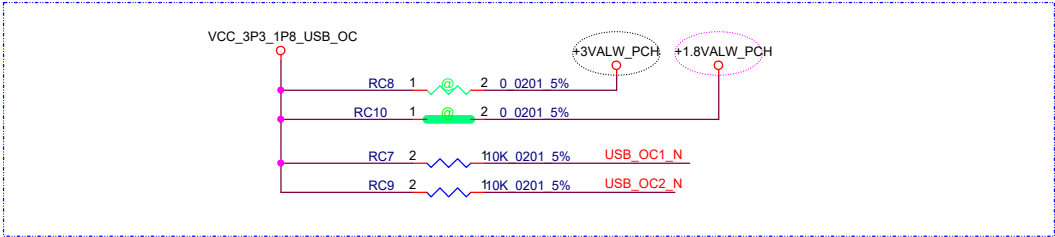
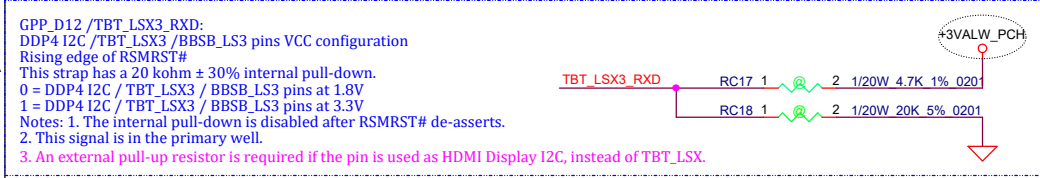
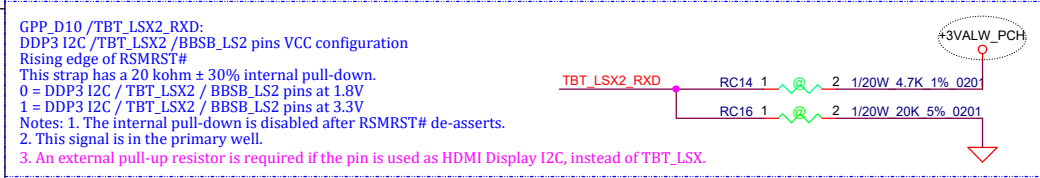
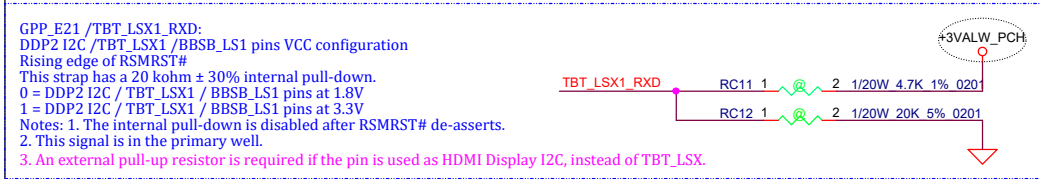
BOM Structure	Description
@	Un-stuff
FHD@	For FHD Panel Part
HDR@	For HDR Panel Part
14@	For GYG41 part
15@	For GYG51 part
8M_ROM@	For 8G BIOS ROM part
AMP@	For external Amplifier solution part
AMP_NS@	For external Amplifier solution un-stuff part
AMP_PWR@	For discrete Amplifier power solution part
ANS@	For power ANS solution part
USM@	For power USM solution part
BL@	For Key Board Backlight part
CD@	For C cost down
CHANNELA@	For Memory Channel A part
DDPA@	For Memory Channel A DDP part
SDPA@	For Memory Channel A SDP part
DDPB@	For Memory Channel B DDP part
SDPB@	For Memory Channel B SDP part
NPI@	For NPI Phase part
MP@	For MP Phase Part
PCIE4@	For PCIE Gen4 part
EMC@	For EMC part
EMC_15@	For GYG51 EMC part
EMC_NS@	For EMC un-stuff part
RF_NS@	For RF Un-Stuff part
AOAC@	For AOAC part
CNVI@	For CNVi WLAN part
ME@	For ME part
OPT@	For NV GPU part
OPTN16@	For NV N16S-GTR GPU part
OPTN17@	For NV N17S-G1 GPU part
TS@	For touch screen part
TP@	For Touch Pad Part
UMA@	For UMA part



the diffrent with Base:
1.S540 no HDMI,S360&V14&V15 Ues HDMI;
2.S540 Use TCP for TBT,S360&V14&V15 no USE
3.S540 Use GPP_E22&E23,S360&V14&V15 no USE
4.S540 Stuff RC12,S360&V14&V15----- @
5.S540 No TP3, S360&V14&V15 reserve



Confirmed with Intel Kaiyin follow TGP EDS 20200103



Primary Well Group A (Per-Pad 1.8 V)
Primary Well Group B (Per-Pad 3.3 V)
Primary Well Group C (Per-Pad 3.3 V)
Primary Well Group D (Per-Pad 3.3 V)
Primary Well Group E (Per-Pad 3.3 V)
Primary Well Group F (Per-Pad 1.8 V)
Primary Well Group H (Per-Pad 3.3 V)
Primary Well Group R (Per-Family 1.8 V)
Primary Well Group S (Per-Family 1.8 V)
Primary Well Group T (Per-Pad 1.8 V or 3.3 V) - Applied to TGL PCH UP3 Only
Primary Well Group U (Per-Pad 1.8 V or 3.3 V) - Applied to TGL PCH UP3 Only
Deep Sleep Well Group (3.3 V Only)(GPD0~GPD11)

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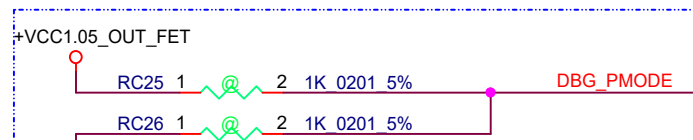
SLOT DIMM



the diffrent with Base:
1.S540 BT42&BT41BU52&CF42*CF44---- NC,
S360&V14V15 use For SO-DIMM

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				C	DDR4-Chanel-A	<RevCode>	
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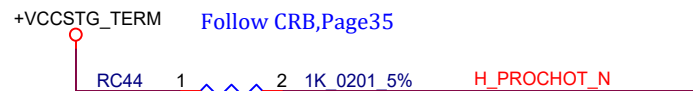


DBG_PMODE(Reserved): Rising edge of RSMRST#
This strap has a 20 kohm $\pm$ 30% internal pull-up.
This strap should sample high. There should NOT be any on-board device driving it to opposite direction during strap sampling.

Notes:

1. The internal pull-up is disabled after RSMRST# deasserts.
2. This signal is in the primary well.

Confirmed with Intel Kaiyin, follow PDG 20200103

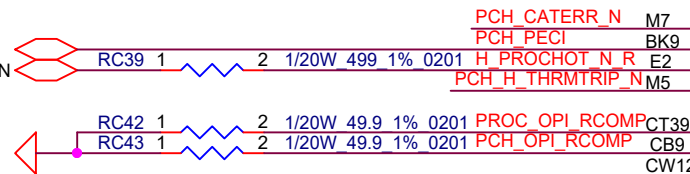


Follow CRB,Page35



[79]
[20,79]

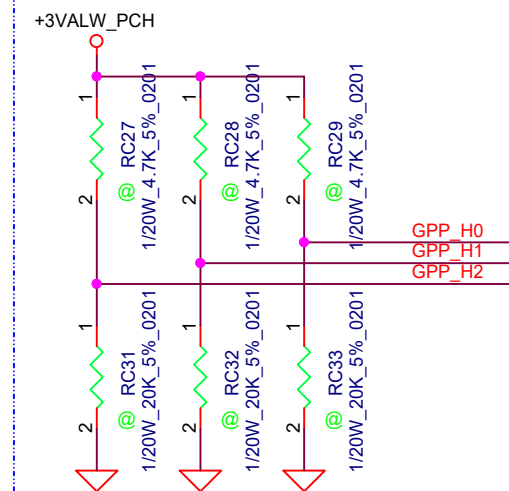
PCH_PECI
H_PROCHOT_N



[79]

EC_PCH_SCI_N

EC_PCH_SCI_N

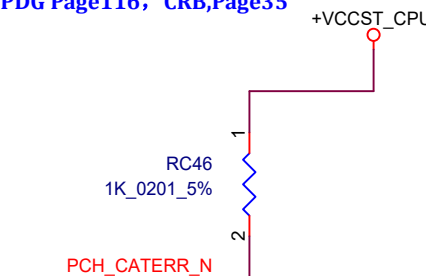


GPP_H0: Boot Strap 1
GPP_H1: Boot Strap 2
GPP_H2: Boot Strap 3
Boot Strap, Rising edge of RSMRST#
These straps has a 20 kohm $\pm$ 30% internal pull-down.
They are bit [3:1] of a total of 4-bit encoded pin straps for boot configuration.
Refer to Boot Strap 0 (on GPP_C5) for the encoding.
Notes: 1. The internal pull-down is disabled after RSMRST# de-asserts.
2. This signal is in the primary well.

0000 = Master Attached Flash Configuration (BIOS / CSME on SPI). eSPI is enabled
0010 = Master Attached Flash Configuration (BIOS / CSME on SPI). eSPI is disabled
0100 = BIOS on eSPI Peripheral Channel; CSME on master attached SPI
1000 = Slave Attached Flash Configuration (BIOS / CSME on eSPI attached device).
1100 = BIOS on eSPI peripheral Channel; CSME on slave attached SPI.

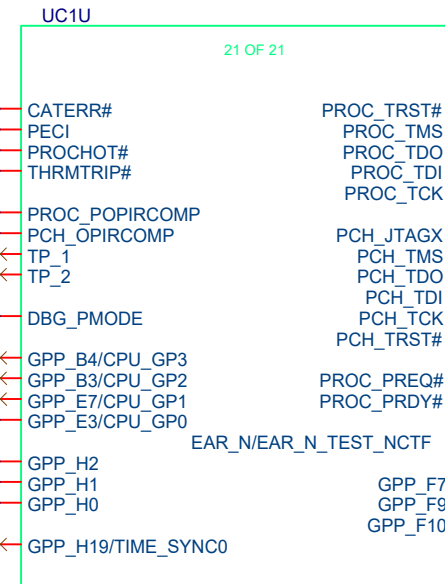
the different with Base:
1.S540 CM39 PIN---- NC,S360&V14V15 Reserve TP9;
2.S540 DB42 PIN For TBT_FORCE_PWR,S360&V14V15 --NC

PDG Page116, CRB,Page35



Place at C/S side for Debug easy

EC can not support, Delete
--20200227



TGLLAKE-U_BGA1449

@

CPU_JTAG_TCK RC48 1 2@ 0_0201_5%
CPU_JTAG_TRST_N RC233 1 2@ 0_0201_5%
CPU_JTAG_TMS RC234 1 2@ 0_0201_5%
CPU_JTAG_TDO RC235 1 2@ 0_0201_5%
CPU_JTAG_TDI RC236 1 2@ 0_0201_5%

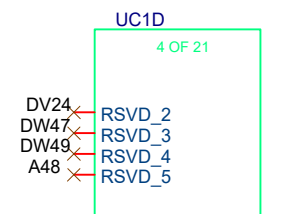
PCH_JTAG_JTAGX RC48 1 2@ 0_0201_5%
PCH_JTAG_TMS RC233 1 2@ 0_0201_5%
PCH_JTAG_TDO RC234 1 2@ 0_0201_5%
PCH_JTAG_TDI RC235 1 2@ 0_0201_5%
PCH_JTAG_TCK RC236 1 2@ 0_0201_5%

PROC_TRST# K4
PROC_TMS B9
PROC_TDO D12
PROC_TDI A12
PROC_TCK B6

PCH_JTAGX D8
PCH_JTAGTMS A9
PCH_JTAGTDO E12
PCH_JTAGTDI B12
PCH_JTAGTCK A7
PCH_JTAGTRST_N H4

PROC_PREQ# C11
PROC_PRDY# D11

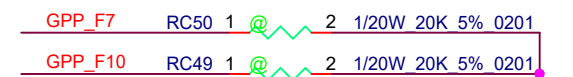
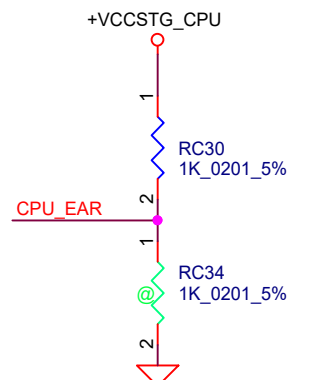
CPU_EAR G1
GPP_F7 DT15
GPP_F9 DR15
GPP_F10 DT14



TGLLAKE-U_BGA1449

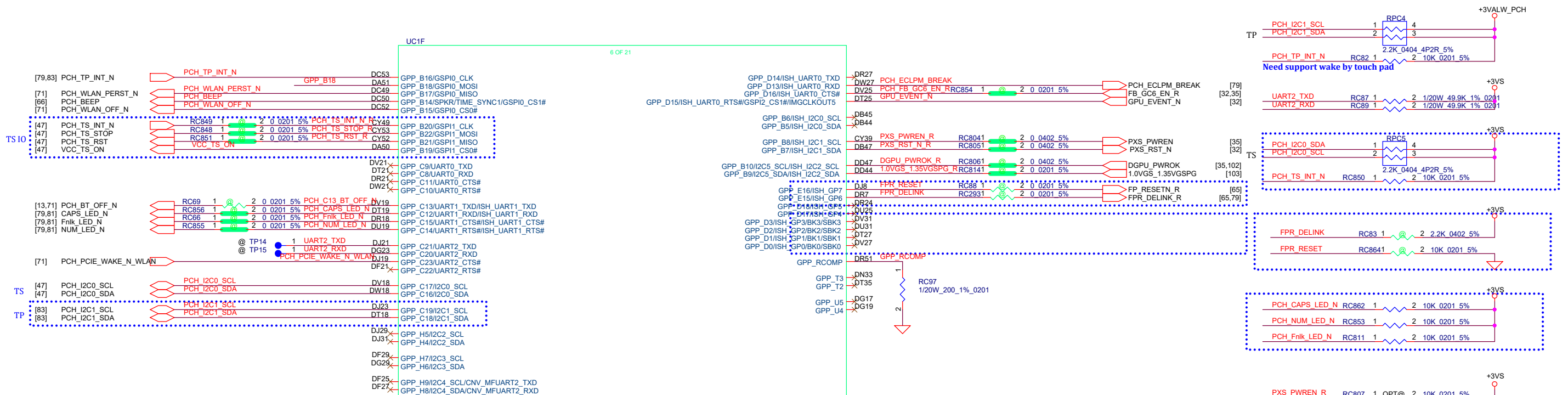
@

EAR
Stall CPU reset sequence until de-asserted:
- 1 = (Default) Normal Operation; No stall.
- 0 = Stall;
Follow PDG Page118

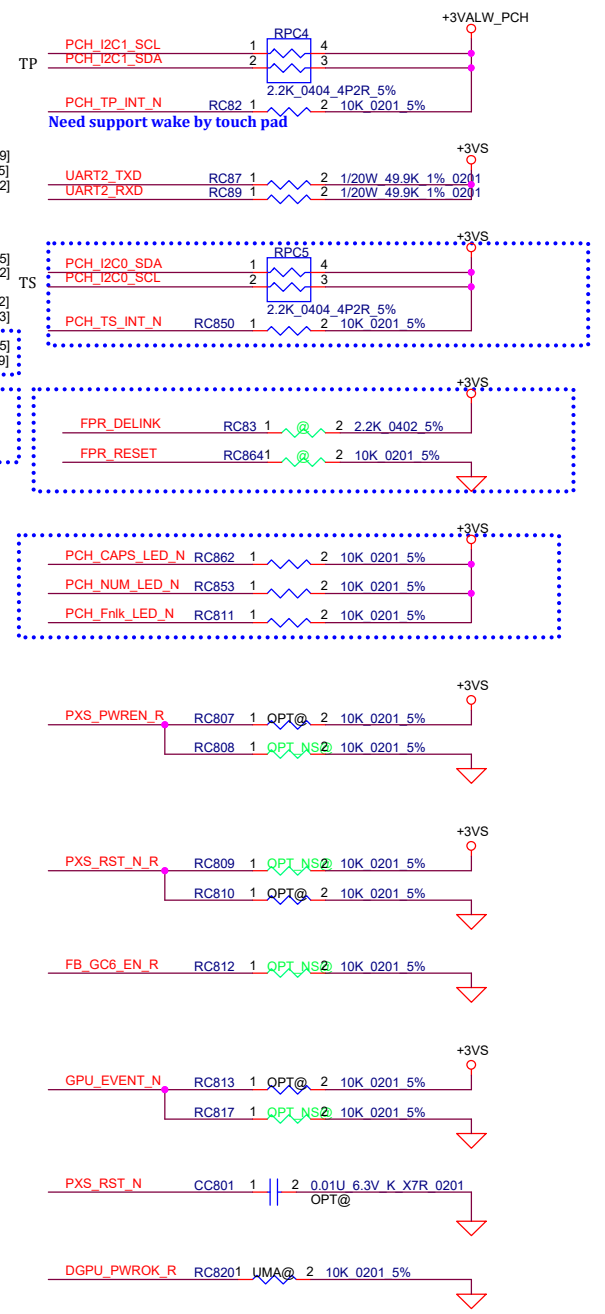
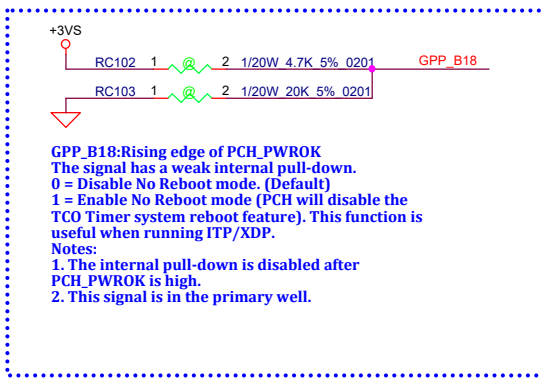
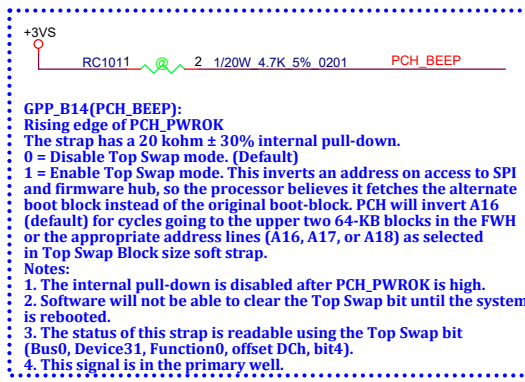
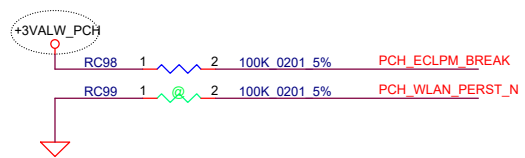


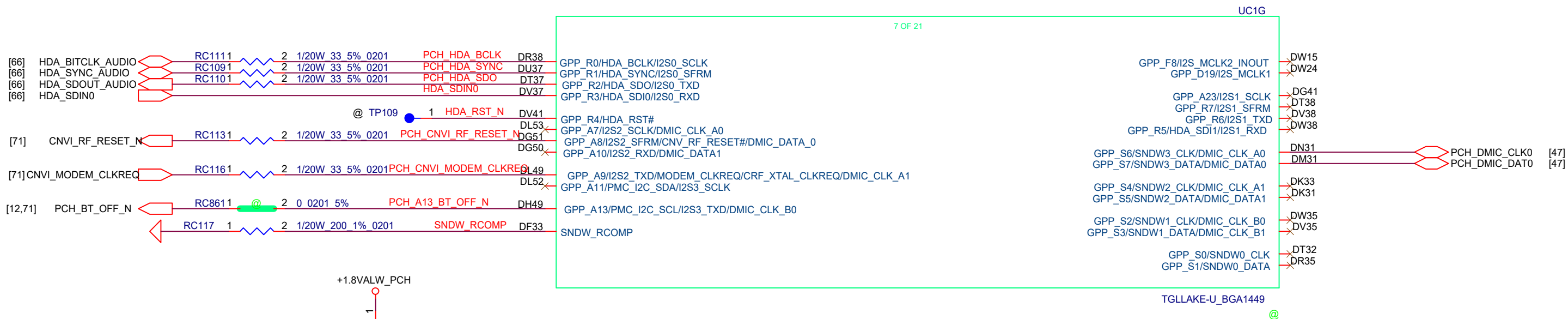
GPP_F7 and GPP_F10:
Reserved, Rising edge of RSMRST#
This strap has a 20 kohm $\pm$ 30% internal pull-down.
This strap should sample LOW. There should NOT be any onboard device driving it to opposite direction during strap sampling.
Notes: 1. The internal pull-down is disabled after RSMRST# de-asserts.
2. This signal is in the primary well.

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								MISC/JTAG		
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the diffrent with Base:
1.S540 NO Touch scrren &I2C for touch, S360&V14V15 use
2.S540 has Tof& I2C for TOF;S360&V14V15 no use
3.S540 NO Fingerprinter, S360&V14V15 use;
4.S540 NO num&fnlink LED, S360&V14V15 use
5.S540 DD44 PIN----1.0VGS_1.25VGSPG, S360&V14V15 DD44 PIN----1.0VGS_1.35VGSPG
6.S540 DV31 PIN----PCH_LID_OD_SW_N,S360&V14V15---NC
7.S540 DJ8 PIN----ISH_GP7_ITS_R,S360&V14V15---FPR_RESET
8.S540 DR7 PIN----NC,S360&V14V15---FPR_DELINK





GPP_R2(HDA_SDO_R):

This signal has a 20K ±30% internal pull-down.

0 = Enable security measures defined in the Flash Descriptor. (Default)
1 = Disable Flash Descriptor Security (override). This strap should only be asserted high using external Pull-up in manufacturing/debug environments ONLY.

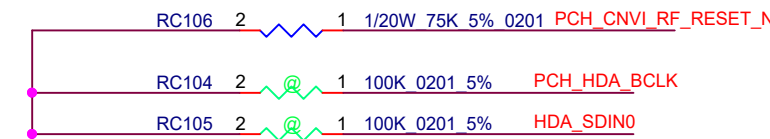
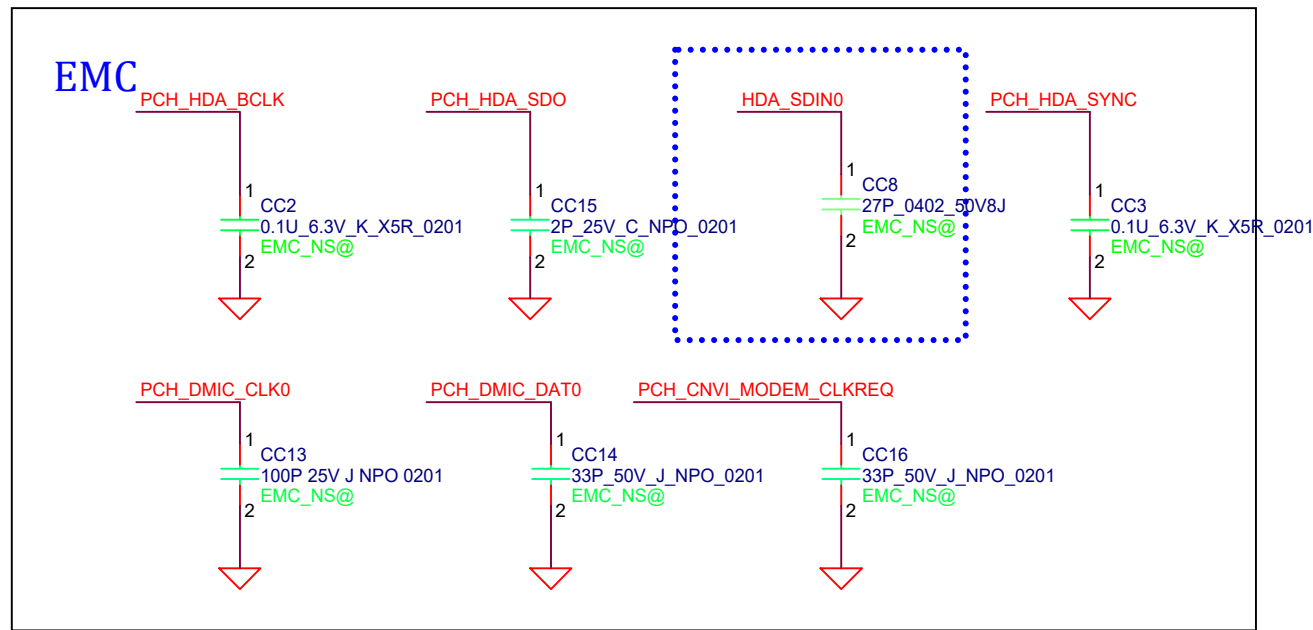
Notes:

1. The internal pull-down is disabled after PCH_PWROK is high.
2. This signal is in the primary well.

the diffrent with Base:

1.S540 PCH_DMIC_CLK0 CC13 Stuff,S360&V14V15---NC:

2.S540 PCH_HDA_SDI no EMC reserve,S360&V14V15 reserve CC8



Glitch Free Requirements:
Pull-down resistor to ensure the stability of the signal during platform bootup

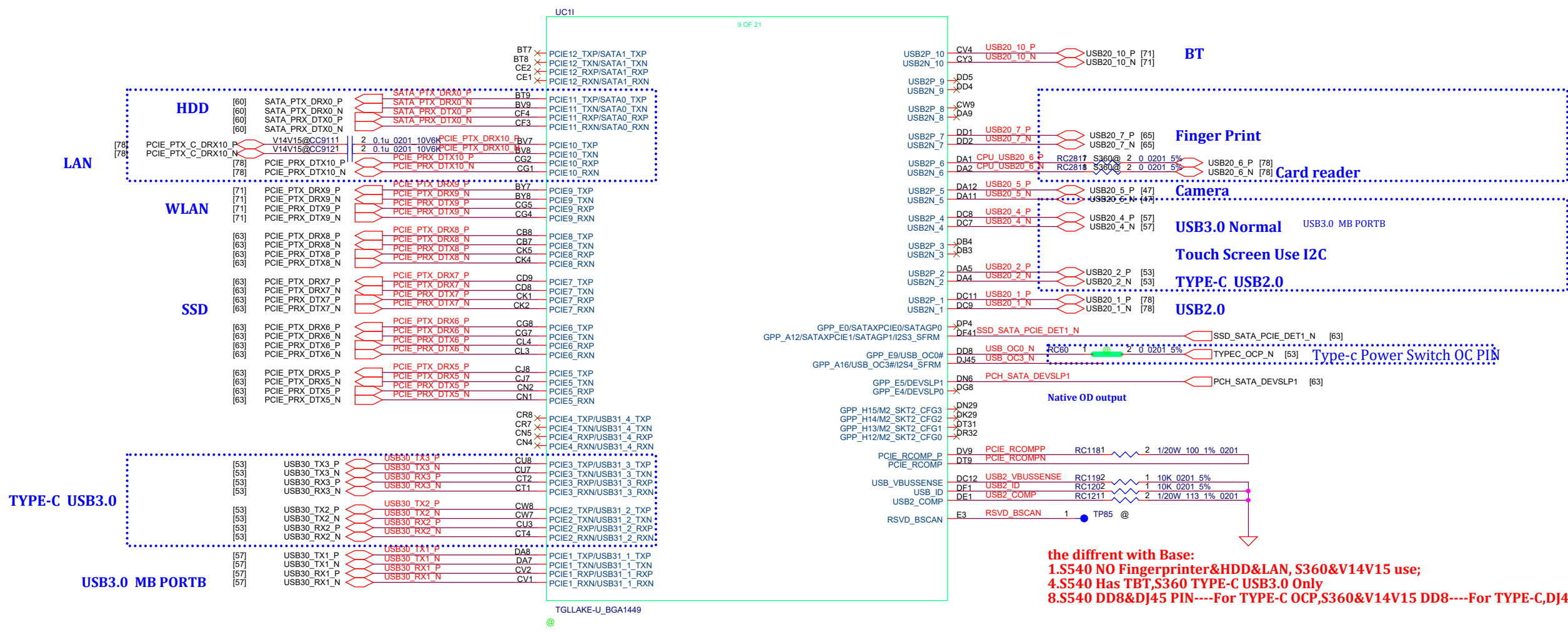
Option 1:Cap Implementation

- NA for 3.3v Ramp Rate from 5-50ms
- NA for 3.3V Ramp Rate Less than 5ms

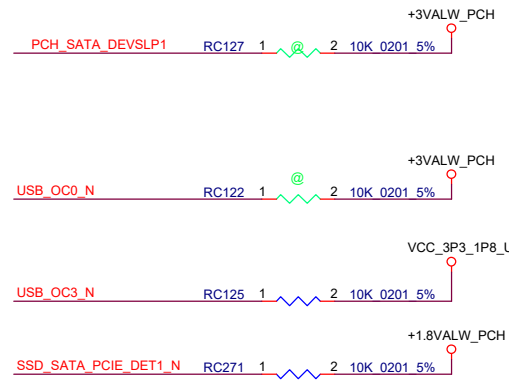
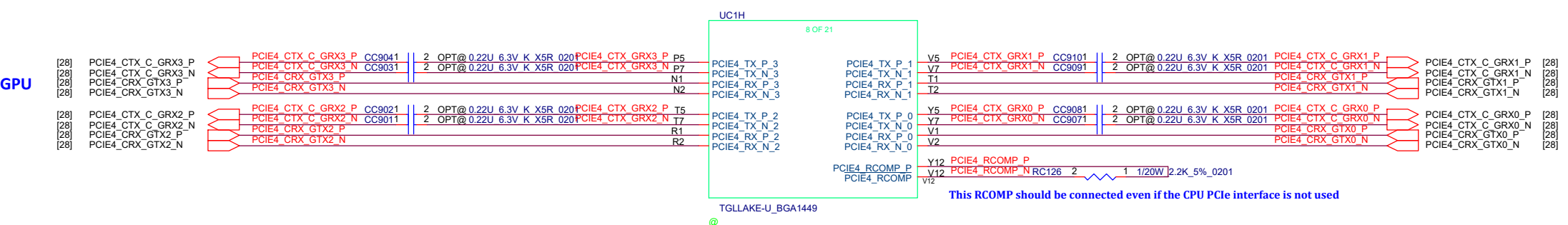
Option 2:Pull-down Resistor Implementation

- NA for 3.3V Signaling Mode
- 75K for 1.8V Signaling Mode

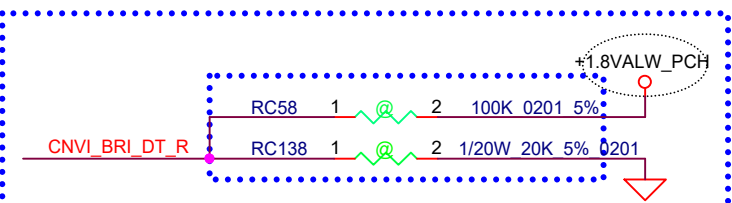
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				AUDIO	
				Date:	Tuesday, November 10, 2020
				Sheet	13 of 110
				Rev	0.1



the diffrent with Base:
1.S540 NO Fingerprinter&HDD&LAN, S360&V14V15 use;
4.S540 Has TBT,S360 TYPE-C USB3.0 Only
8.S540 DD8&DJ45 PIN----For TYPE-C OCP,S360&V14V15 DD8----For TYPE-C,DJ45-pull up



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				PCIE/USB/SATA	0.1
				Date: Tuesday, November 10, 2020	Sheet 14 of 110



PU at M.2 side
Follow CNVI Design Guide
Confirm with Intel

CNVI RGI_DT_R

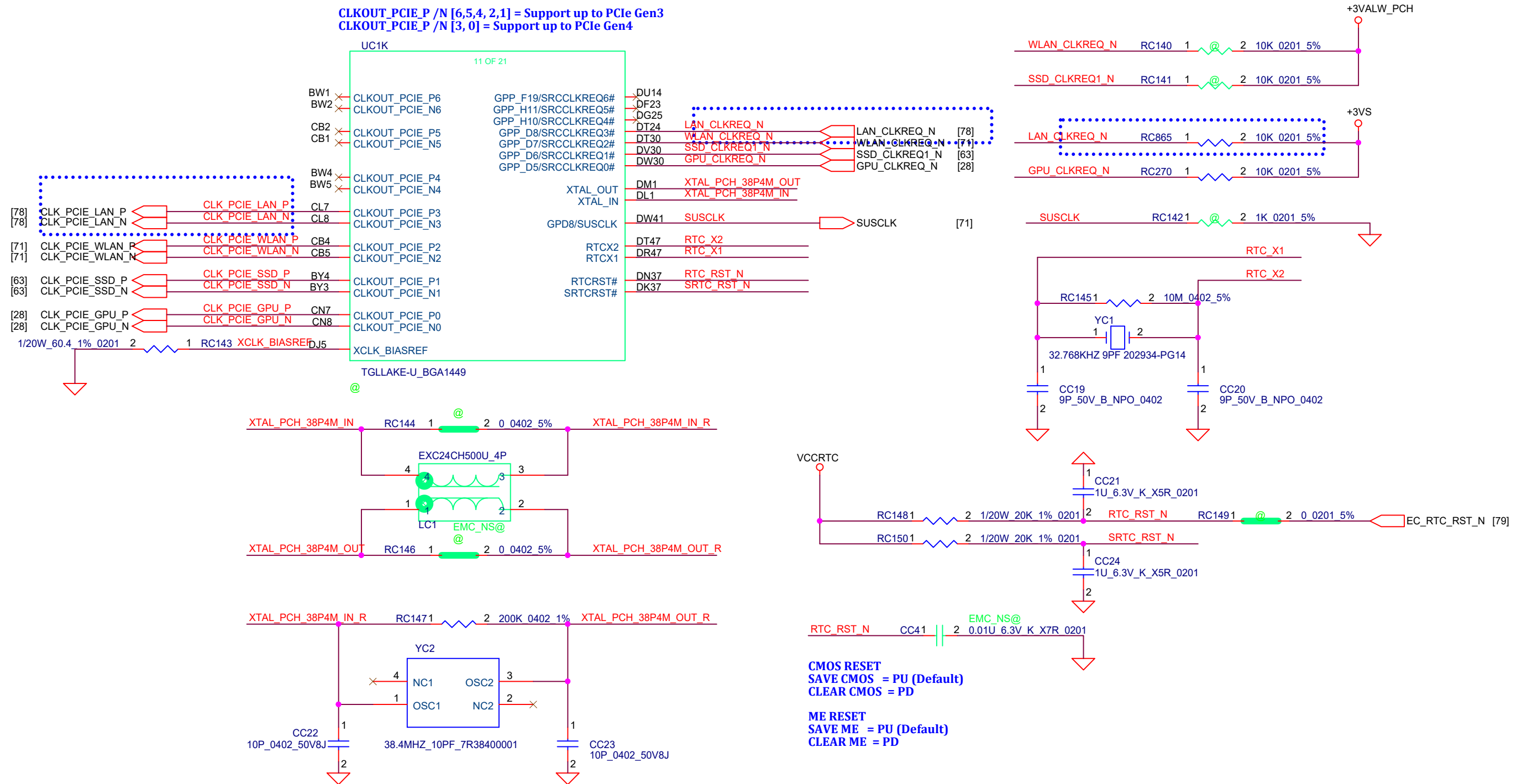
RC132 1 @ 2 100K 0201 5%

RC133 1 @ 2 1/20W 4.7K 5% 0201

+1.8VALW_PCH

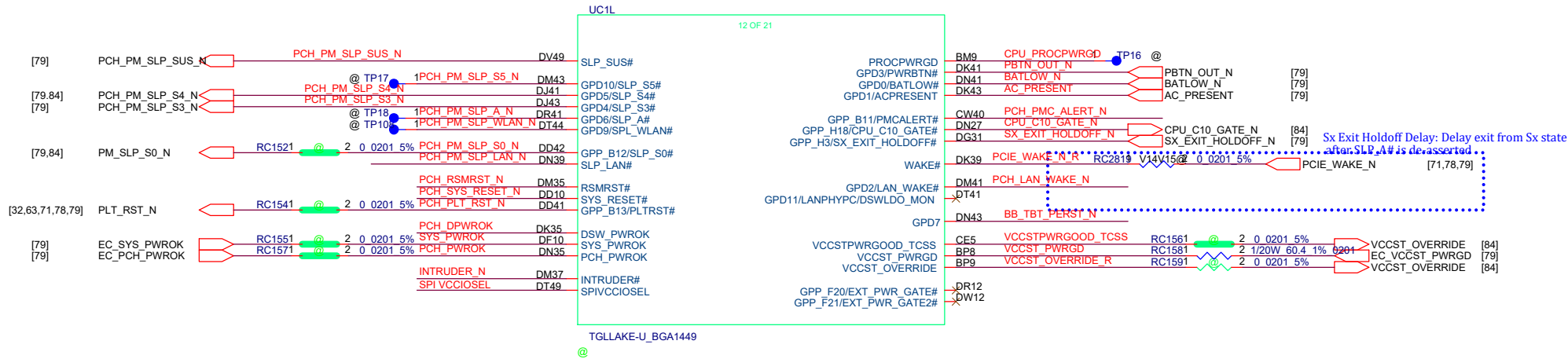
GPP_F2 /CNV_RGL_DT /UART0_TXD:
M.2 CNVi MODES, Rising edge of RSMRST#
A weak external pull-up is required.
0 = Integrated CNVi enabled.
1 = Integrated CNVi disabled.
Note:
When a RF companion chip is connected to the PCH CNVi interface,
the device internal pulldown resistor will pull the strap low to
enable CNVi interface.

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Date:		Tuesday, November 10, 2020		Sheet 15 of 110	



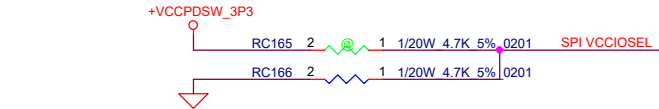
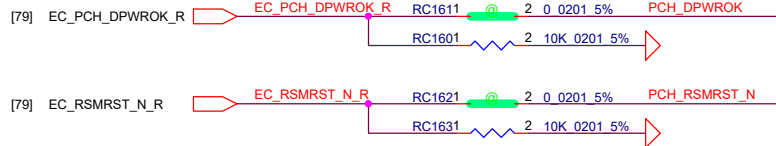
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				CLOCK SIGNALS	
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LCFC



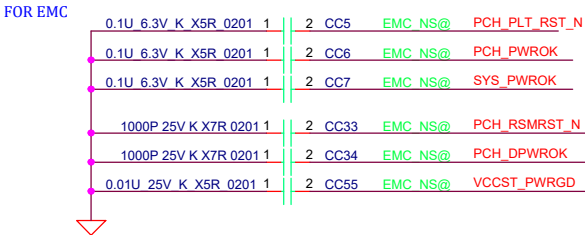
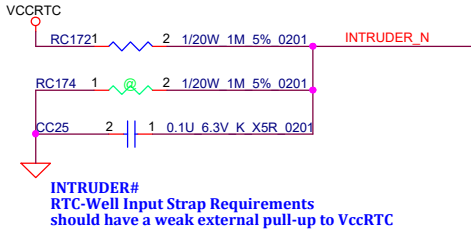
the diffrent with Base:
1.S540 DM41 no LAN, S360&V14V15 Lan to page78;
2.1.S540 PCH_PM_SLP_WLAN_N no reserve,S360&V14V15 reserve RC186

DSW_PWROK and RSMRST# are always separate power good signals



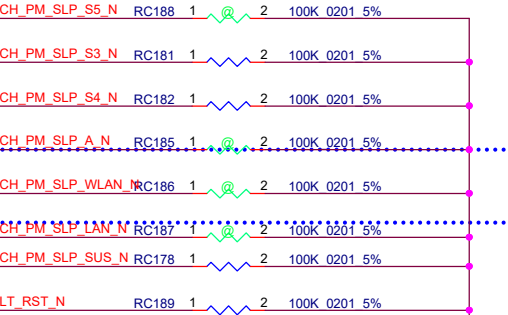
The VCCSPI voltage (3.3V or 1.8V) is selected via a strap on SPIVCCIOSEL:
This strap sets the SPI interface signaling voltage at the rising edge of DSW_PWROK.
Designers should strap this pin to match the expected interface operational voltage for their target SPI device as follows.
0 = SPI voltage is 3.3V (4.7K ohm pull-down to GND);
1 = SPI voltage is 1.8V (4.7K ohm pull-up to VCCDSW_3P3)

CAD NOTE:
INPUT3VSEL: 3V SELECT STRAP
LOW-> 3.3V +/-5%
HIGH->3.0V +/-5%

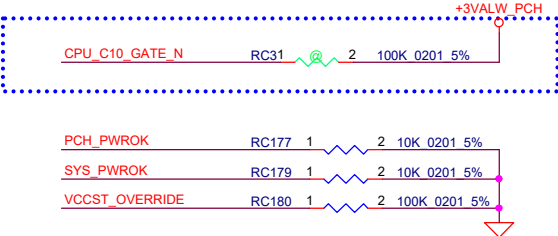
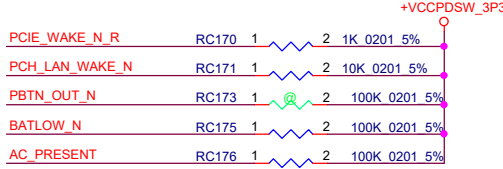
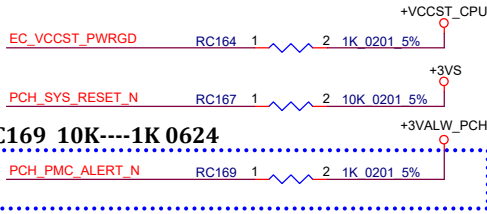


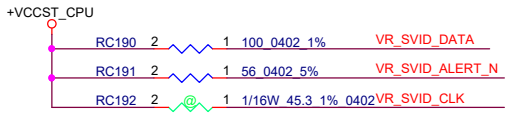
Glitch Free Requirements:
Pull-up resistor is required if a device is monitoring SLP_S0#
before RSMRST# de-assertion
100K for 3.3V Signaling Mode
75K for 1.8V Signaling Mode

BB_TBT_PERST#: Rising edge of DSW_PWROK
This signal has a 20K+-30% internal pull-down.
This strap should sample LOW. There should NOT be any on-board device driving it to opposite direction during strap sampling
Notes:
1. The internal pull-down is disabled after DSW_PWROK is high;
2. This signal is in the DSW well

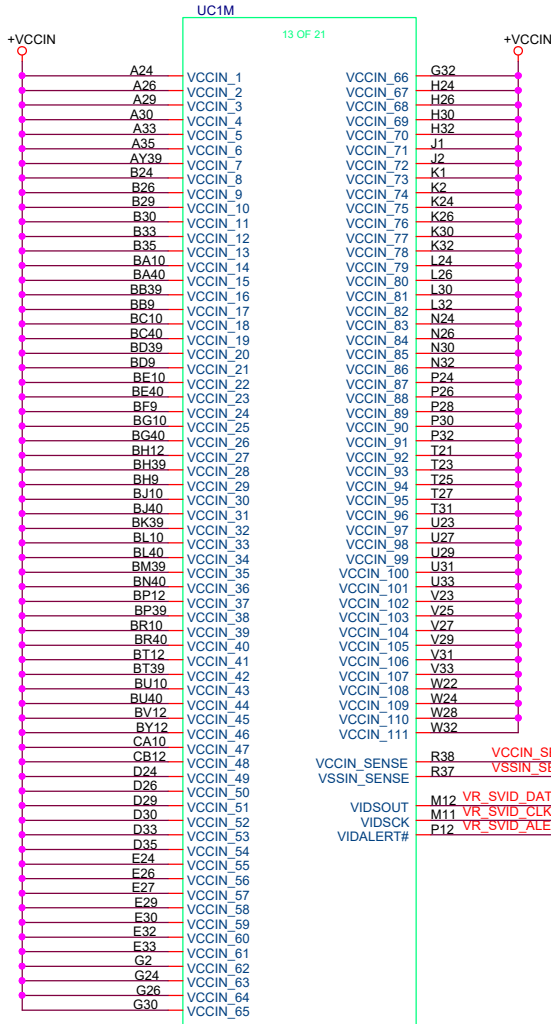
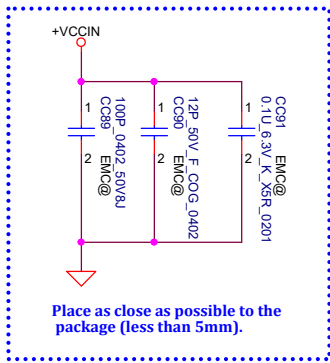
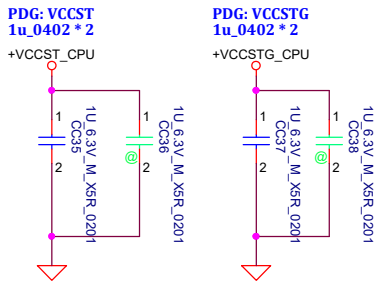
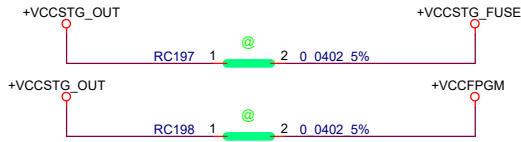
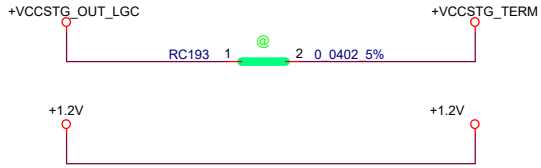


Glitch Free Requirements:
Cap or pull-down resistor is required
Option 1:Cap Implementation
330 nF for 3.3v Ramp Rate from 5-50ms
33 nF for 3.3V Ramp Rate Less than 5ms
Option 2:Pull-down Resistor Implementation
100K for 3.3V Signaling Mode
75K for 1.8V Signaling Mode

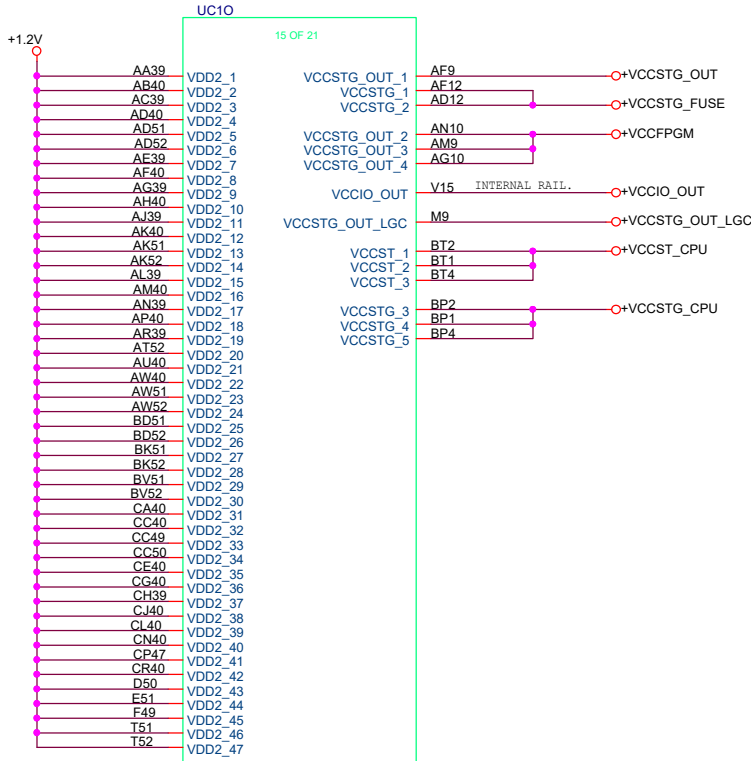




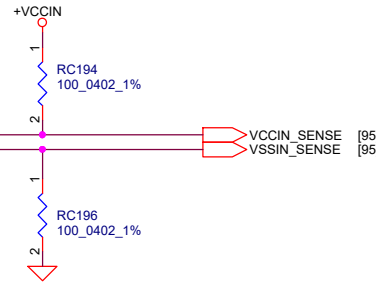
CAD NOTE:
Alert signal must be routed between Clk and Data signals
to minimize Cross-Talk.



TGLLAKE-U_BGA1449

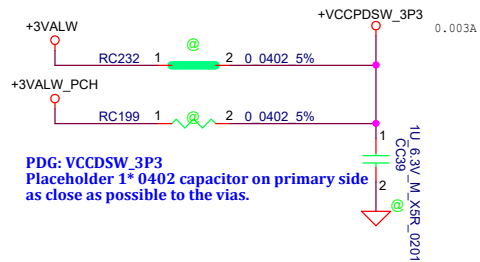
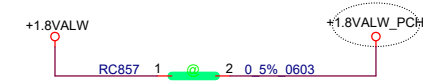


TGLLAKE-U_BGA1449

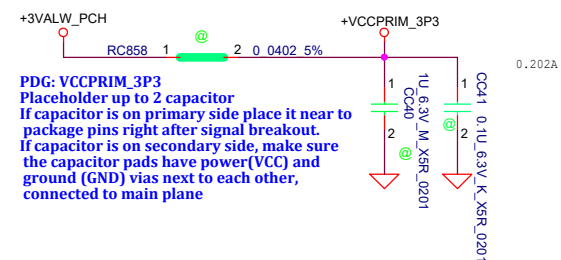


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				CPU Power (1/2)	0.1
				Date:	Tuesday, November 10, 2020
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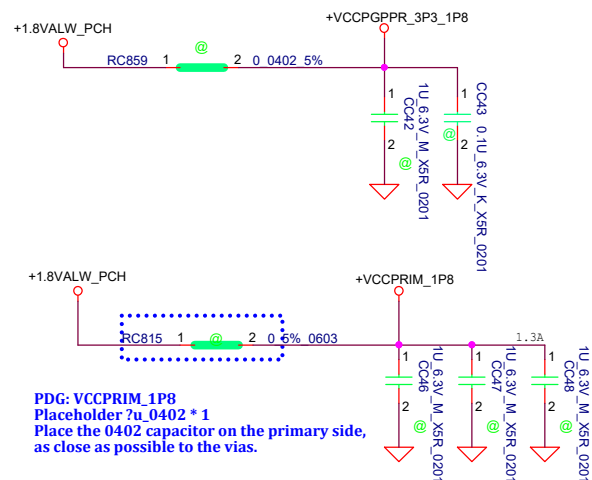
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				C	CPU Power (2/2)	Rev 0.1
				Date:	Tuesday, November 10, 2020	Sheet 19 of 110



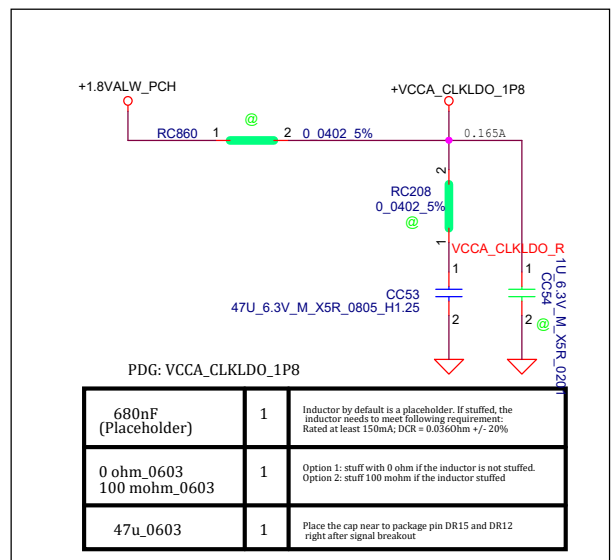
PDG: VCCDSW_3P3
Placeholder 1* 0402 capacitor on primary side as close as possible to the vias.



PDG: VCCPRIM_3P3
Placeholder up to 2 capacitor
If capacitor is on primary side place it near to package pins right after signal breakout.
If capacitor is on secondary side, make sure the capacitor pads have power(VCC) and ground (GND) vias next to each other, connected to main plane

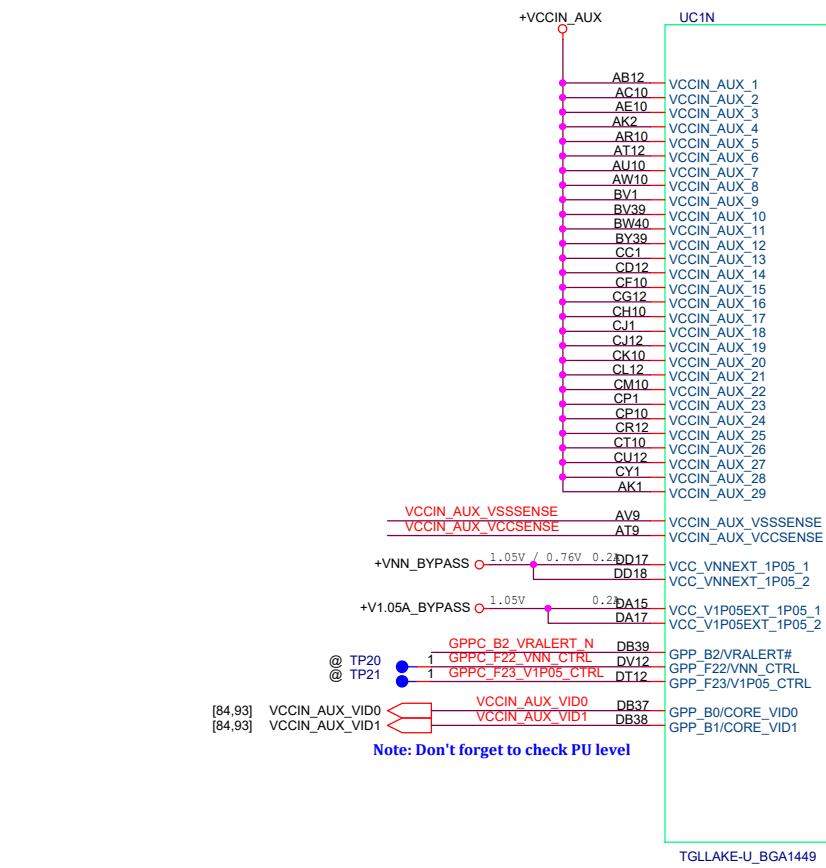


PDG: VCCPRIM_1P8
Placeholder 2u 0402 * 1
Place the 0402 capacitor on the primary side, as close as possible to the vias.



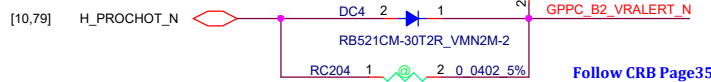
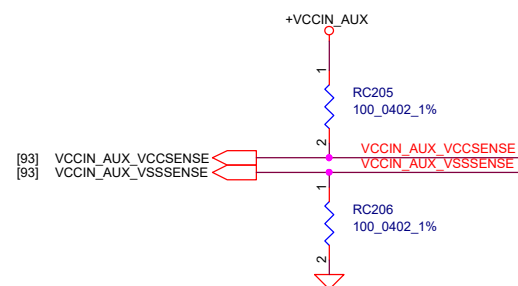
PDG: VCCA_CLKLDO_1P8

680nF (Placeholder)	1	Inductor by default is a placeholder. If stuffed, the inductor needs to meet following requirement: Rated at least: 150mA; DCR = 0.0550mΩ +/- 20%.
0 ohm_0603 100 mohm_0603	1	Option 1: stuff with 0 ohm if the inductor is not stuffed. Option 2: stuff 100 mohm if the inductor is stuffed.
47u_0603	1	Place the cap near to package pin DR15 and DR12 right after signal breakout

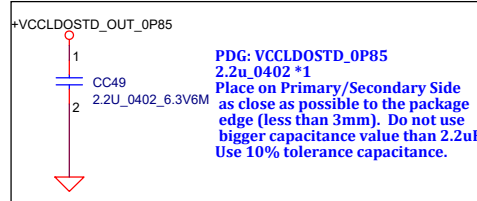


Note: Don't forget to check PU level

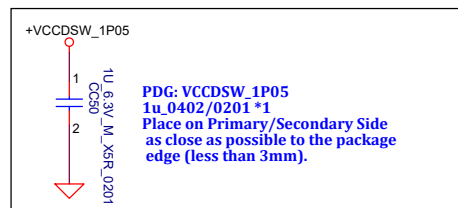
the diffrent with Base:
1.S540 +1.8VALW_PCH To +VCCPRIM_1P8 directly , S360&V14V15 use RC815;



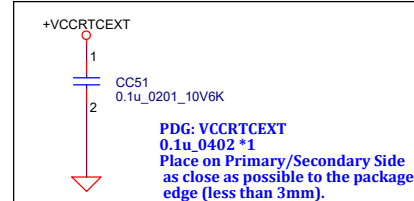
Follow CRB Page35



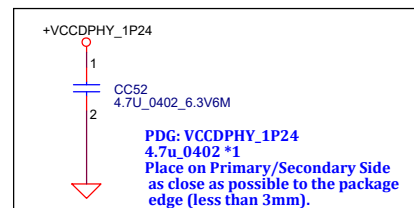
PDG: VCCLDOSTD_0P85
2.2u_0402 *1
Place on Primary/Secondary Side as close as possible to the package edge (less than 3mm). Do not use bigger capacitance value than 2.2uF. Use 10% tolerance capacitance.



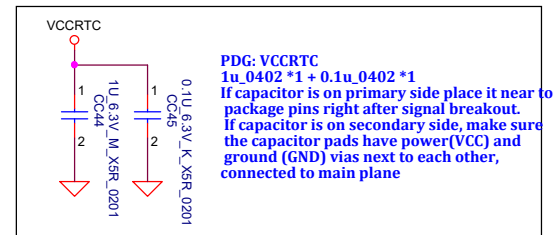
PDG: VCCDSW_1P05
1u_0402/0201 *1
Place on Primary/Secondary Side as close as possible to the package edge (less than 3mm).



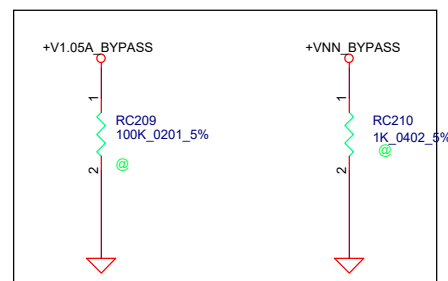
PDG: VCCRTC
0.1u_0402 *1
Place on Primary/Secondary Side as close as possible to the package edge (less than 3mm).



PDG: VCCDPHY_1P24
4.7u_0402 *1
Place on Primary/Secondary Side as close as possible to the package edge (less than 3mm).



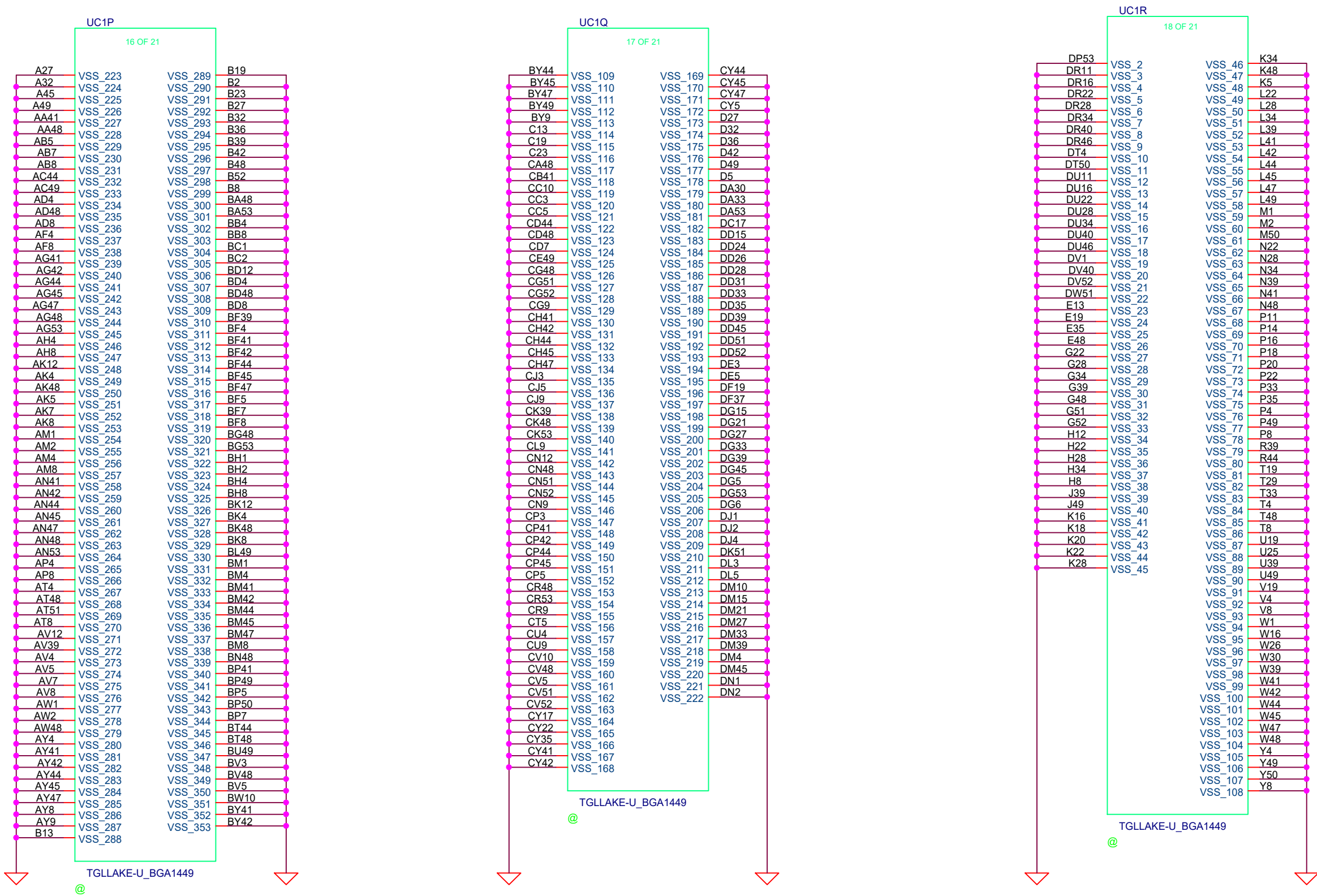
PDG: VCCRTC
1u_0402 *1 + 0.1u_0402 *1
If capacitor is on primary side place it near to package pins right after signal breakout.
If capacitor is on secondary side, make sure the capacitor pads have power(VCC) and ground (GND) vias next to each other, connected to main plane



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			Document Number
			PCH Power
			Rev 0.1
			Date: Tuesday, November 10, 2020
			Sheet 20 of 110

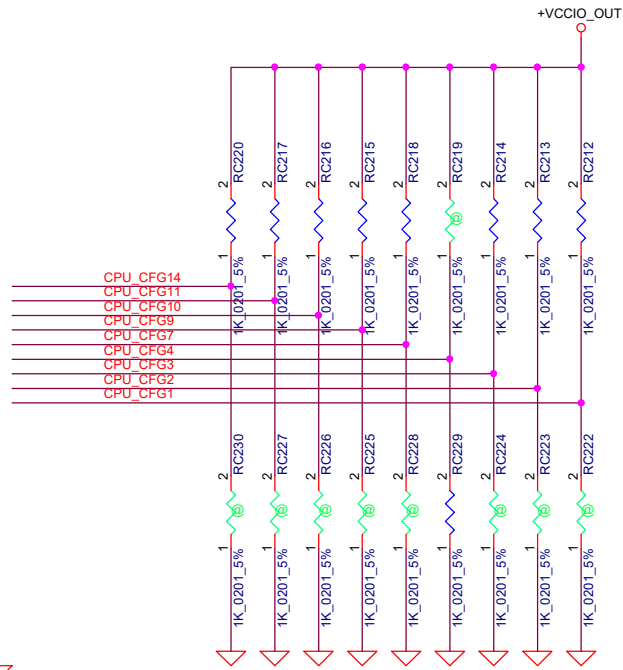
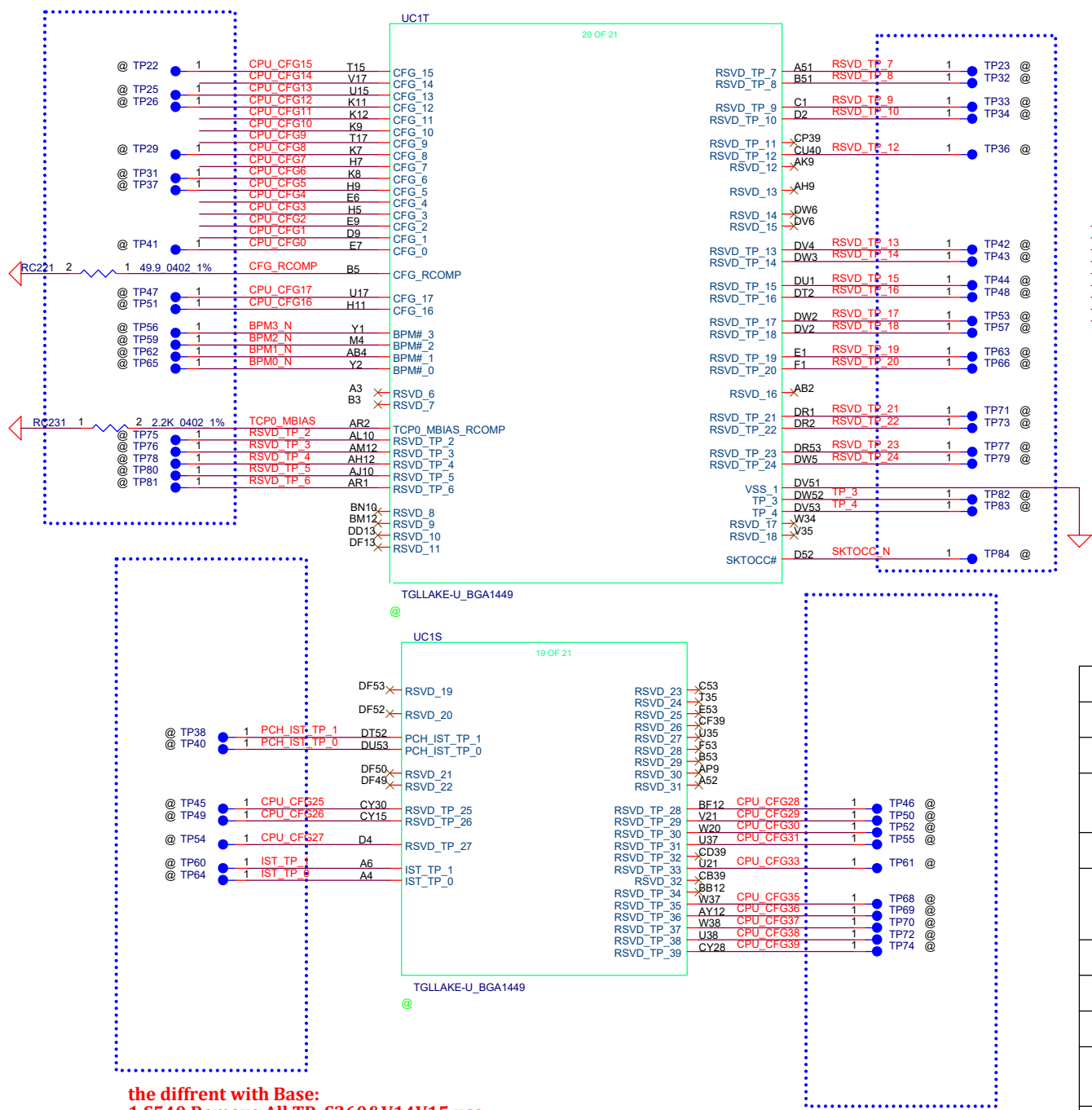
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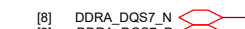


Pin Name	Strap Description	Configuration	Default Value
CFG[0]	RSVD	None	
CFG[3:1]	RSVD	Pull-up to VCCIO	1Kohm
CFG[4]	eDP enable strap 1 = Disabled 0 = Enabled	Pull-up to VCCIO / Pull-down Platform design dependent	1Kohm
CFG[6:5]	RSVD	None	
CFG[7]	PEG deferred link training 1 = (default) PEG Trainimmediately following RESET# de-assertion. 0 = PEG Wait for BIOS for training.	Pull-up to VCCIO / Pull-down Platform design dependent	1Kohm
CFG[8]	RSVD	None	
CFG[11:9]	RSVD	Pull-up to VCCIO	1Kohm
CFG[13:12]	RSVD	None	
CFG[14]	PEG60 Lane Reversal 1 = Normal(Default) 0 = Reversed	Pull-up to VCCIO / Pull-down Platform design dependent	1Kohm
CFG[17:15]	RSVD	None	

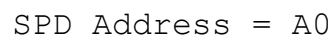
CPU PCIe Gen4 Bifurcation and Lane Reversal Mapping

Bifurcation	CFG Signals	Lanes			
	CFG [14]	0	1	2	3
1x4	1	0	1	2	3
1x4(Reversed)	0	3	2	1	0

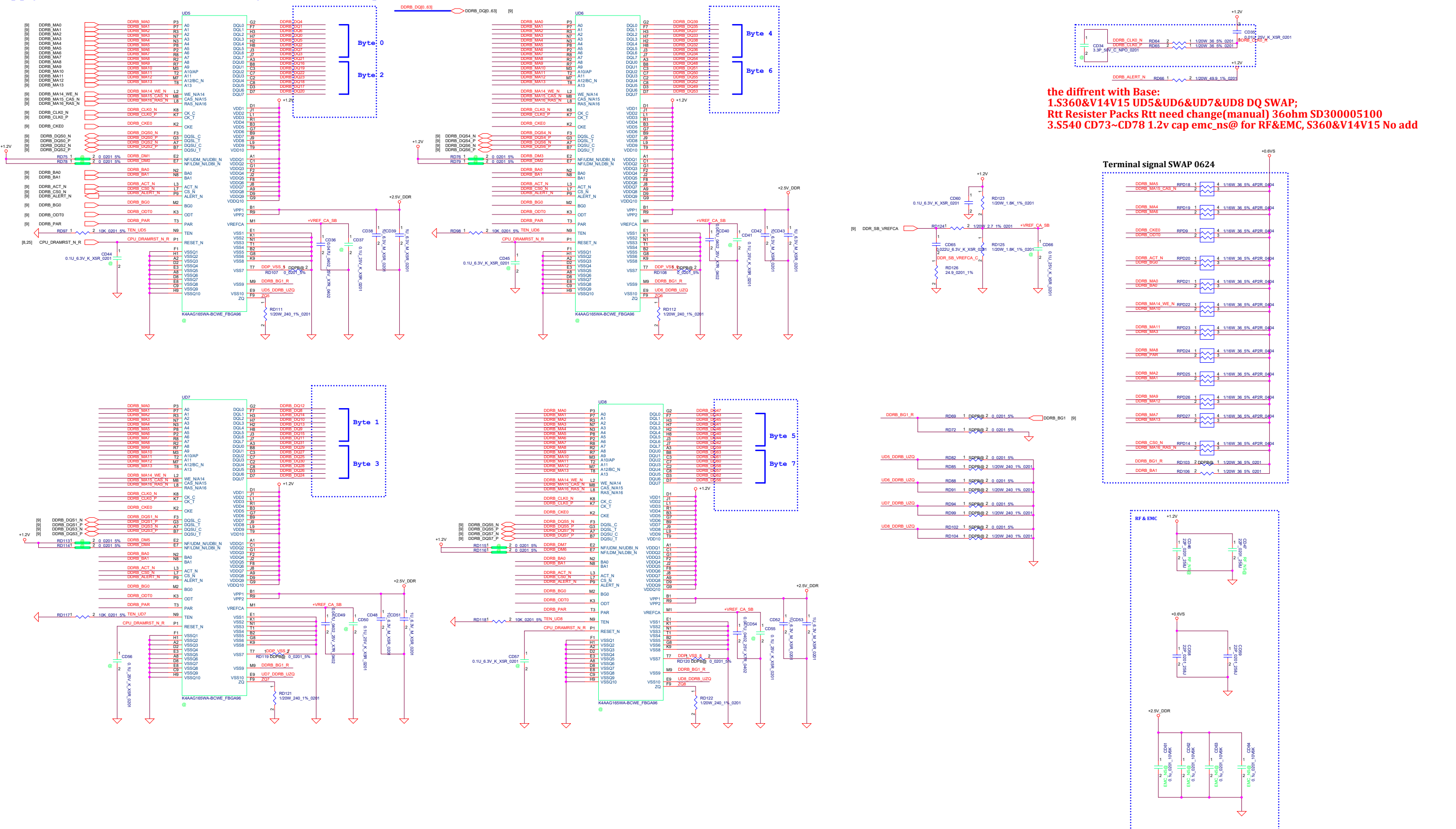
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SPD address setting Need confirm with BIOS

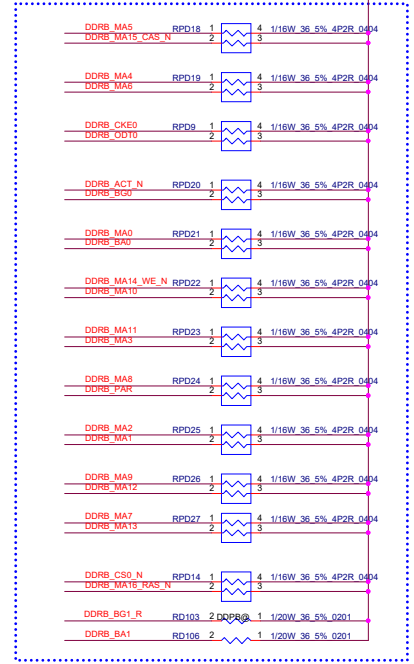


Apply X76 BOM to control SDP Memory Down stuff components!

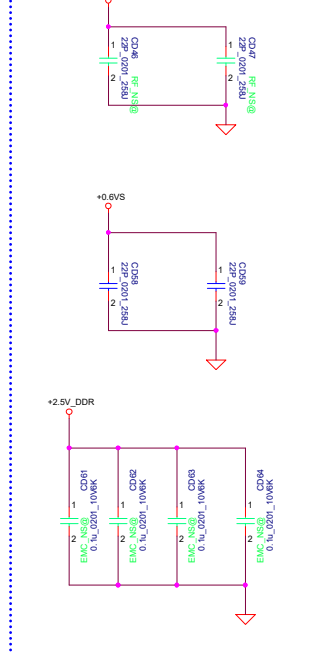


the diffrent with Base:
1.S360&V14V15 UD5&UD6&UD7&UD8 DQ SWAP;
Rtt Resister Packs Rtt need change(manual) 36ohm SD300005100
3.S540 CD73-CD78 1.2v cap emc_ns@ for RF&EMC, S360&V14V15 No add

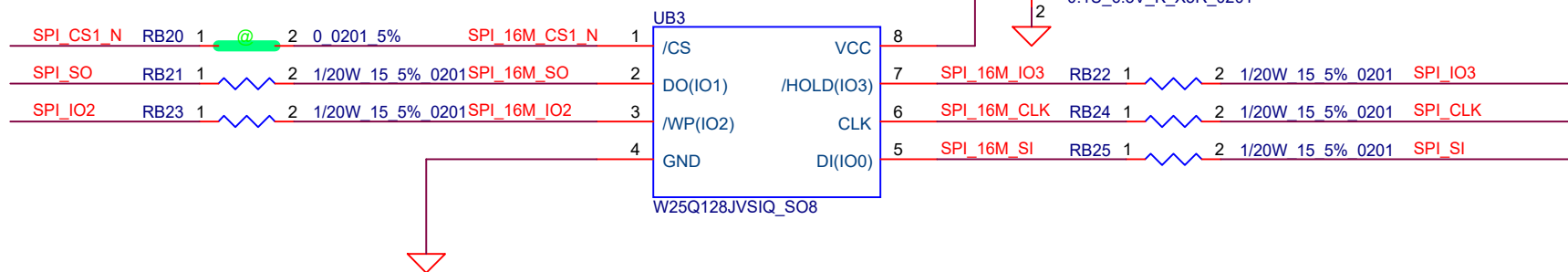
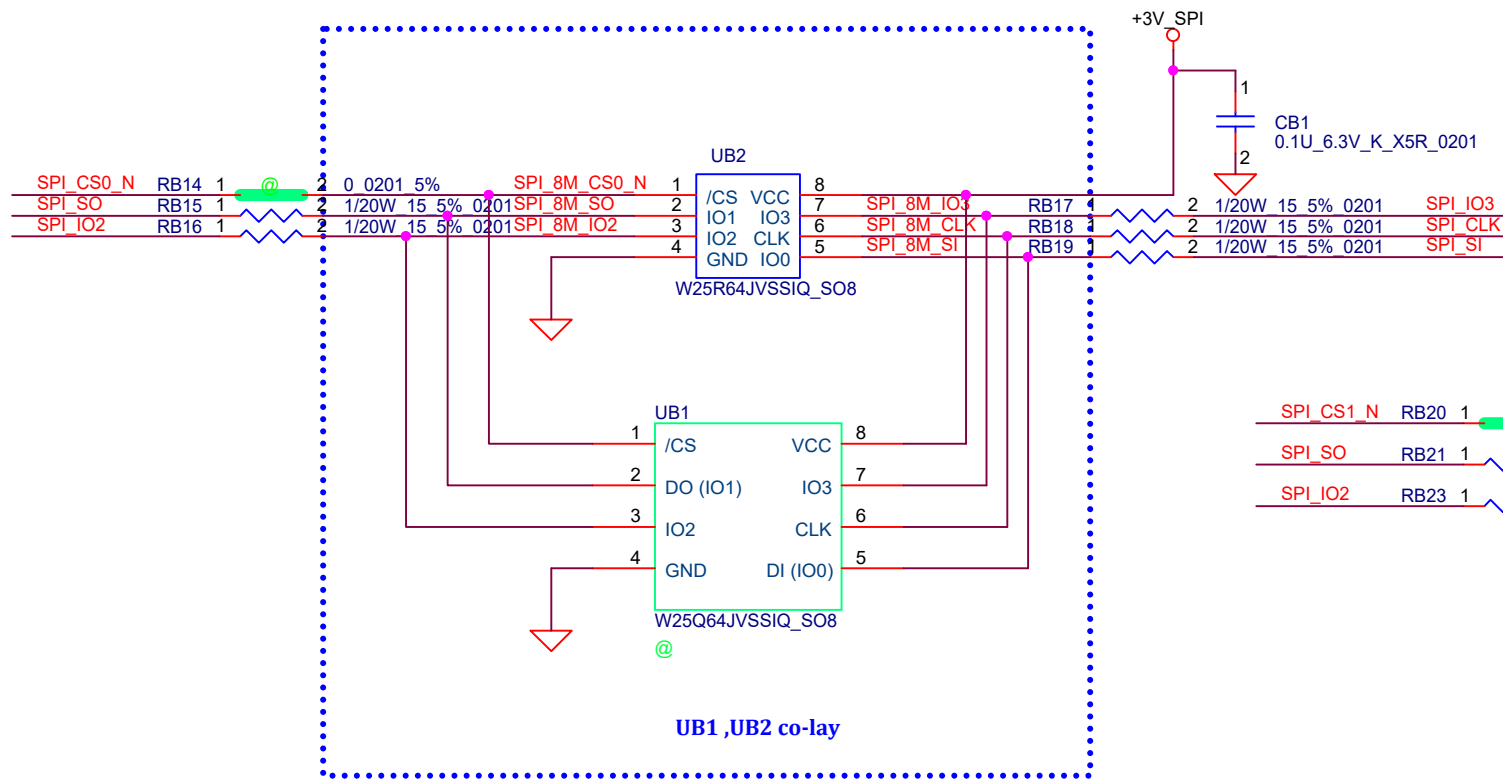
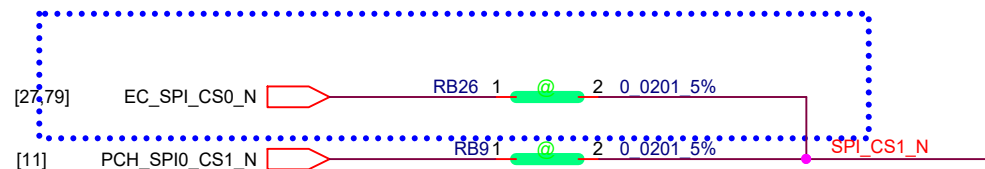
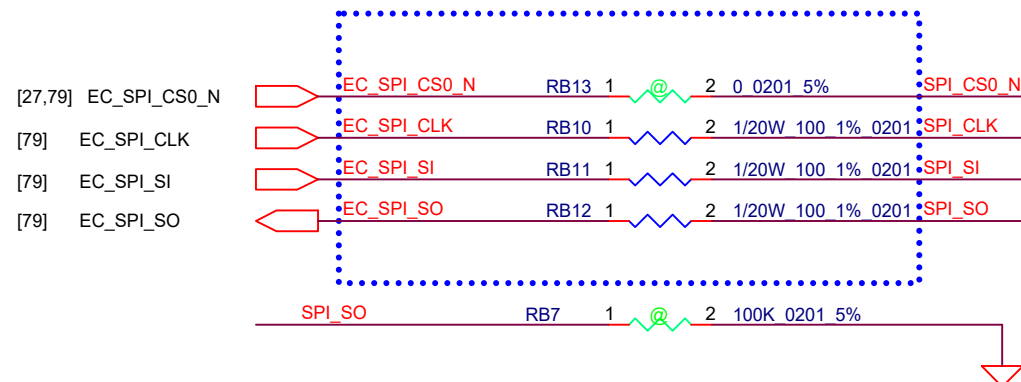
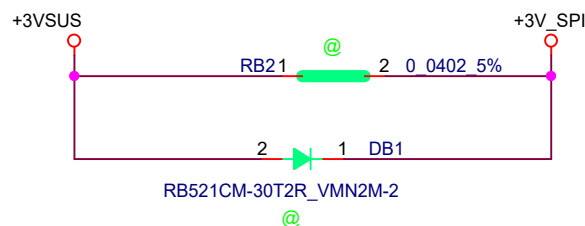
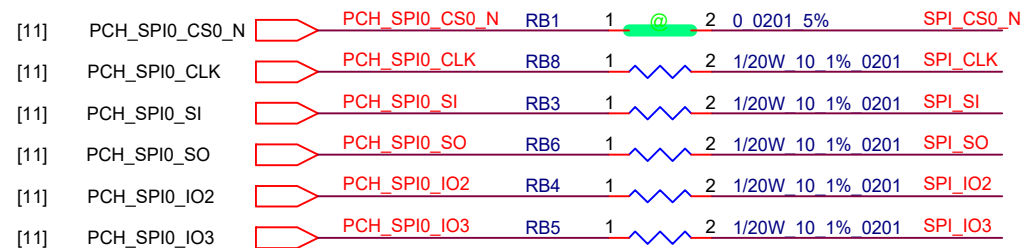
Terminal signal SWAP 0624



RF & EMC



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			Document Number
			Rev
			Date

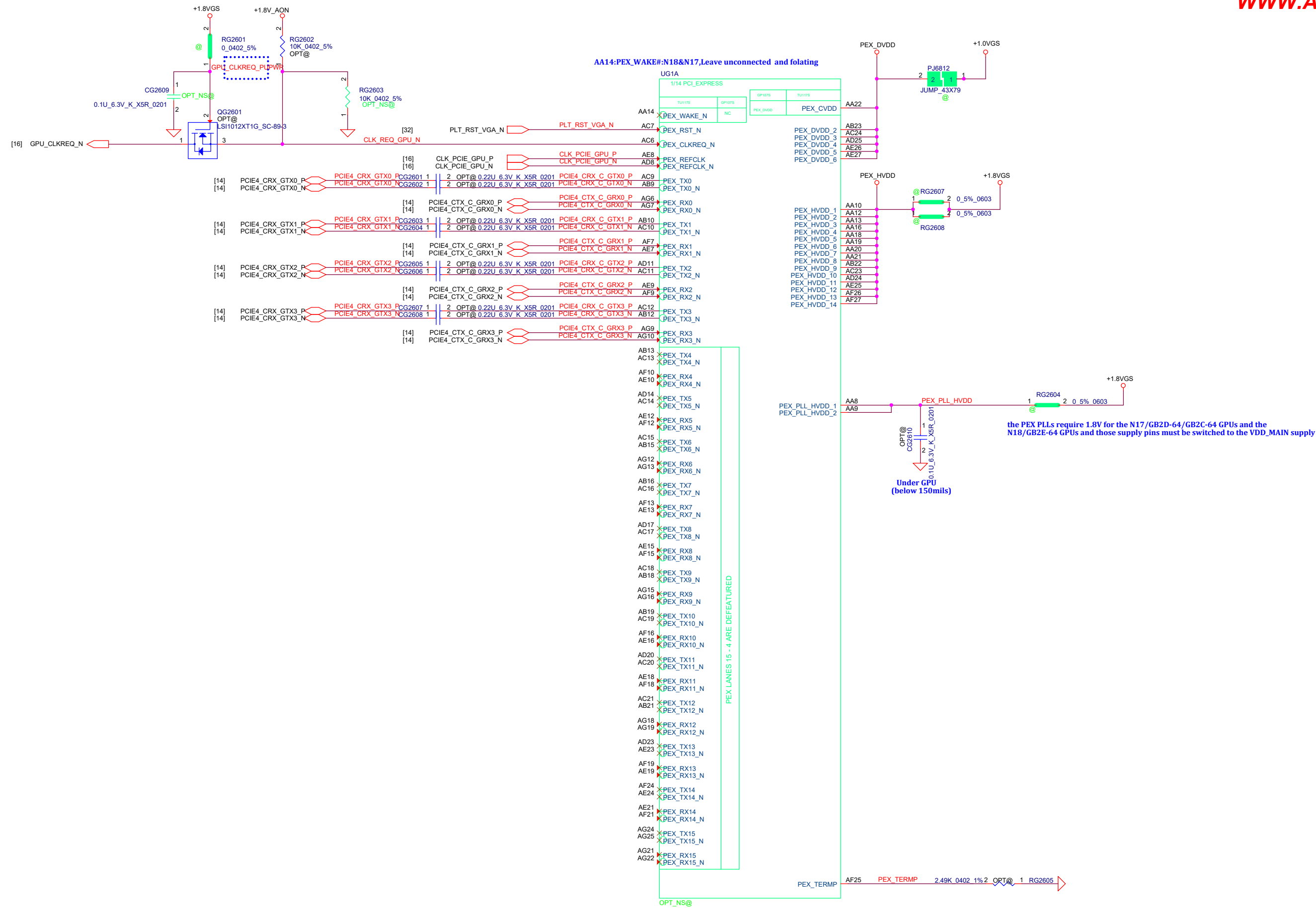


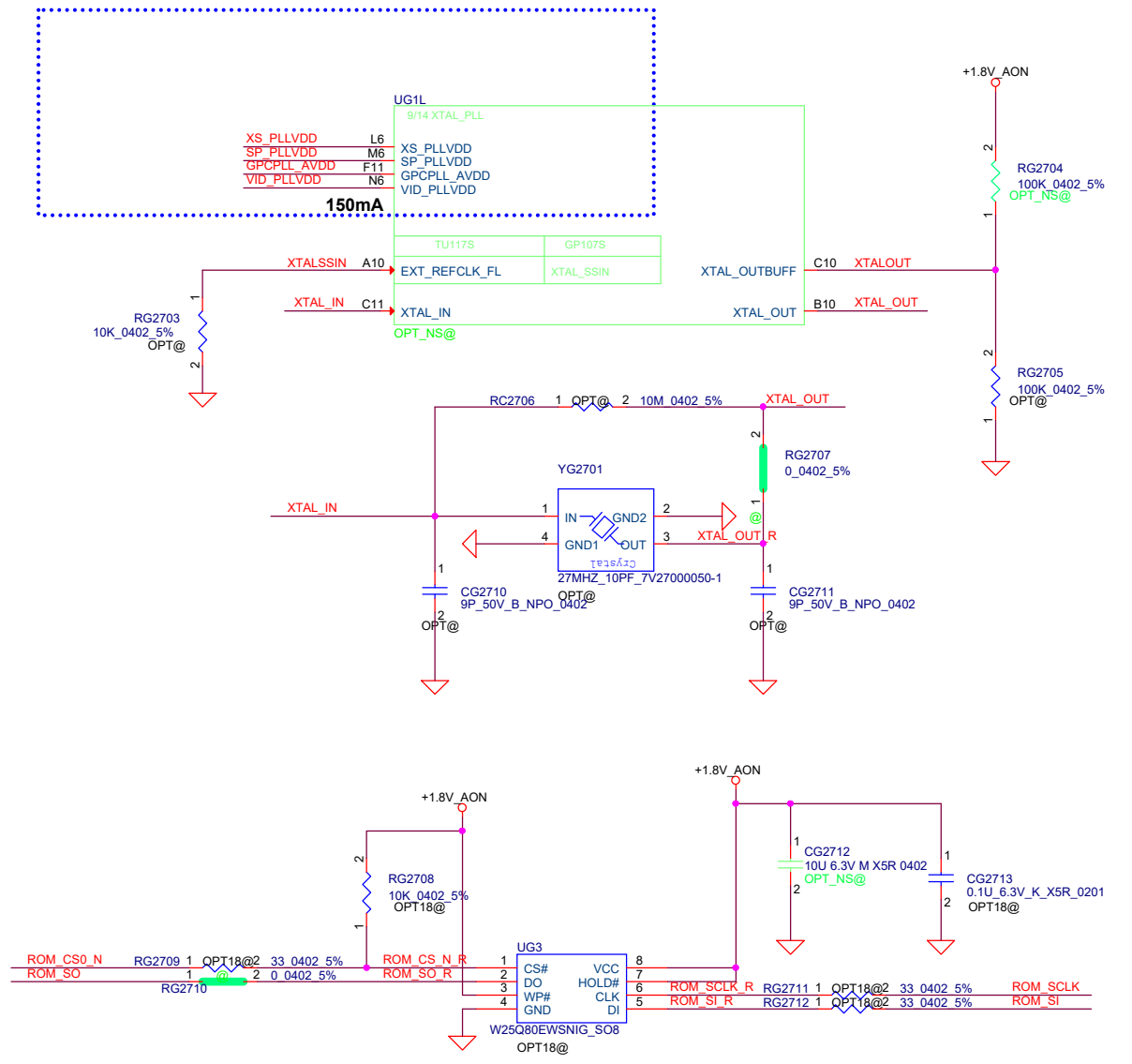
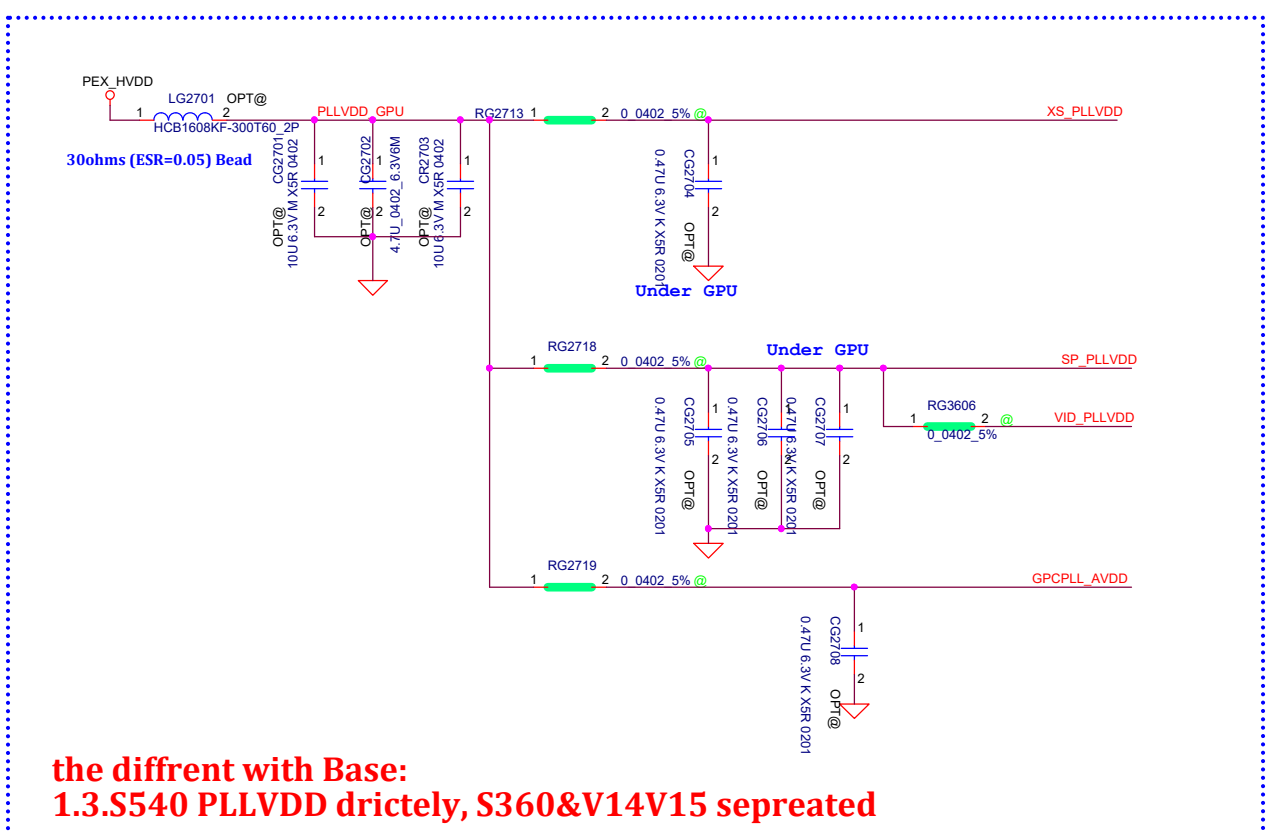
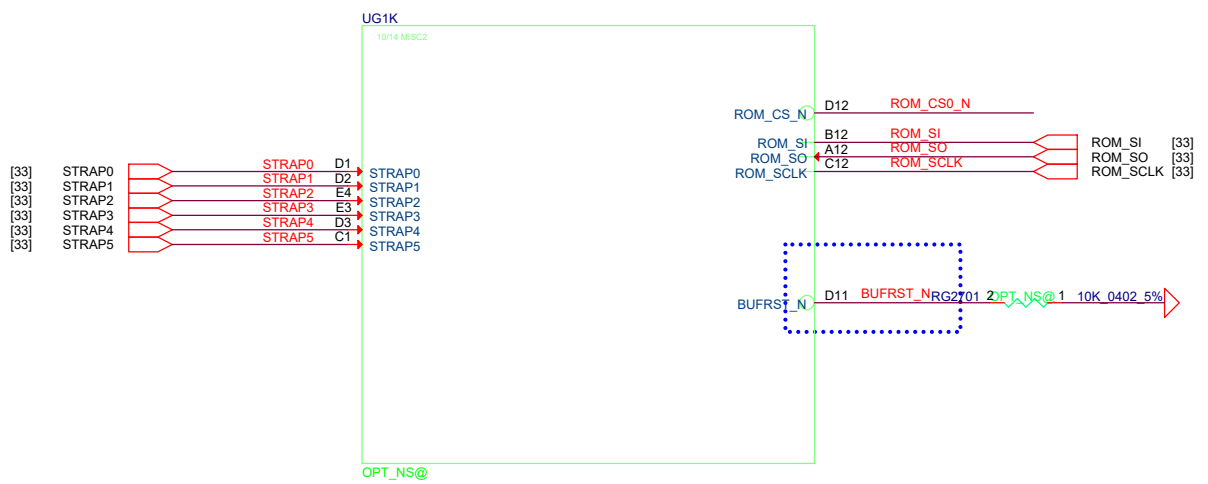
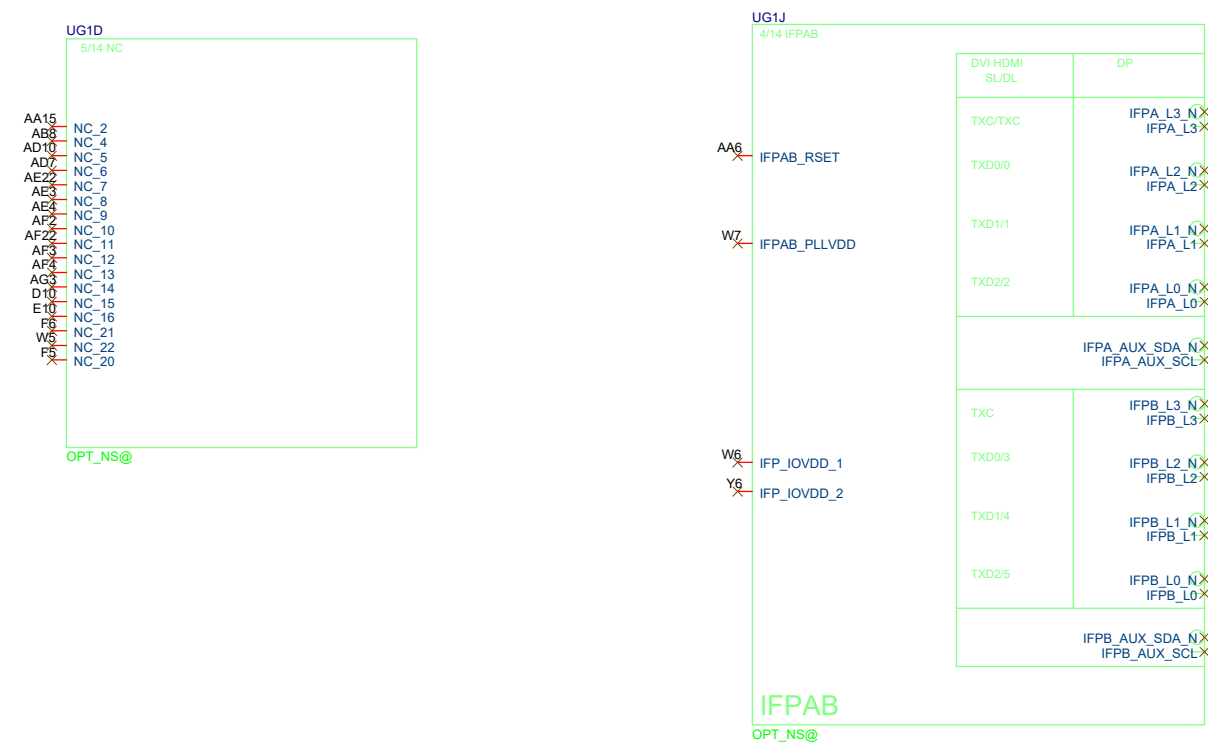
the diffrent with Base:

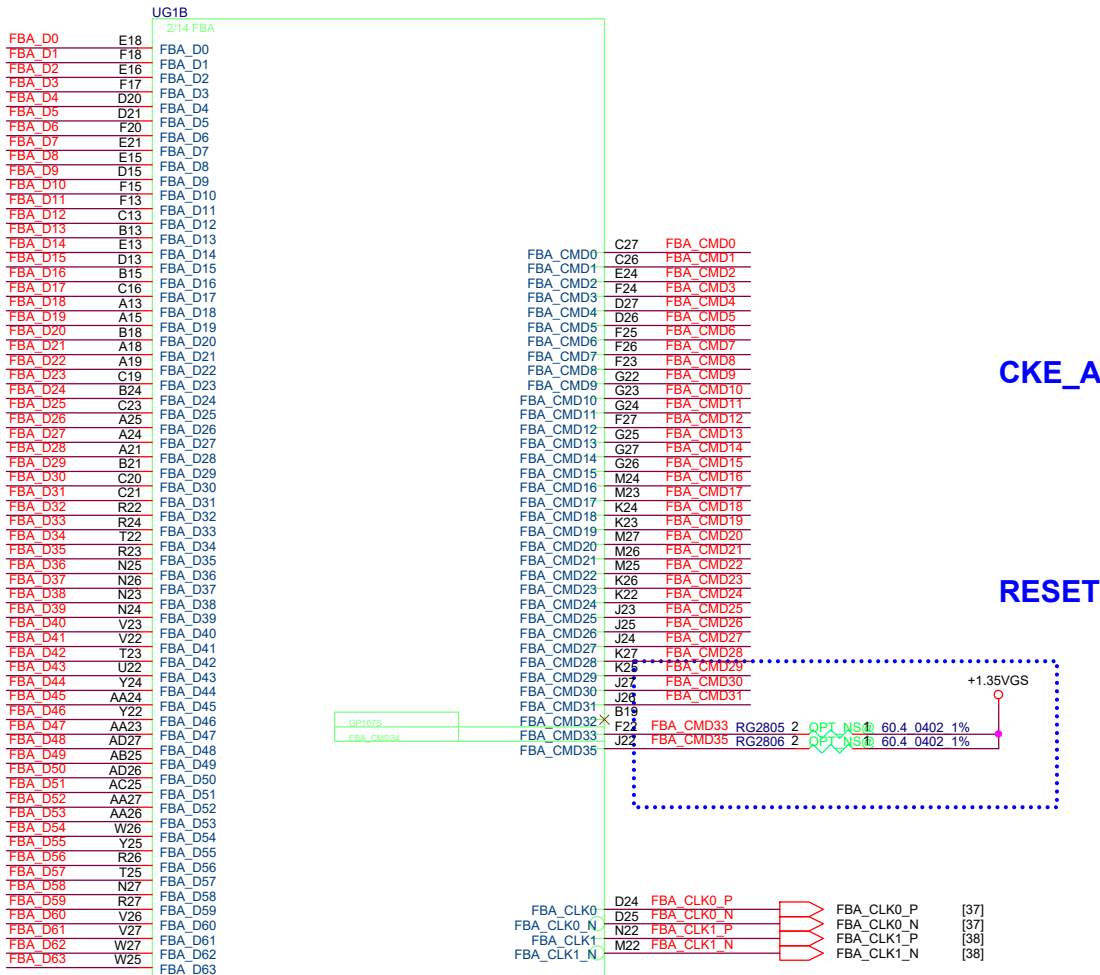
1.UB1&UB2 ROM CO-LAYOUT

2.RB13,RB26 Stuff different as co-layout different

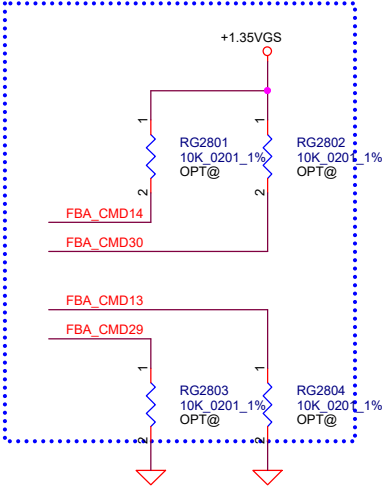
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				SPI ROM/TPM	
				Date:	Rev
				Tuesday, November 10, 2020	0.1
				Sheet 27 of 110	







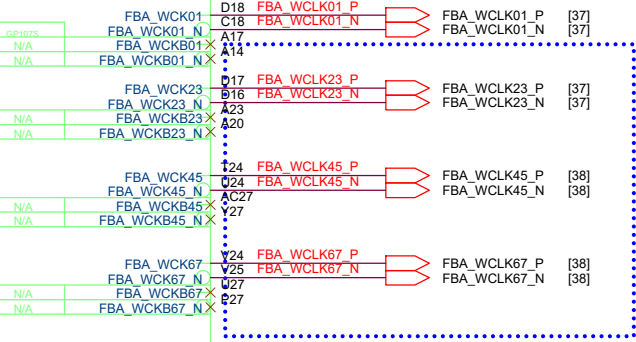
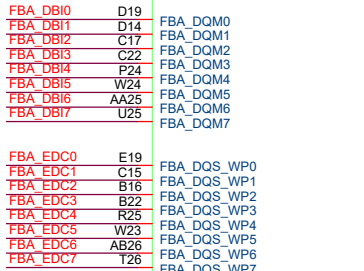
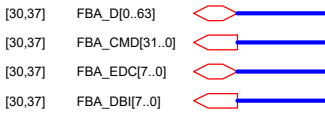
Note: GDDR5&GDDR6 different



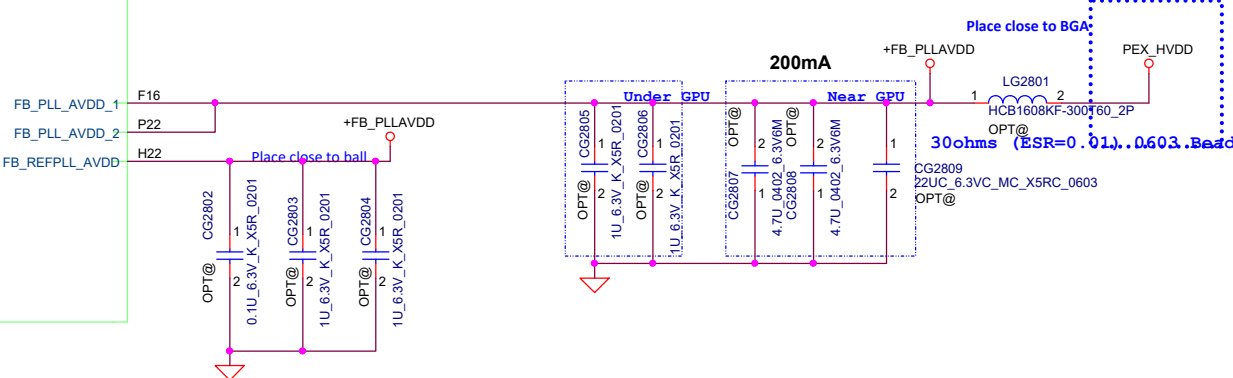
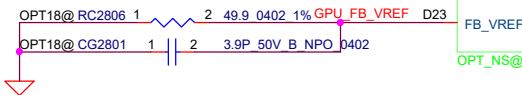
the diffrent with Base:

1. S540 FBA_WCKB0x_N&P connect GDDR6 NC pin, S360&V14V15 NC
2. S540 LG2801 connect 1.8VGS, S360&V14V15 LG2801 connect PEX_HVDD
3. S540 B19&F22&J22 Pin Connect different with S360&V14V15
4. S540 CMD Pin Connect different with S360&V14V15, PULLUP power Different

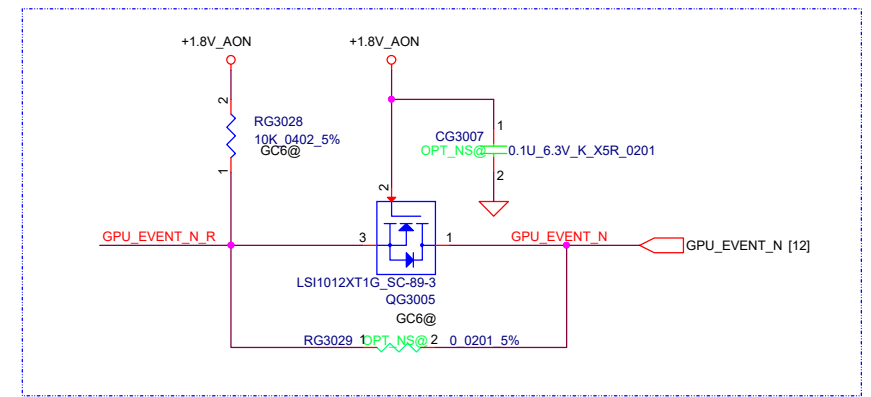
A14:N18:FBA_WCKB01_N;N17 N/A: N18 Leave unconnected and folating
A17:N18:FBA_WCKB01;N17 N/A: N18 Leave unconnected and folating
A20:N18:FBA_WCKB23_N;N17 N/A: N18 Leave unconnected and folating
A23:N18:FBA_WCKB23;N17 N/A: N18 Leave unconnected and folating
Y27:N18:FBA_WCKB45_N;N17 N/A: N18 Leave unconnected and folating
AC27:N18:FBA_WCKB45;N17 N/A: N18 Leave unconnected and folating
P27:N18:FBA_WCKB67_N;N17 N/A: N18 Leave unconnected and folating
U27:N18:FBA_WCKB67;N17 N/A: N18 Leave unconnected and folating



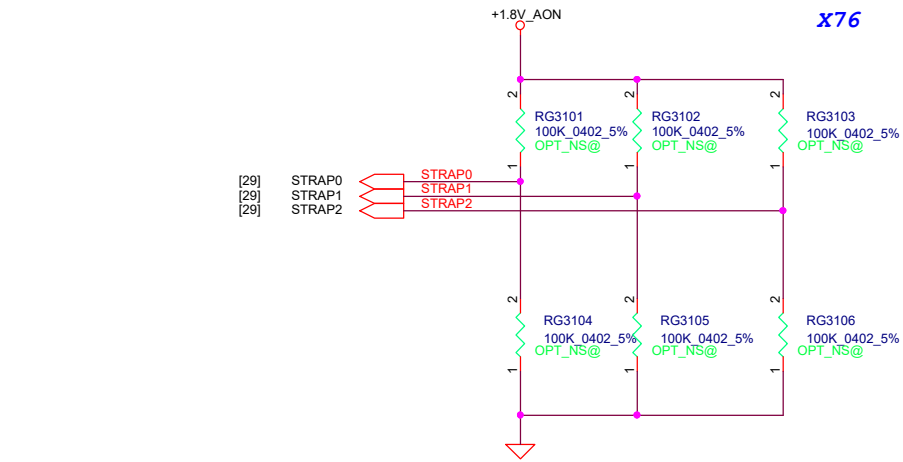
These balls are shown as GND in the N18 GeForce Data Sheet. Please follow the below guidelines:
a. If doing N18/GB2E-64 alone, connect these balls to GND.
b. If doing N18/GB2E-64 and N17/GB2D-64/GB2C-64 co-layout, leave these balls unconnected and floating.
c. If doing N17/GB2D-64/GB2C-64 (GDDR5) alone, leave these balls unconnected and floating.



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				Date:	Tuesday, November 10, 2020	Sheet	31 of 110

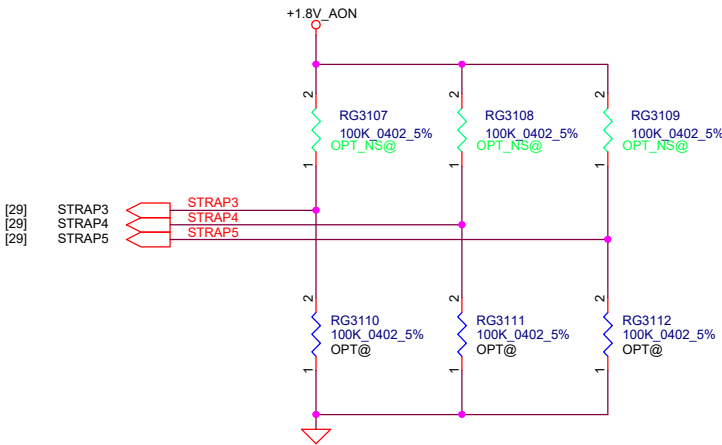


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Size C		Document Number		GPU_GPIO/JTAG				Rev 0.1	
Date:		Tuesday, November 10, 2020				Sheet		32 of 110	



VRAMCFG

GPU	FB Memory (GDDR5)		STRAP	STRAP2	STRAP1	STRAP0
N18S	Micron 8Gb	MT51J256M32HF-80:B	0x1	L	L	H
	Hynix 8Gb	H5GC8H24AJR-R2C	0x2	L	H	L
	Samsung 8Gb	K4G80325FC-HC25	0x4	H	L	L
GPU	FB Memory (GDDR5)		STRAP	STRAP2	STRAP1	STRAP0
N17S	Micron 8Gb	MT51J256M32HF-80:B	0x9	L	M	L
	Hynix 8Gb	H5GC8H24AJR-R2C	0xA	L	M	H
	Samsung 8Gb	K4G80325FC-HC25	0xB	L	H	M



STRAP5	STRAP4	STRAP3	SMB_ALT_ADDR	DEVID_SEL	PCIE_CFG	VGA_DEVICE
L	L	L	0	0	0	0

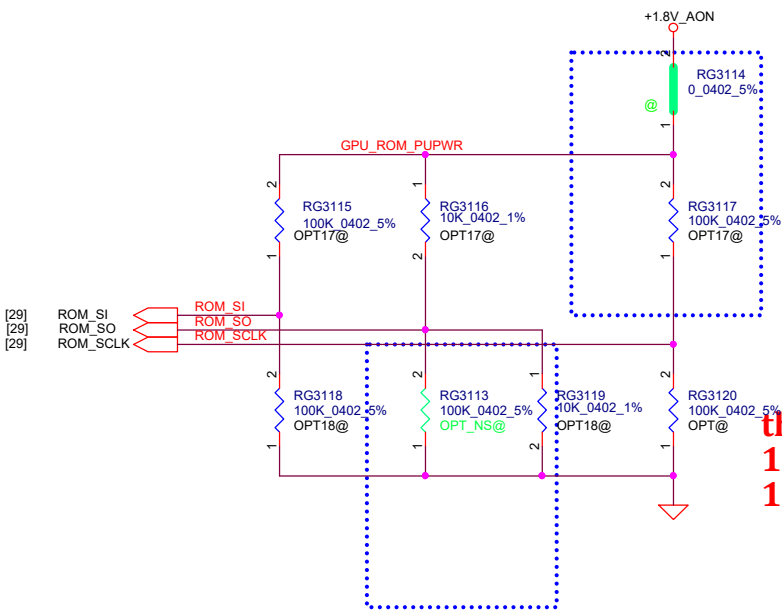
- 1: SMB_ALT_ADDR ENABLE
0: SMB_ALT_ADDR DISABLE
- 1: DEVID_SEL REBRAND
0: DEVID_SEL ORIGNAL
- 1: PCIE_CFG LOW POWER
0: PCIE_CFG HIGH POWER
- 1: VGA_DEVICE ENABLE
0: VGA_DEVICE DISABLE

DEVID_SEL	
0	(Default)
1	

SMBUS_ALT_ADDR	
0	0x9E (Default)
1	0x9C (Multi-GPU usage)

PCIE_CFG	
0	(Default)
1	

VGA_DEVICE	
0	3D Device (Class Code 302h)
1	VGA Device (Default)

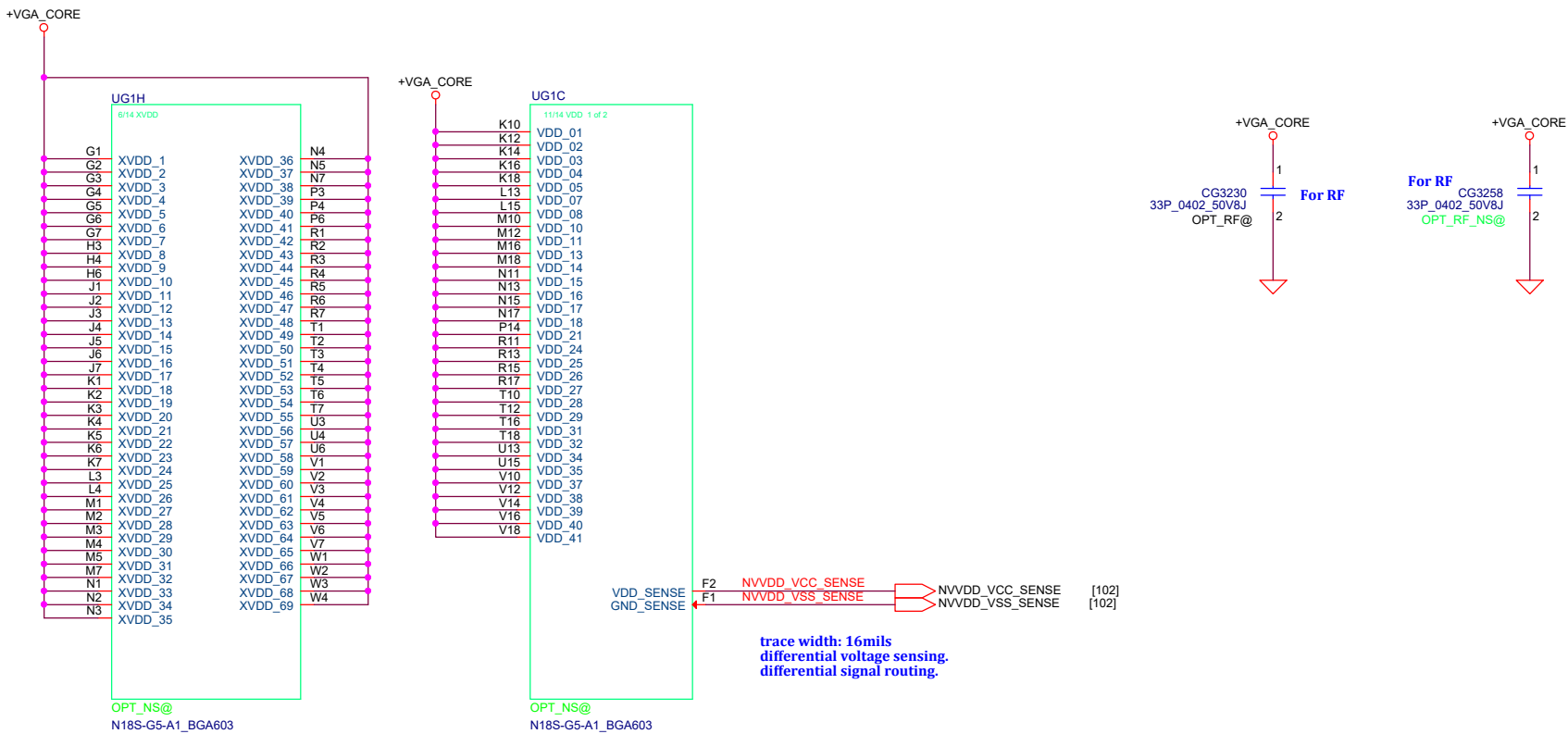


N18:ROM_SO&ROM_SI&ROM_SCLK Pull down to Enable The failsafe_OVERT* function
N17:SOR_EXPOSED[3:0]

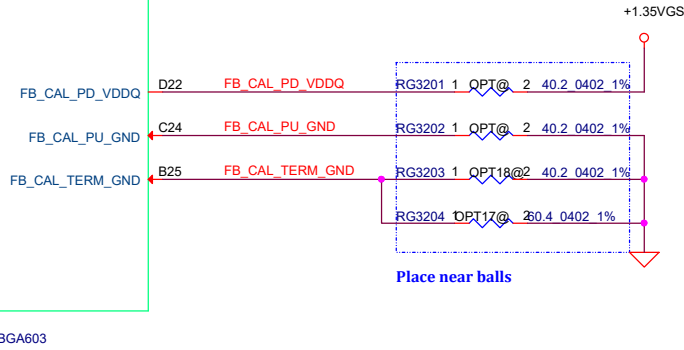
	ROM_SO	ROM_SI	ROM_SCLK	SOR_EXPOSED[3:0]
N18S-G5	L	L	L	XXX0
N17S-G5	H	H	M	0000

1:ENABLE 0:DISABLE SOR0/1/2/3 DISABLE
1::DISABLE SOR0/1/2/3 DISABLE

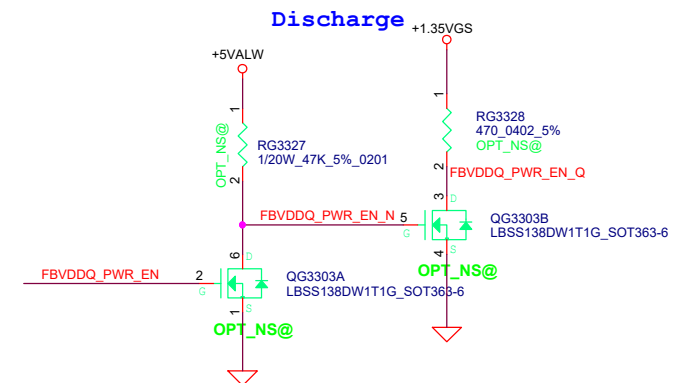
the diffrent with Base:
1.S540 RG3117---- OPT_NS@,S360&V14V15 RG3117---- OPT17@
1.S540 No RG3113,S360&V14V15 RG3113---- OPT17@



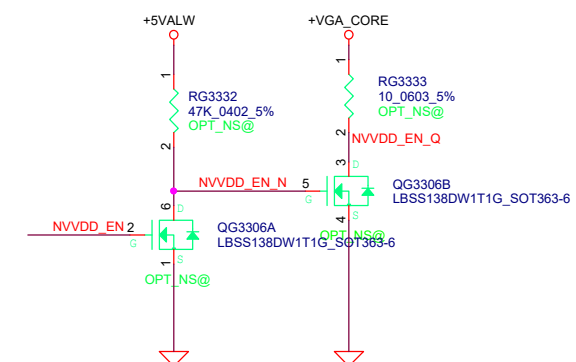
the diffrent with Base:
1.S540No RG3204,S360&V14V15 RG3204---- OPT17@



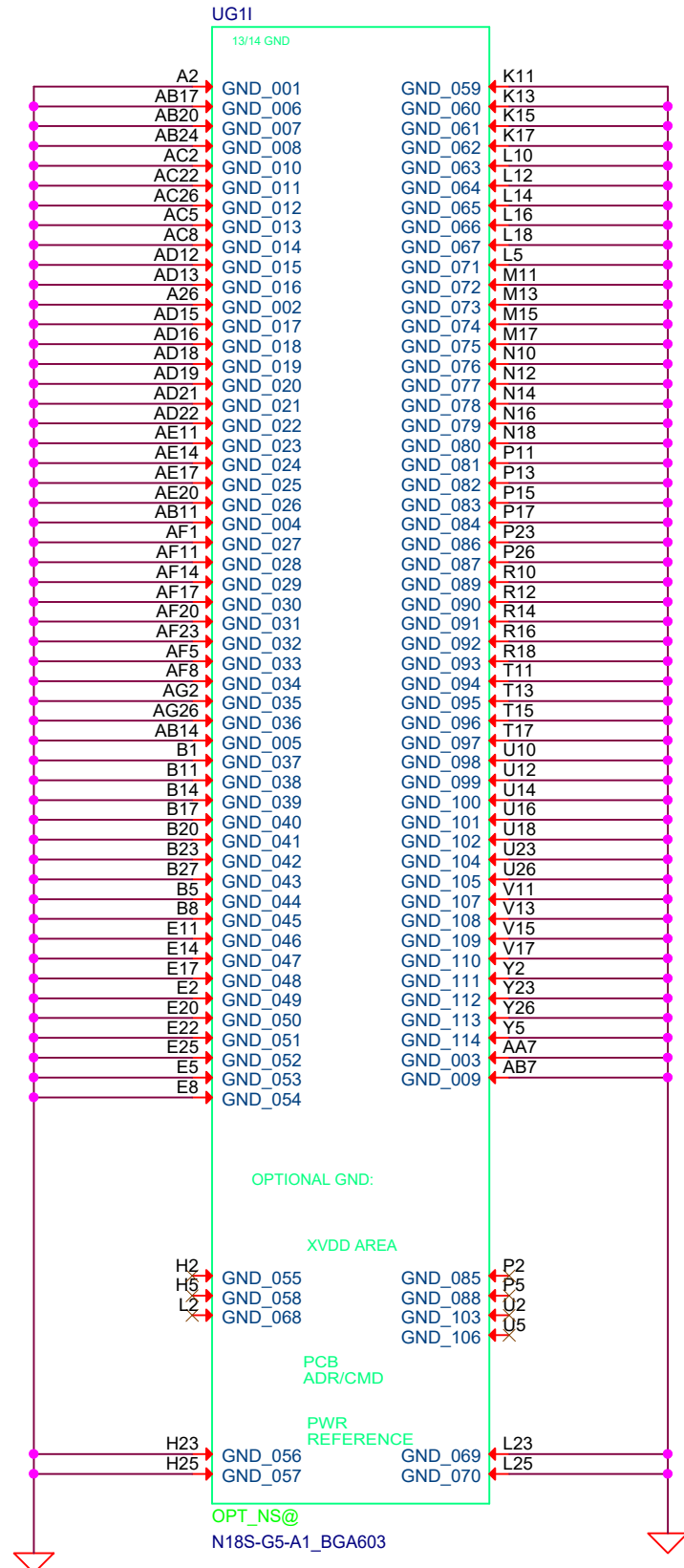
CALIBRATION PIN	GDDR5
FB_CAL_x_PD_VDDQ	40.2ohm
FB_CAL_x_PU_GND	40.2ohm
FB_CAL_xTERM_GND	60.4ohm



the difference between the 1.540 and 1.540R versions of the EVK

[illegible]

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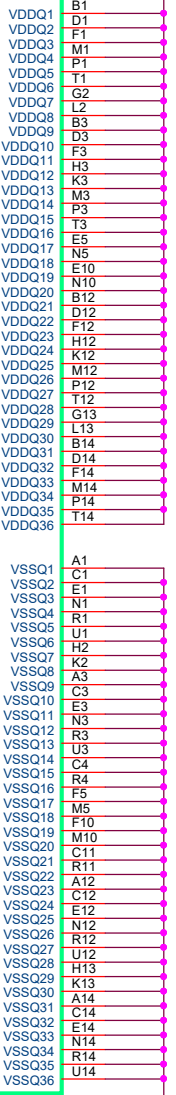
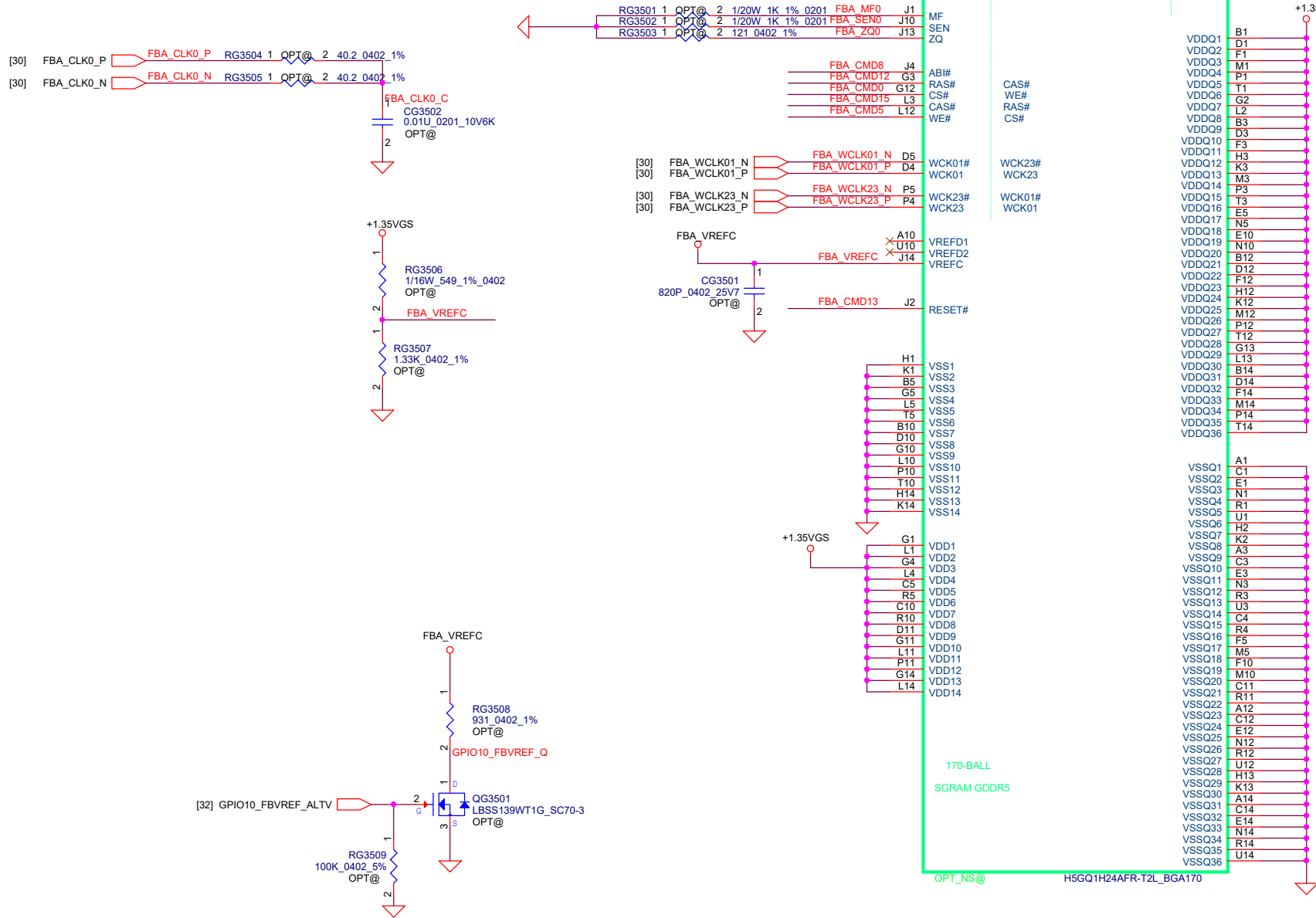
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Size B		Document Number GPU_GND							
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Lower 32 bits

N18S_GDDR5_A[31_0]

the diffrent with Base:
1.S540 GDDR6,S360&V14V15 GDDR5

MF=0 No Mirror



BYTE0

BYTE1

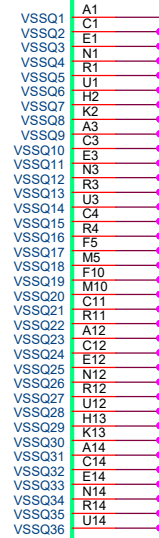
BYTE2

BYTE3

upper 32 bits

170-BALL
SGRAM GDDR5

.....



	BYTE7
--	-------

Title			
S360-TGL			
Size C	Document Number	Rev	
	VRAM A [64:32]	0.1	
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Date:		Tuesday, November 10, 2020		Sheet		39 of 110	

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				Size B	Document Number VRAM_B_[64:32]		Rev 0.1
				Date:	Tuesday, November 10, 2020		Sheet 40 of 110

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						Date:	Tuesday, November 10, 2020		Sheet 41 of 110	

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Size B	Document Number VRAM_C_[64:32]					Rev 0.1		
Date:	Tuesday, November 10, 2020				Sheet	42	of	110

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Size B	Document Number					VRAM_D_[31:0]	Rev 0.1
Date:		Tuesday, November 10, 2020			Sheet	43	of 110

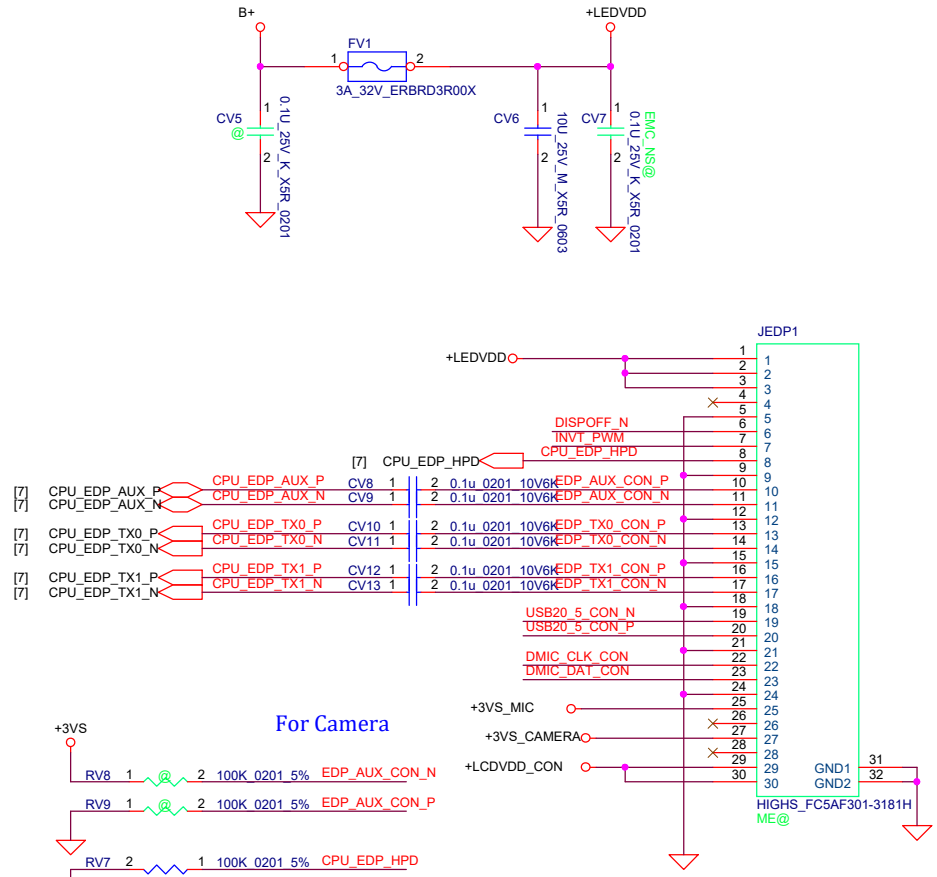
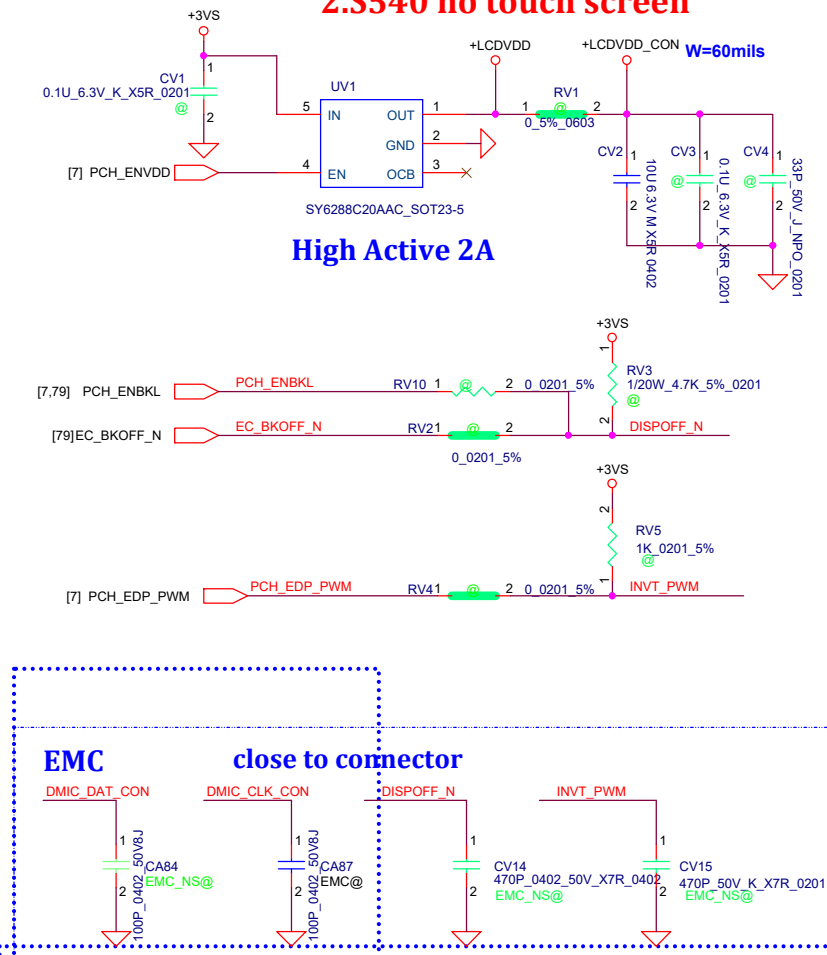
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Issued Date	2012/07/01	Deciphered Date	2014/07/01				
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				Size B	Document Number VRAM_D_[64:32]	Rev 0.1	
				Date:	Tuesday, November 10, 2020	Sheet 44 of 110	

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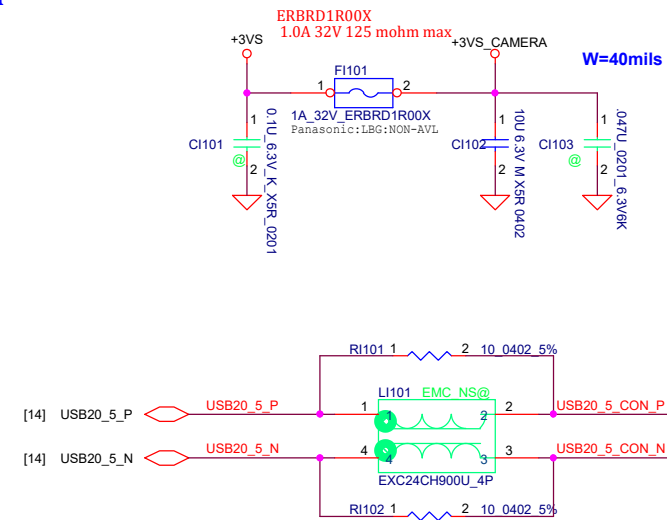
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Size B		Document Number NV DDS LOGIC						Rev 0.1	
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LCD EDP Display

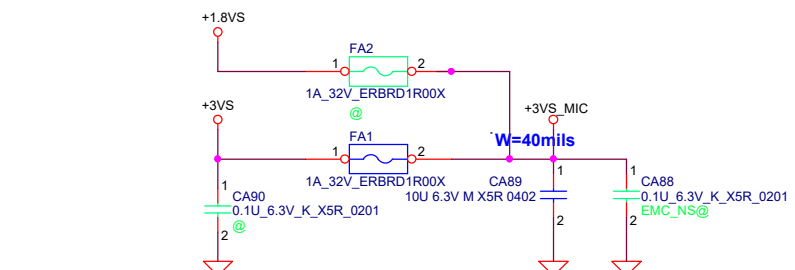
the diffrent with Base:
1.S540 EDP connector pindefine different with S360&V14V15
2.S540 no touch screen



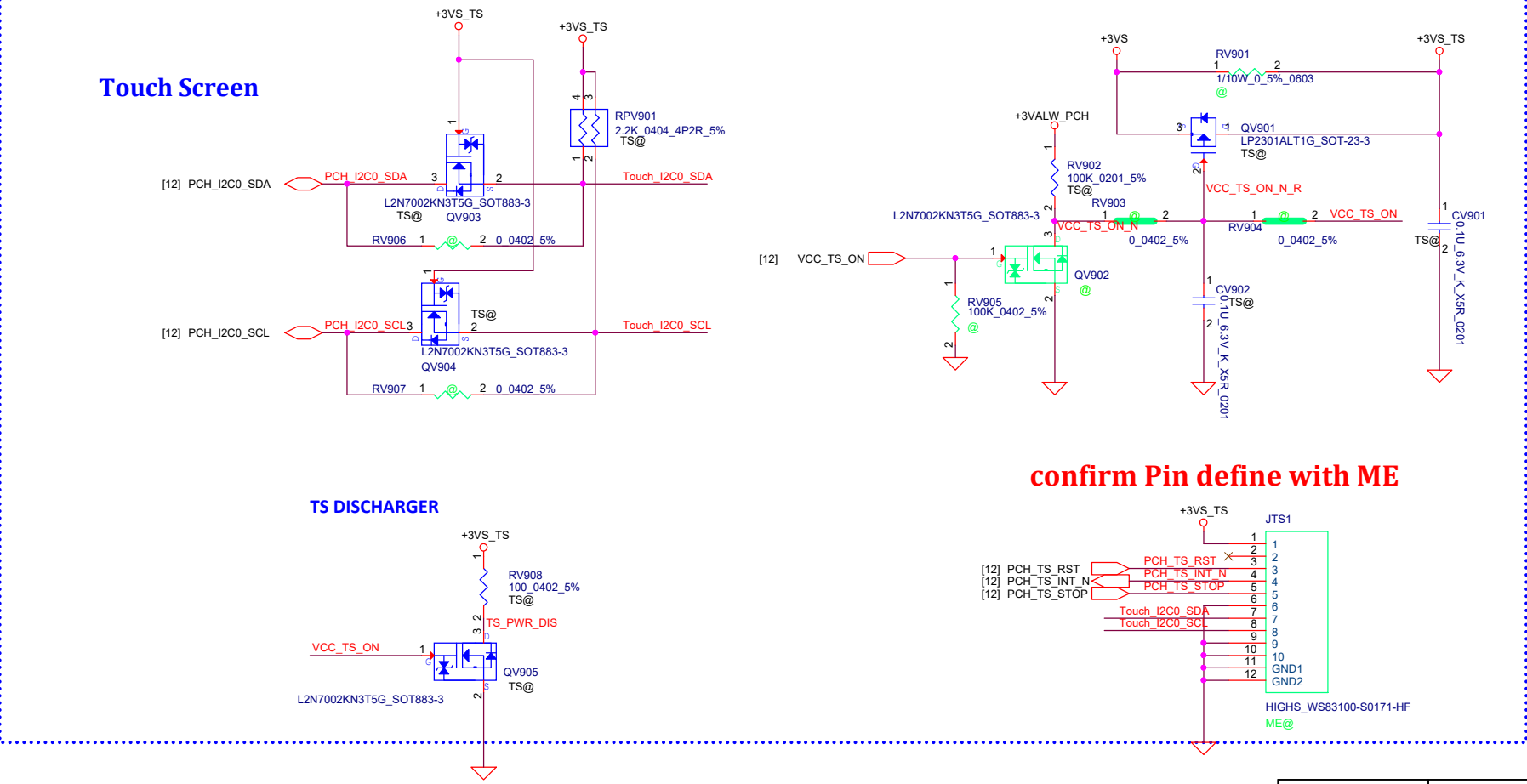
Camera



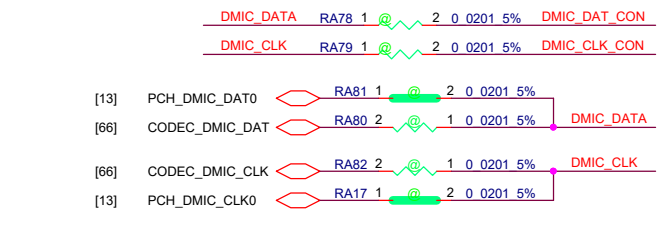
DMIC



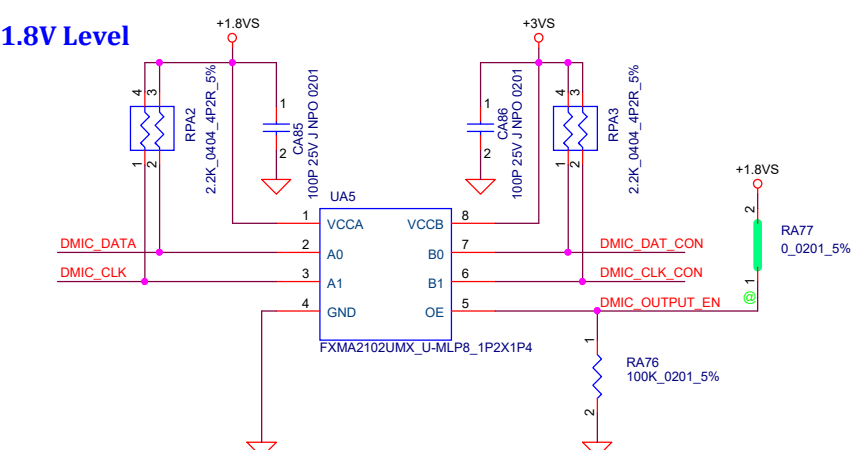
Touch Screen



DMIC 3.3V Level



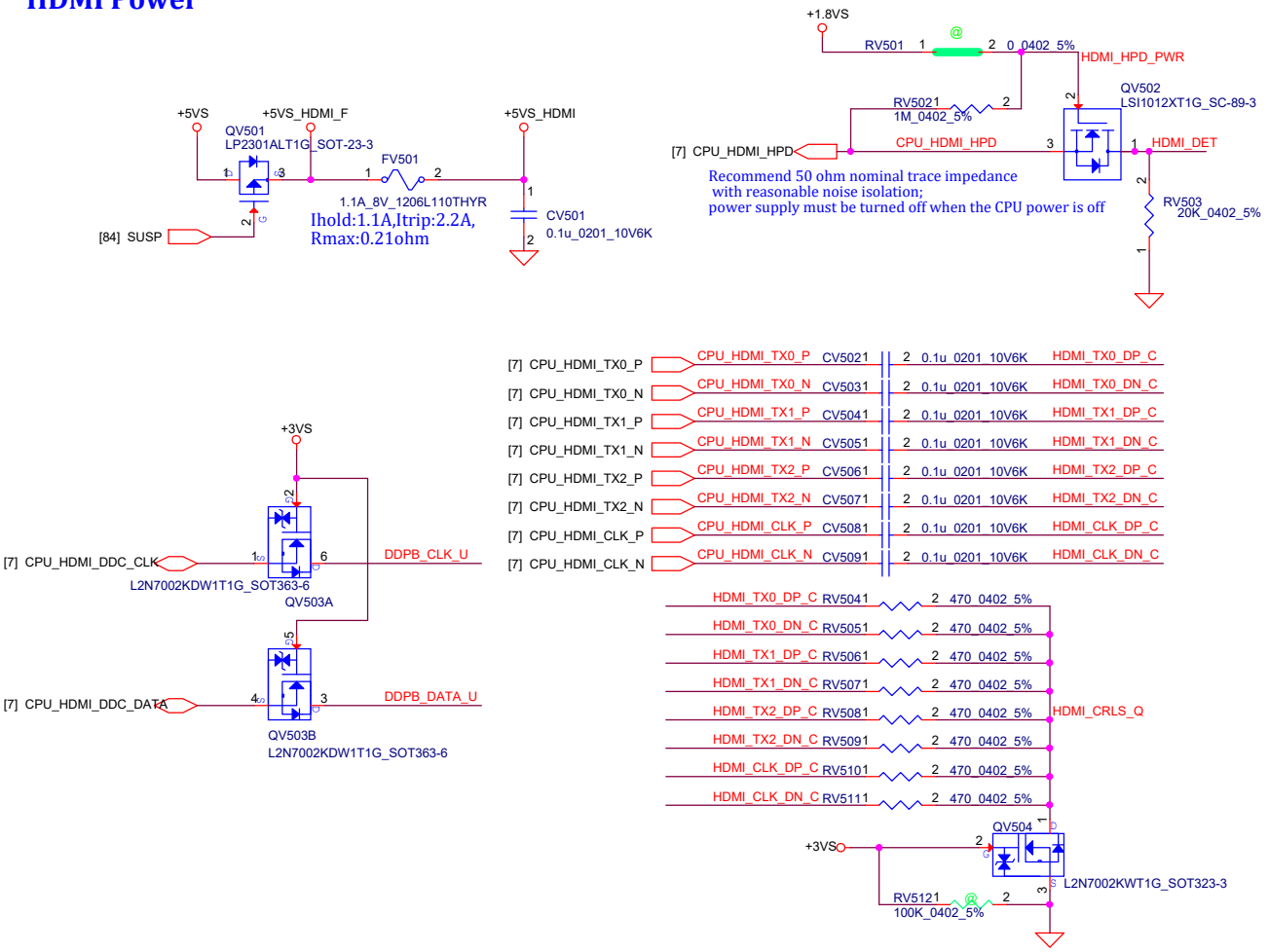
DMIC 1.8V Level



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				Date: Tuesday, November 10, 2020	Sheet 48 of 110		

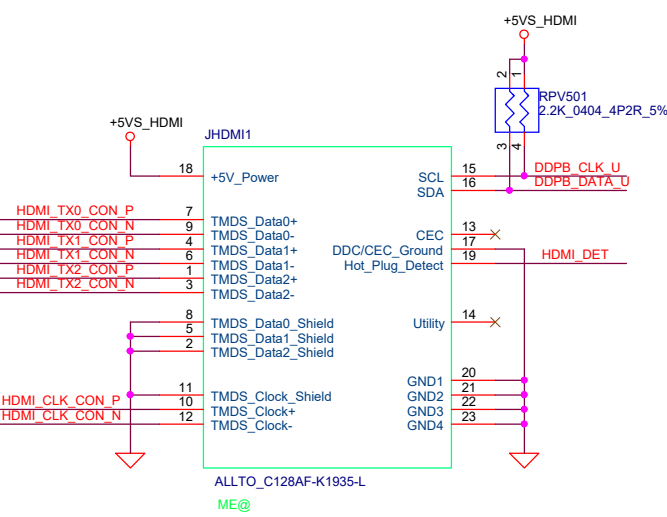
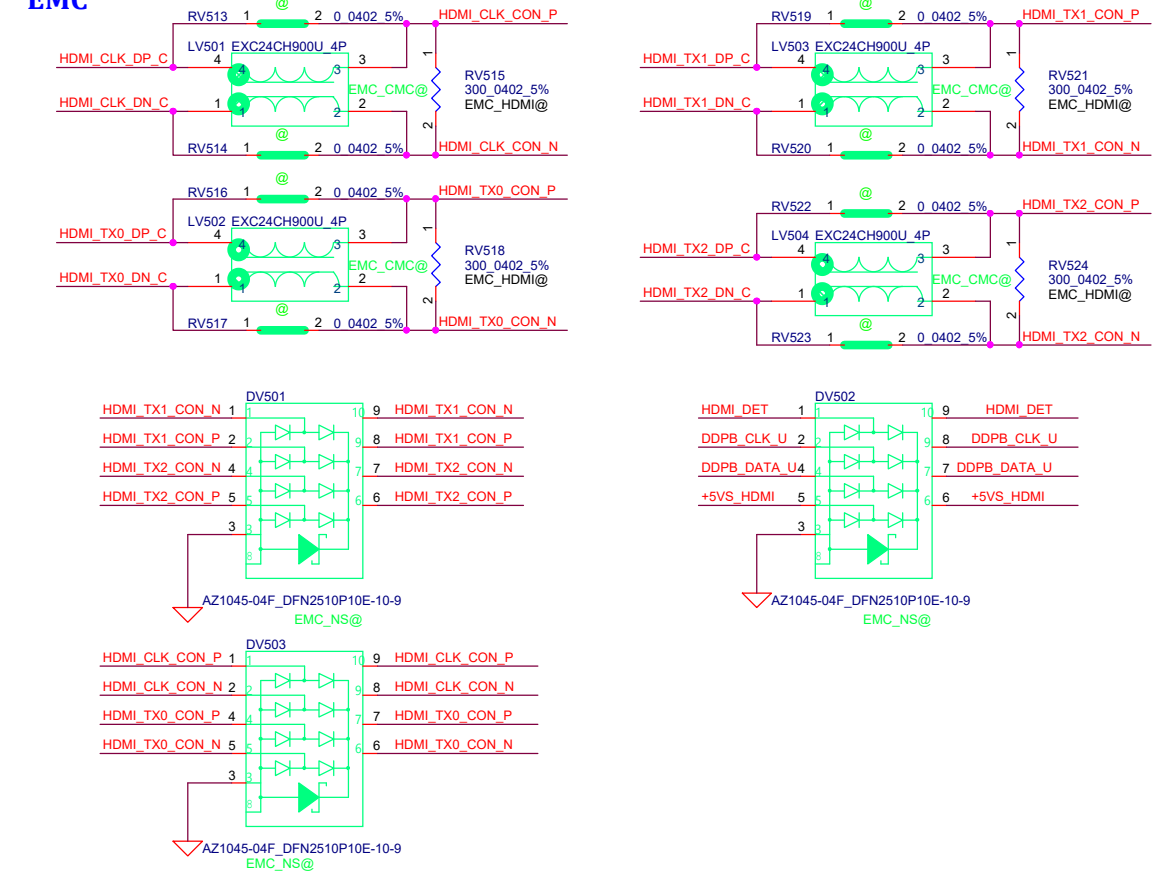
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				Size B	Document Number HDMI _ RETIMER		Rev 0.1
				Date:	Tuesday, November 10, 2020		Sheet 49 of 110

HDMI Power



the diffrent with Base:
1.S540 No HDMI,S360&V14V15 have

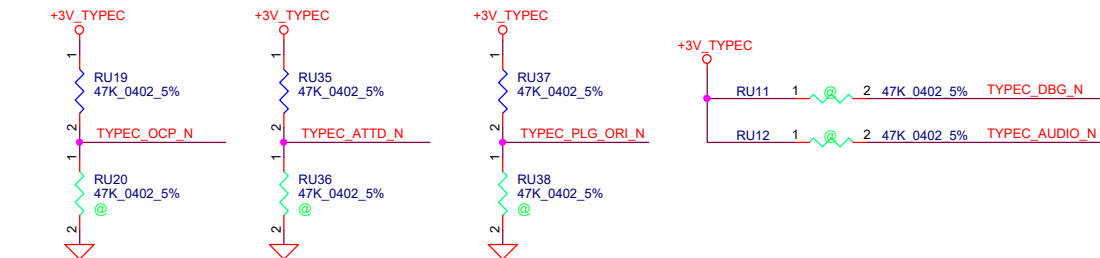
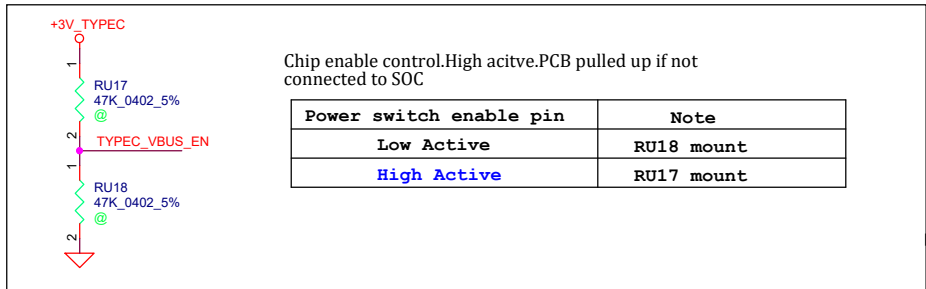
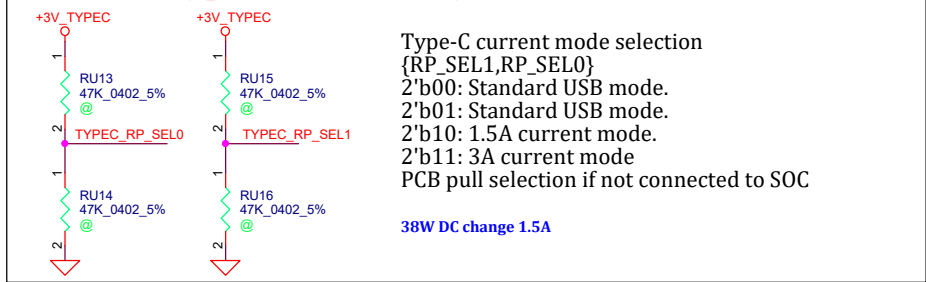
EMC



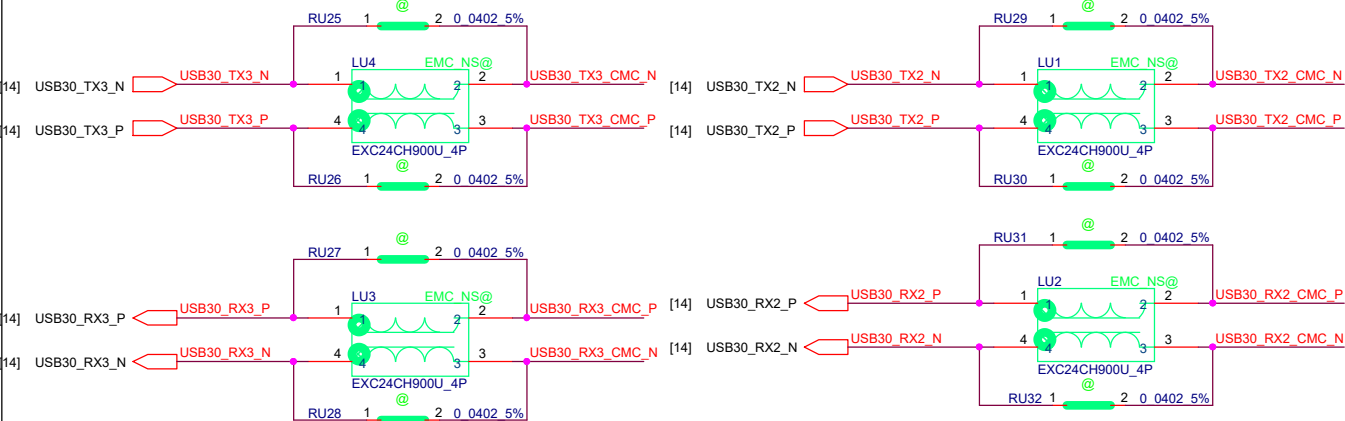
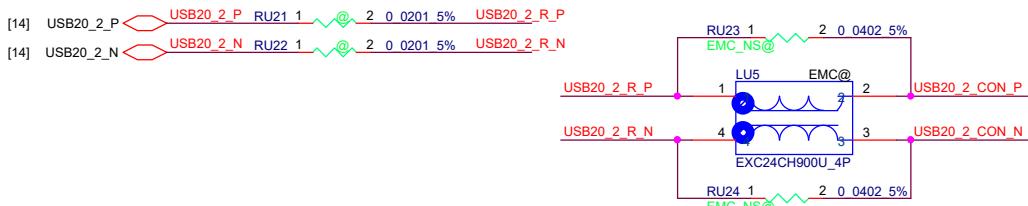
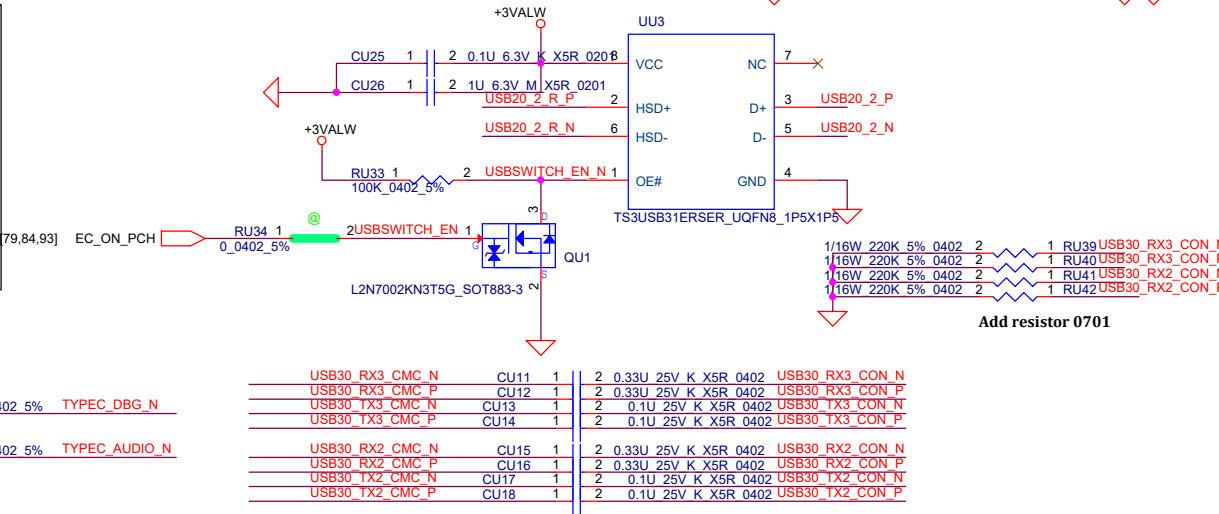
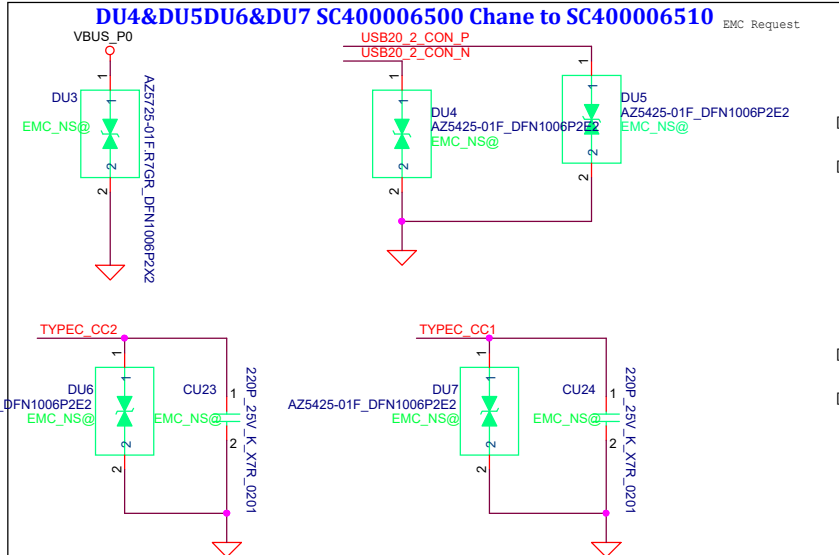
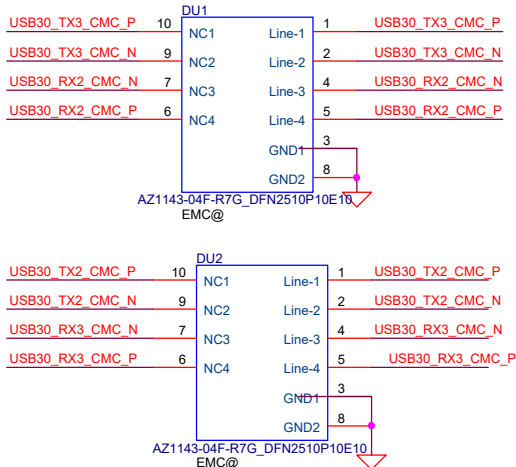
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Issued Date		2012/07/01		Deciphered Date								2014/07/01	
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								B	TYPEC Controller Porta			0.1	
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Date:		Tuesday, November 10, 2020				Sheet	52	of	110

the diffrent with Base:
1.S540 No type-c USB3.0 only,S360&V14V15 have



VBUS power fault indicator.
Output Hi-z: no fault.
Drive Low: VBUS power switch is in current limit state or IC is in over temperature condition.
PCB pulled up(1)
if connected to external controller.
Leave floating if not used.

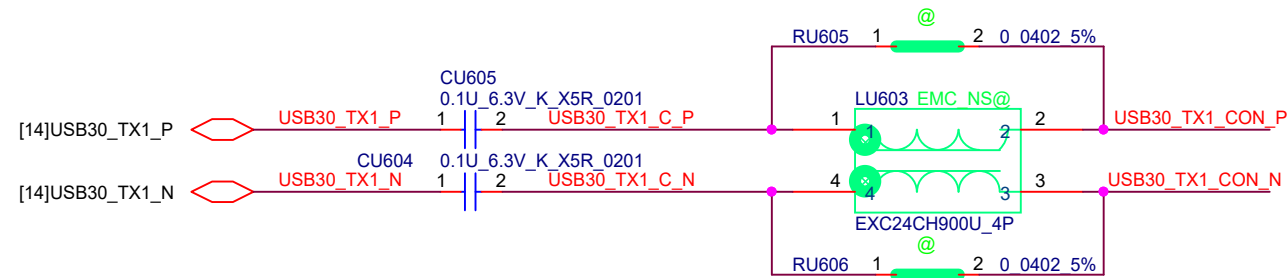
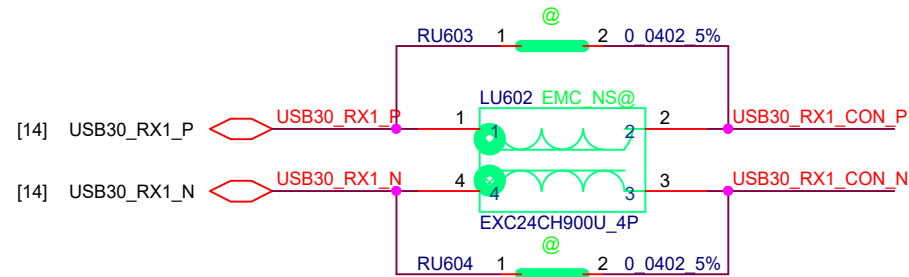
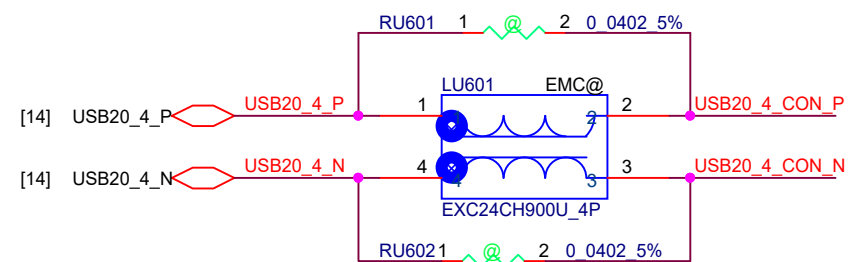
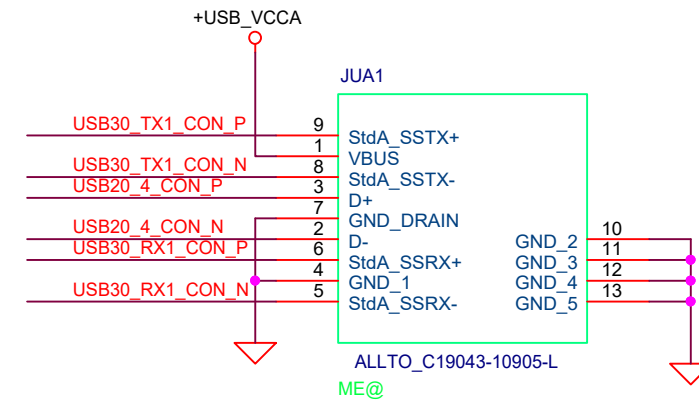
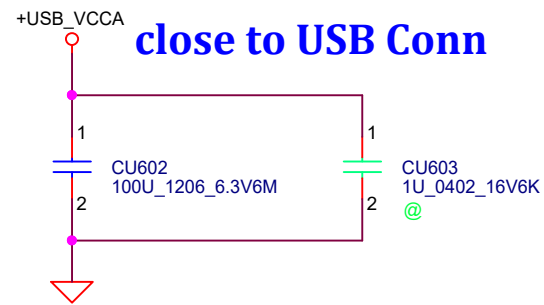
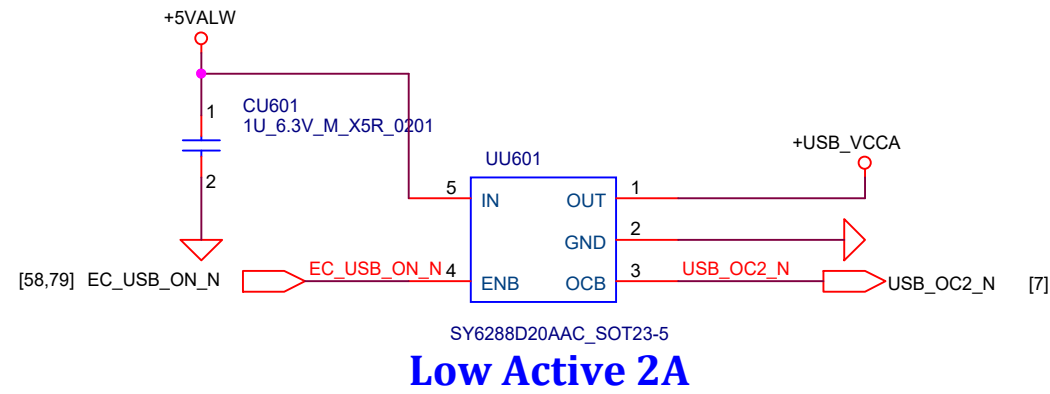


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							Size B	Document Number TYPEC Controller PortB				Rev 0.1
							Date:	Tuesday, November 10, 2020			Sheet	54

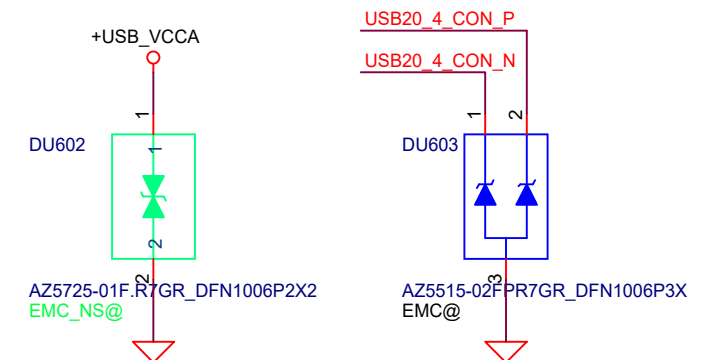
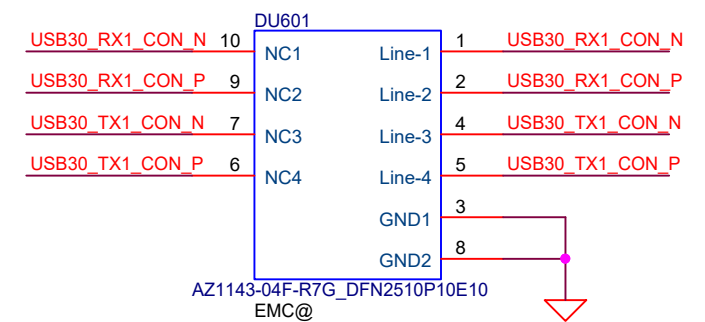
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Size B		Document Number						Rev 0.1	
		TYPEC MUX PortB							
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						Date:	Tuesday, November 10, 2020		Sheet 56 of 110	

USB3.0 PORT



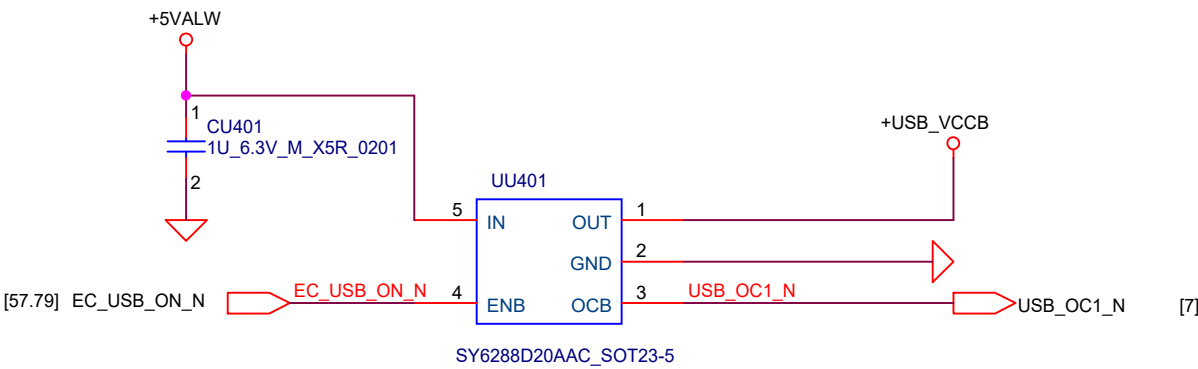
EMC close to USB Conn



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						Date:	Tuesday, November 10, 2020	Sheet 57 of 110	

USB2.0 Port and CMC&TVS&CAP MOVE to SUB Board

USB2.0 PORT



Low Active 2A

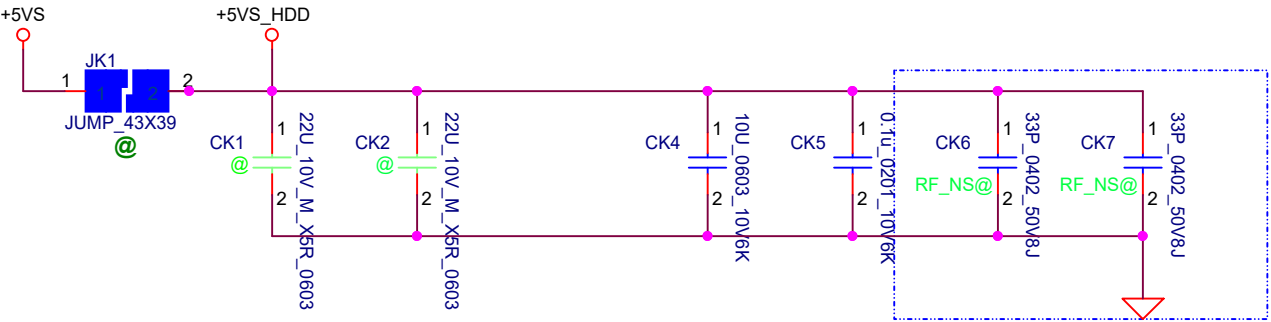
the diffrent with Base:
1.S540 No USB2.0 only,S360&V14V15 have

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							USBA PortB						
						Date:		Tuesday, November 10, 2020		Sheet 58 of 110			

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		USBA PortC									
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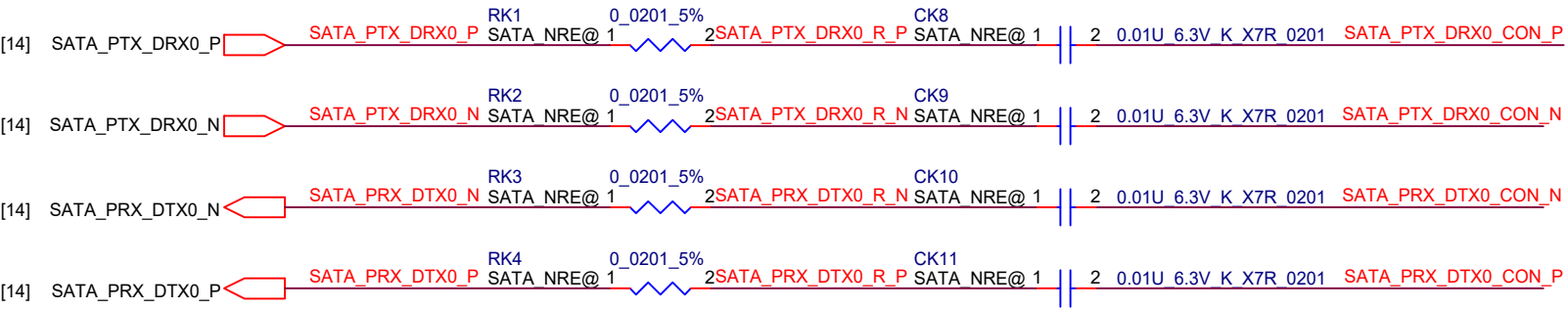
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Size B		Document Number								Rev 0.1	
		USB HUB									
Date:		Tuesday, November 10, 2020						Sheet 60 of 110			

HDD Power



the diffrent with Base:
1.S540 No HDD,S360&V14V15 have

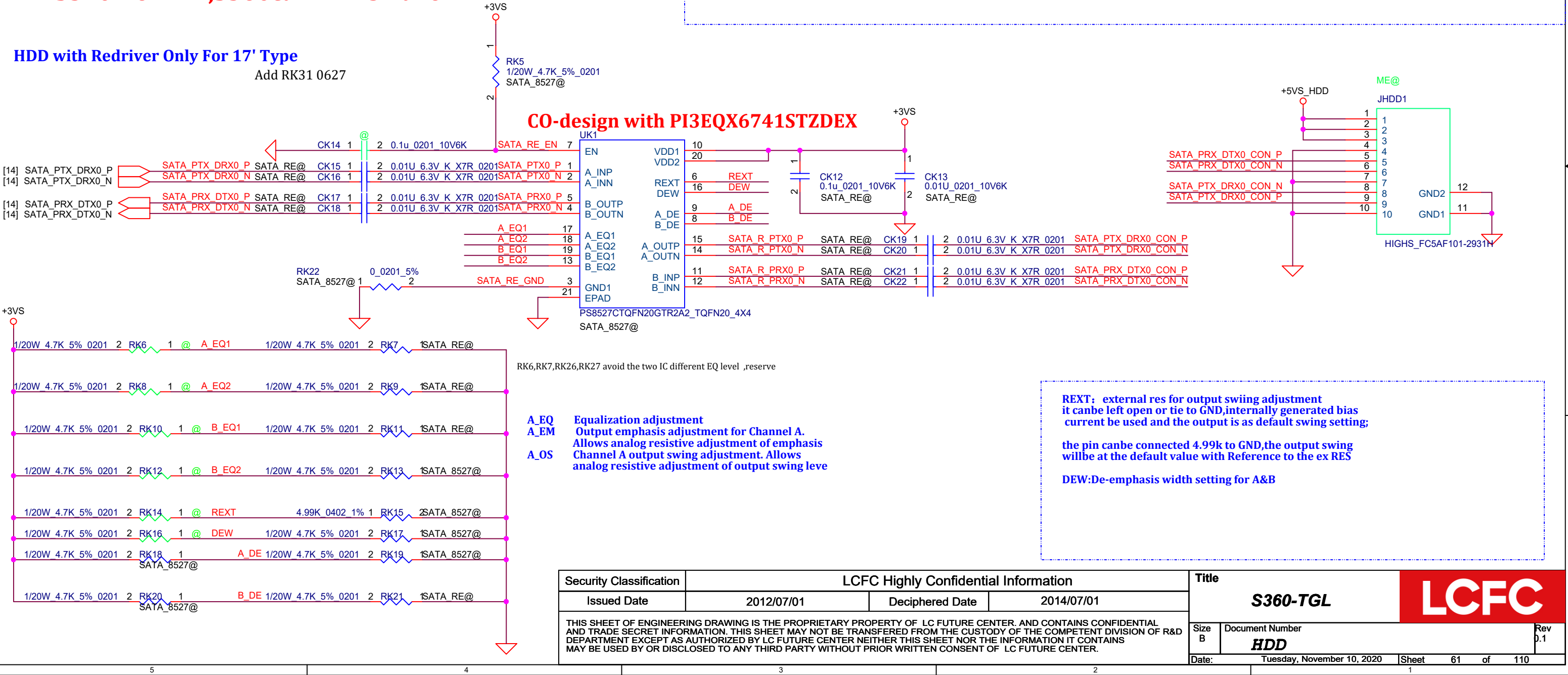
HDD Without Redriver



HDD with Redriver Only For 17' Type

Add RK31 0627

CO-design with PI3EQX6741STZDEX



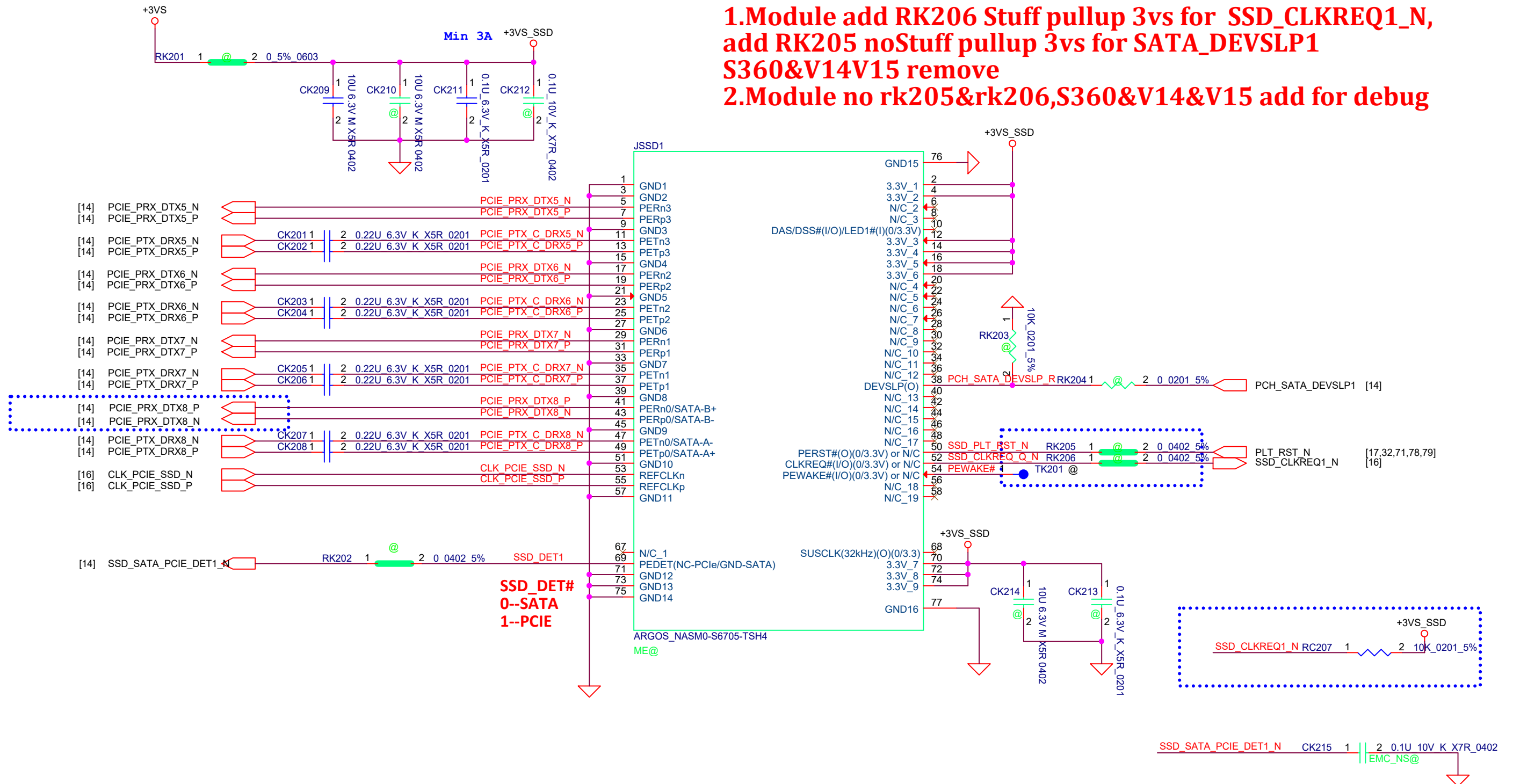
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the diffrent with Base MODULE:

**1.Module add RK206 Stuff pullup 3vs for SSD_CLKREQ1_N,
add RK205 noStuff pullup 3vs for SATA_DEVSLP1**

S360&V14V15 remove

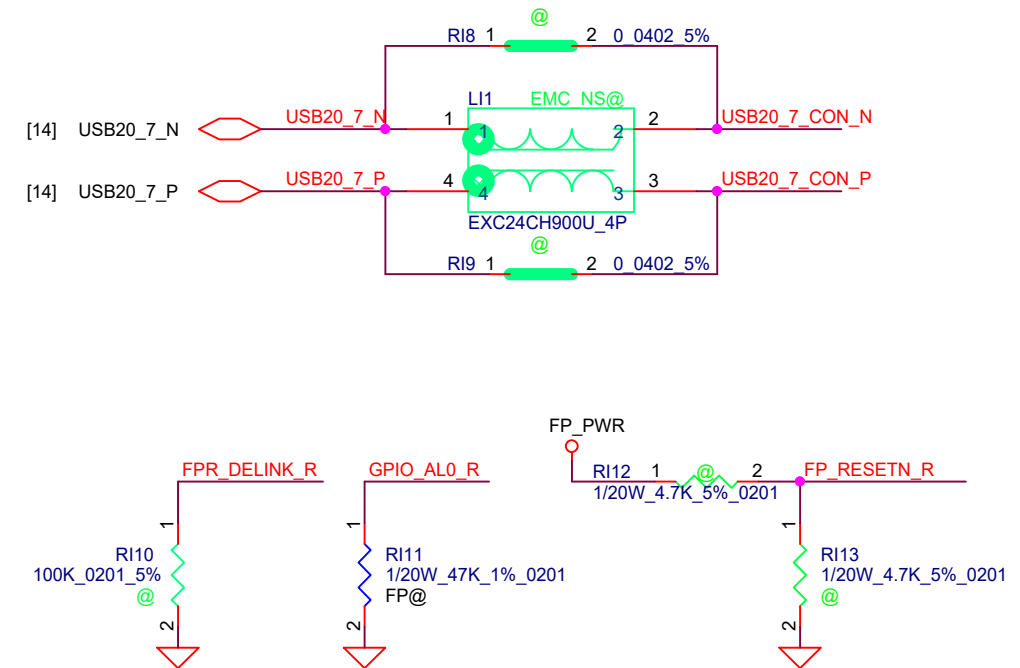
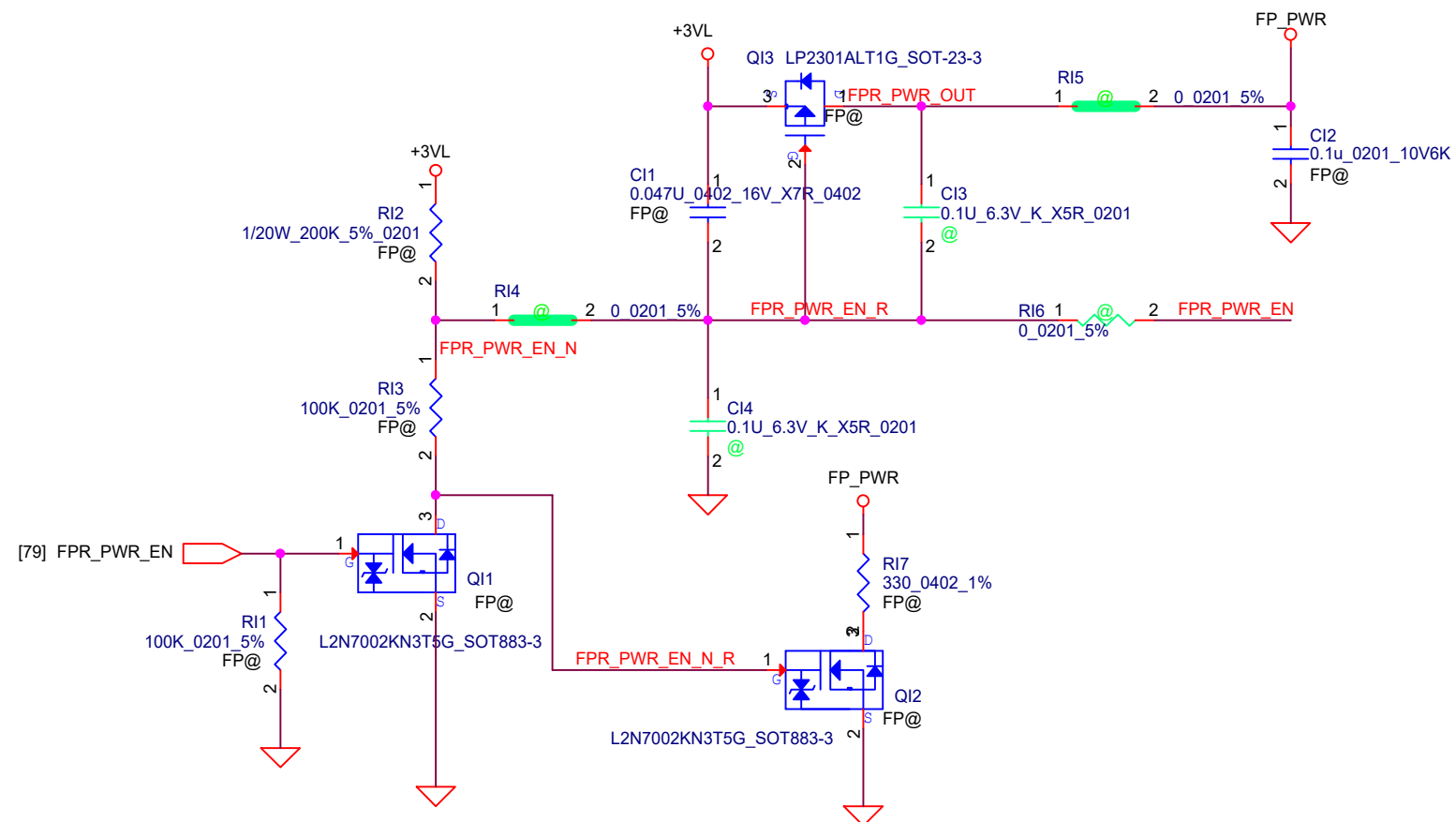
2.Module no rk205&rk206,S360&V14&V15 add for debug



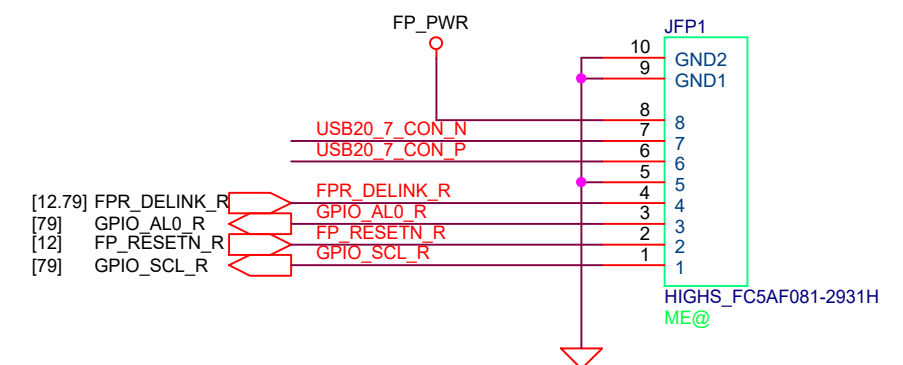
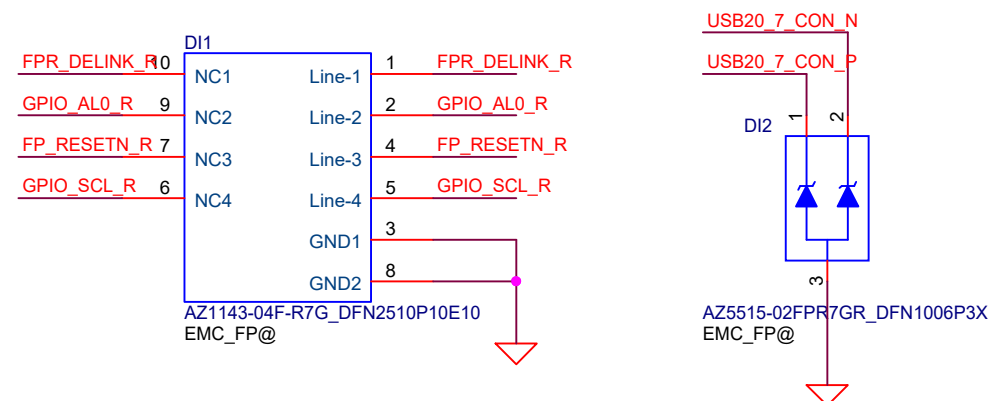
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SSD											
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				Date:	Tuesday, November 10, 2020		Sheet

FP Power Control



the diffrent with Base:
1.S540 No Fingerprinter,S360&V14V15 have

EMC
close to Conn

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		FPR							
Date:		Tuesday, November 17, 2020				Sheet 65 of 110			

S360-TGL

LCFC

FPR

Date:

Tuesday, November 17, 2020

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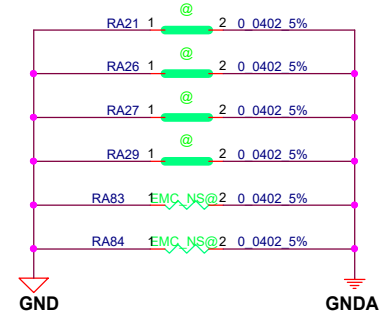
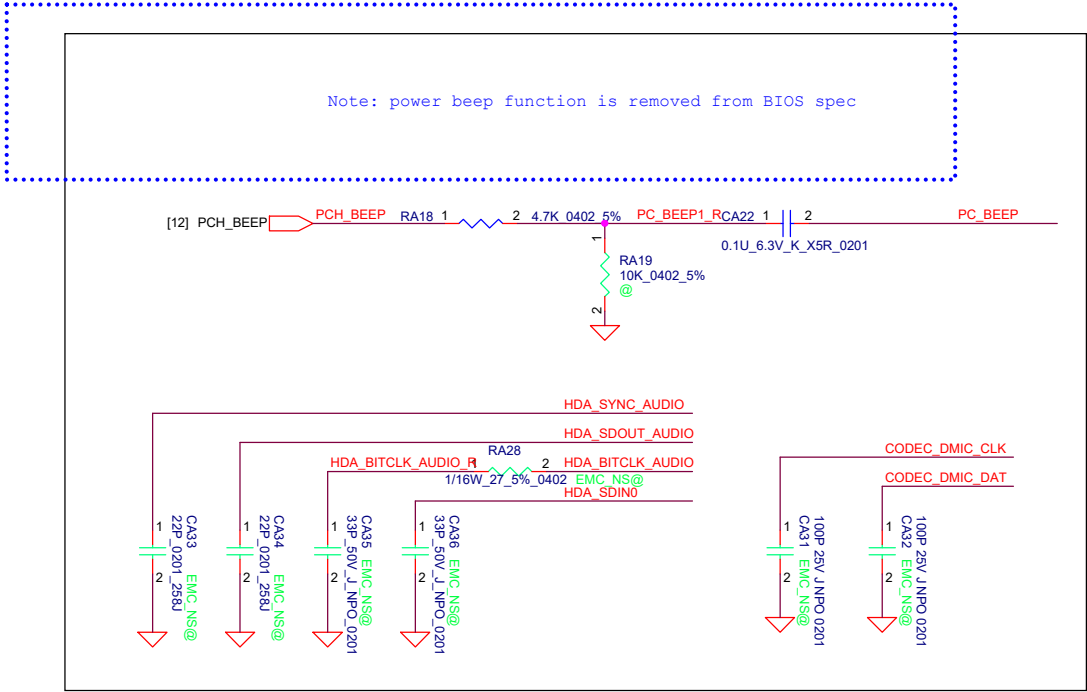
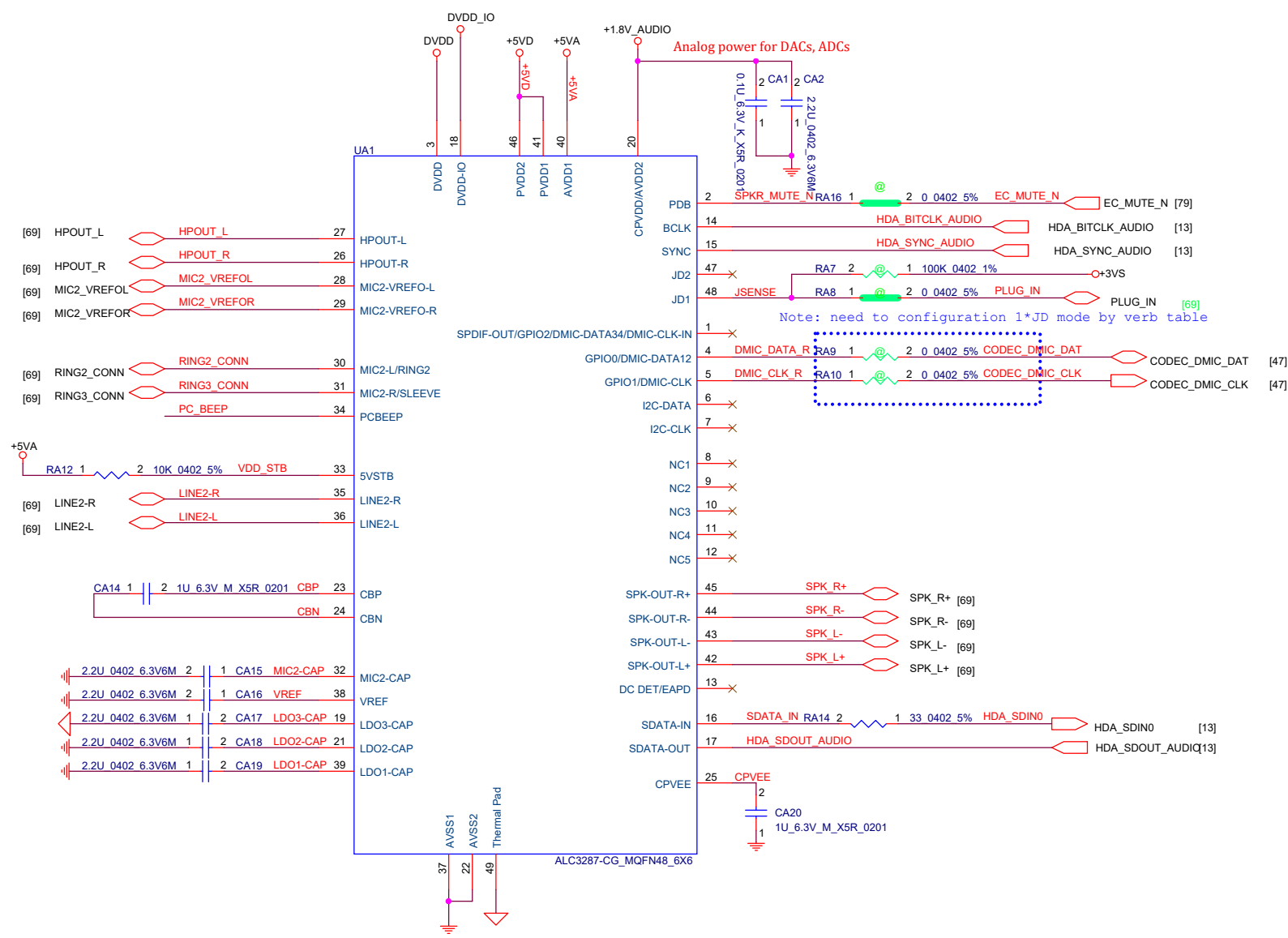
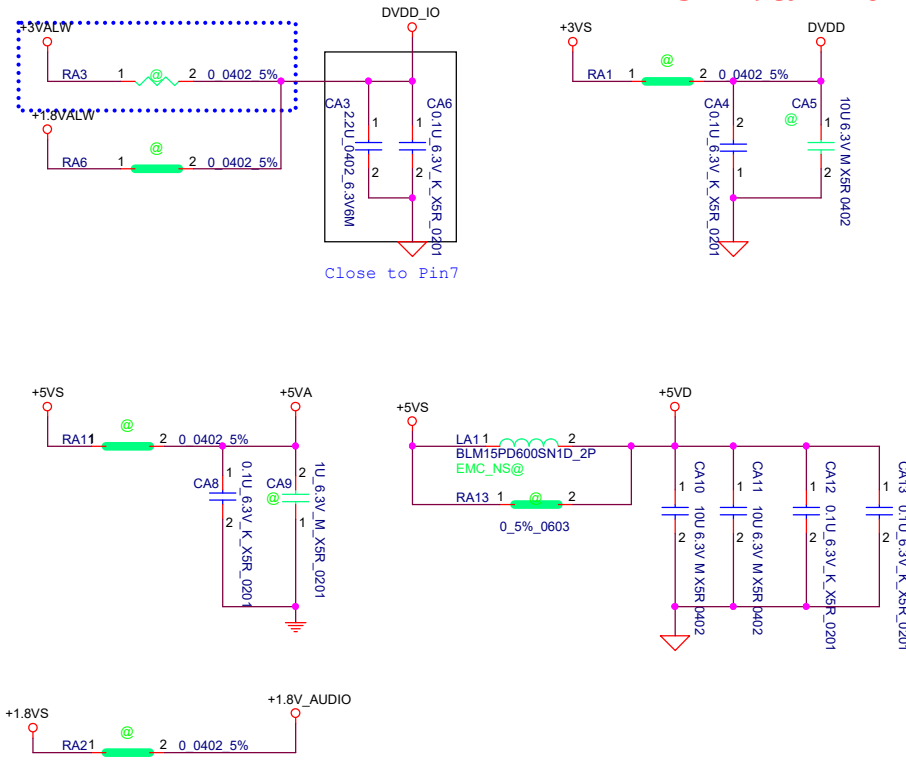
of

110

1

the diffrent with Base MODULE:
1.Module no ra3 to DVDD_IO,S360&V14&V15 reserve
2.Module no ra5 CA21 to EC_BEEP,S360&V14&V15 reserve
3.RA9&RA10 BOMStructure different with module

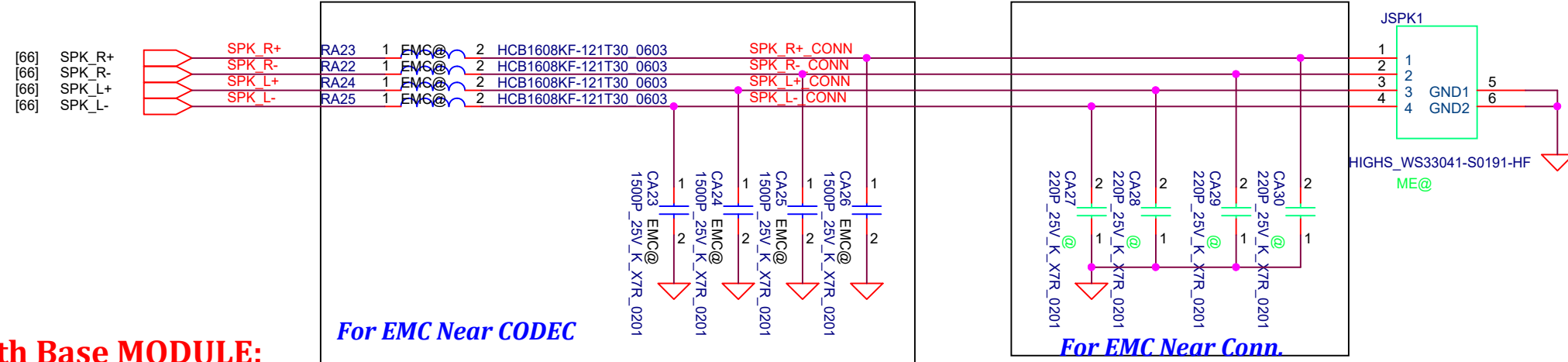
Note: DVDD-IO must be equal to or smaller than DVDD



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							Audio Amplifier						
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Size B		Document Number								Rev 0.1	
		Audio DSP									
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Speaker

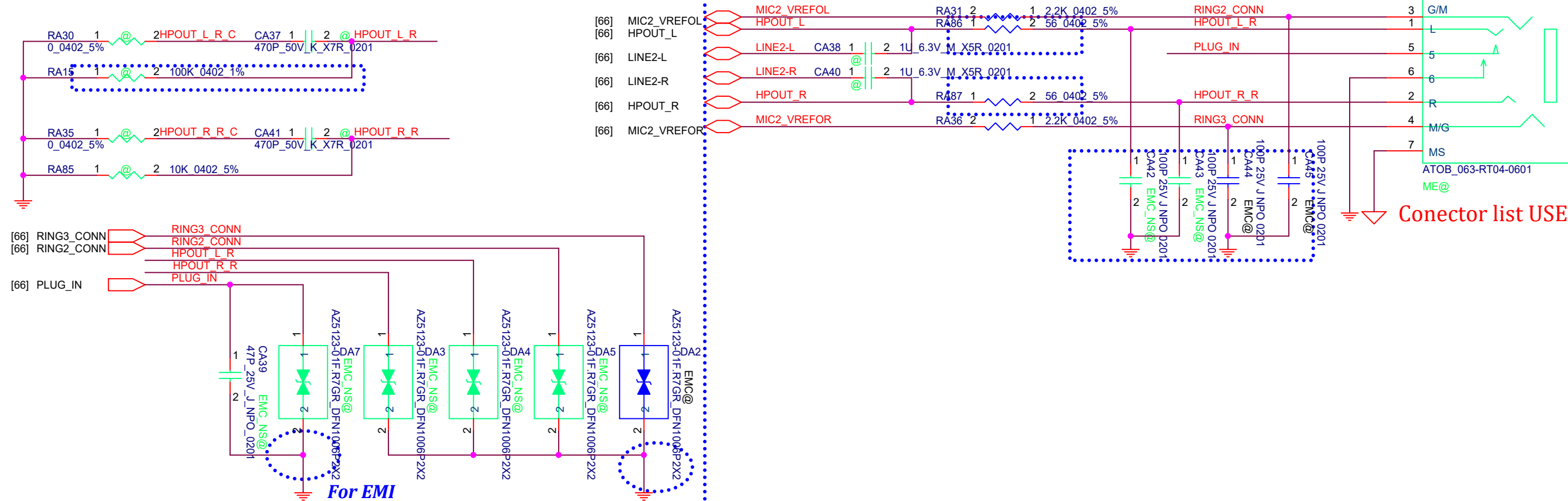


the diffrent with Base MODULE:

- 1.S360&V14&V15 RA86&RA87 VALUE different with module, Module have RA55&RA88,S360&V14&V15 Remove
- 2.for EMC TVS EMC_NS@ type different with module
- 3.GND type different
- 4.module no CA37&CA41 part circuit,,S360&V14&V15 reserve

6pin&7pin audio jack comon design,stuff 6pin default;
need vitrual symbol control

Audio Jack



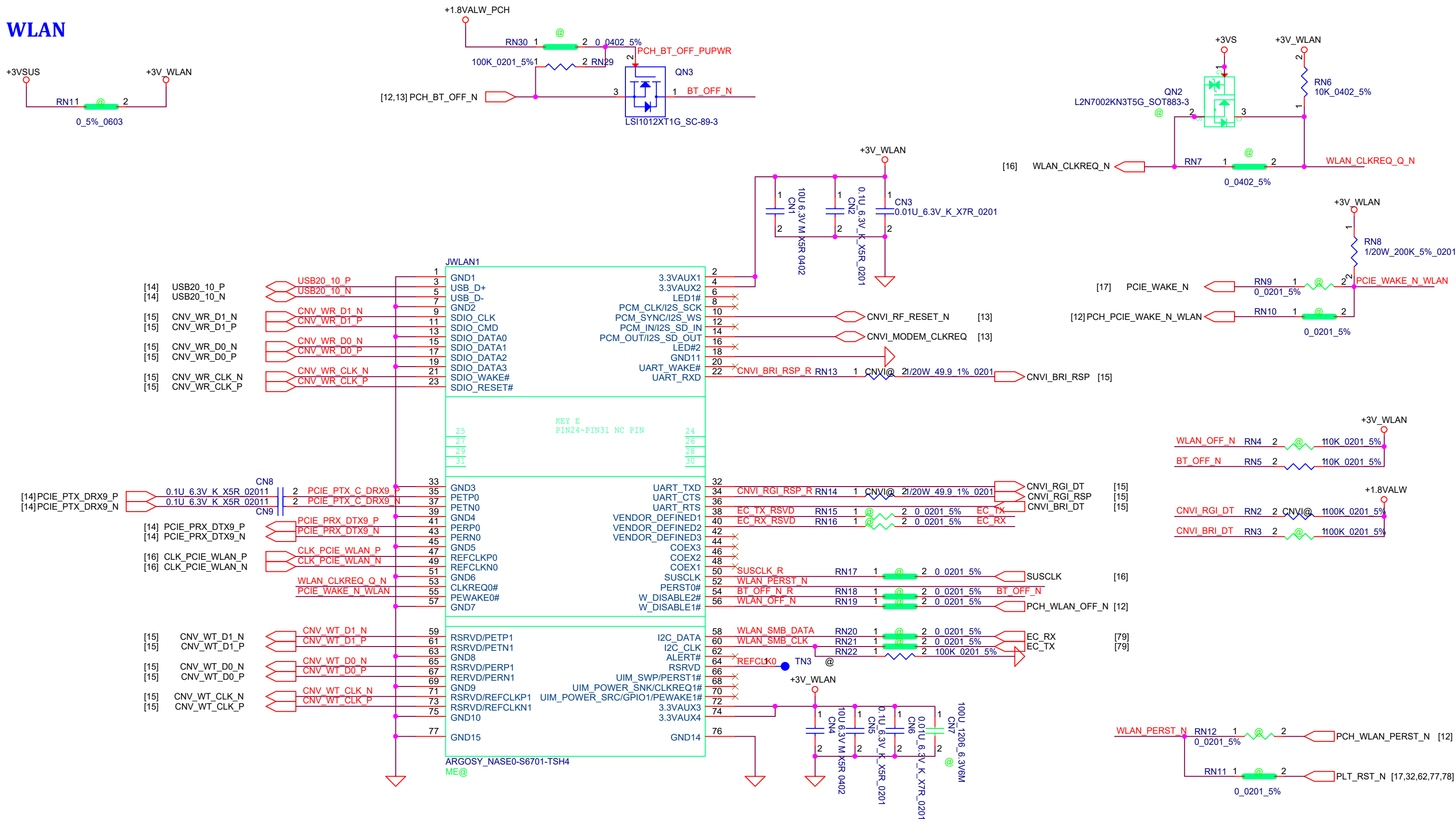
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Issued Date	2012/07/01	Deciphered Date	2014/07/01	S360-TGL	
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				Document Number	Audio SPK/Jack
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LCFC

Card reader Circuit MOVE to SUB Board S360

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				Date: Tuesday, November 10, 2020		Sheet 70 of 110	1

WLAN



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LAN Circuit MOVE to SUB Board in V360 DB

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Size B		Document Number LAN Chipset						Rev 0.1	
Date:		Tuesday, November 10, 2020				Sheet 73 of 110			

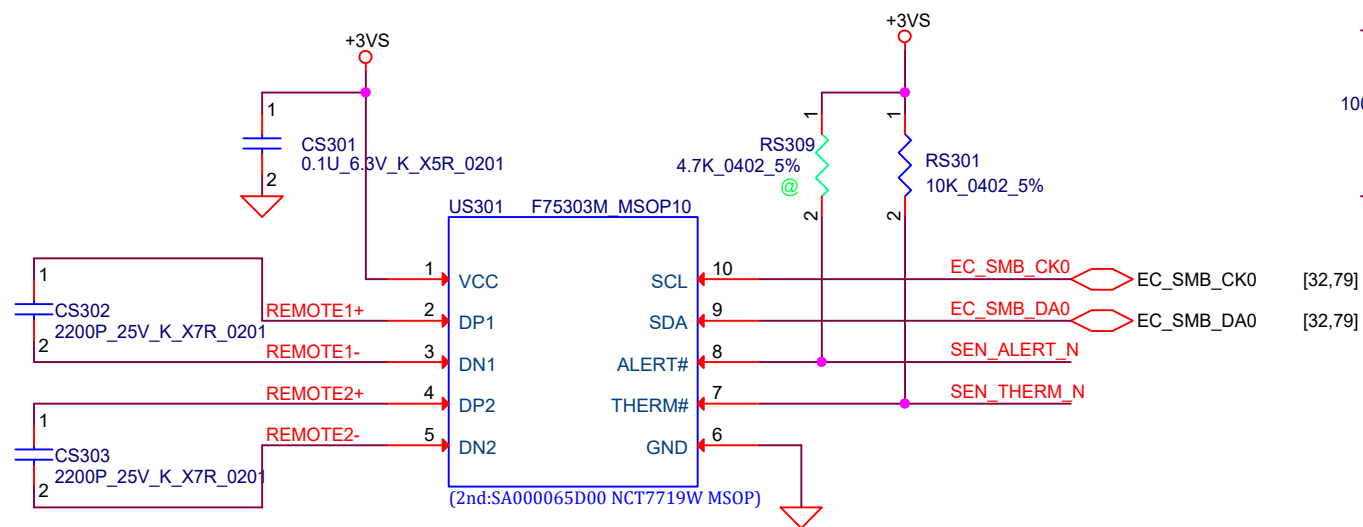
LAN Circuit MOVE to SUB Board in V360 DB

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Issued Date		2012/07/01		Deciphered Date								2014/07/01	
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							LAN Transformer						
Date:		Tuesday, November 10, 2020				Sheet		74		of 110			

LAN Circuit MOVE to SUB Board in V360 DB

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				Date:	Tuesday, November 10, 2020	Sheet 75 of 110

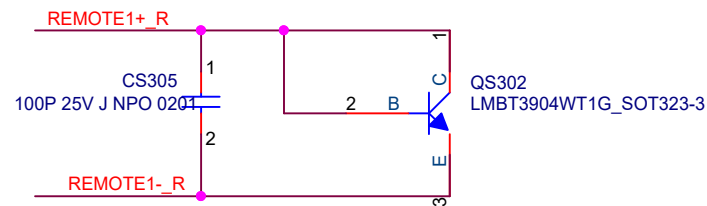
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Issued Date	2012/07/01	Deciphered Date	2014/07/01				
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				Date:	Tuesday, November 10, 2020		Sheet 76 of 110



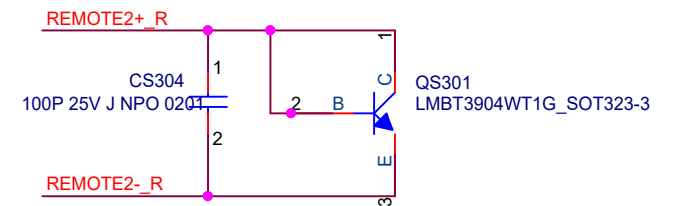
REMOTE+/-_R, REMOTE1+/-, REMOTE2+/-:
Trace width/space:10/10 mil
Trace length:<8"

the different with Base MODULE:
1.Module no RS309,S360 &V14V15 add RS309 location

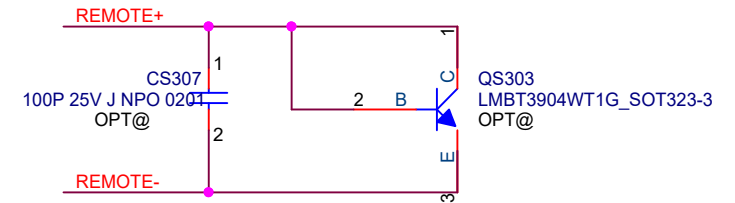
Near CHARGER



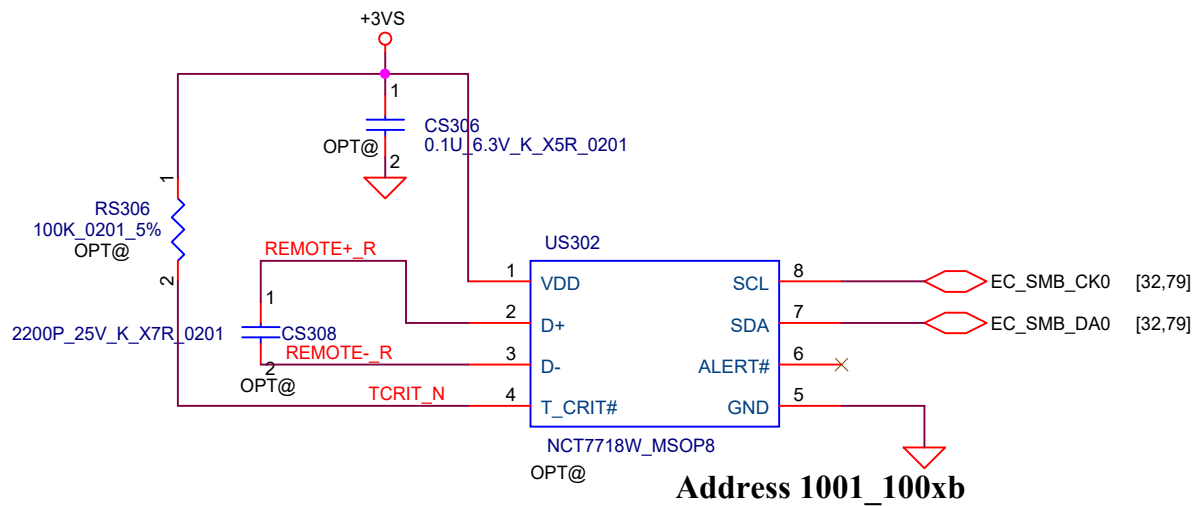
Near CPU



Near GPU



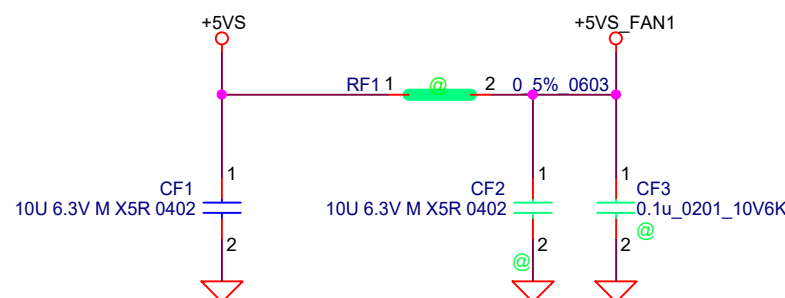
REMOTE1+_R	RS304	1	@	2	0	0402	5%	REMOTE1+_R
REMOTE1-_R	RS305	1	@	2	0	0402	5%	REMOTE1-_R
REMOTE2+_R	RS302	1	@	2	0	0402	5%	REMOTE2+_R
REMOTE2-_R	RS303	1	@	2	0	0402	5%	REMOTE2-_R
REMOTE+_R	RS307	1	@	2	0	0402	5%	REMOTE+_R
REMOTE-_R	RS308	1	@	2	0	0402	5%	REMOTE-_R



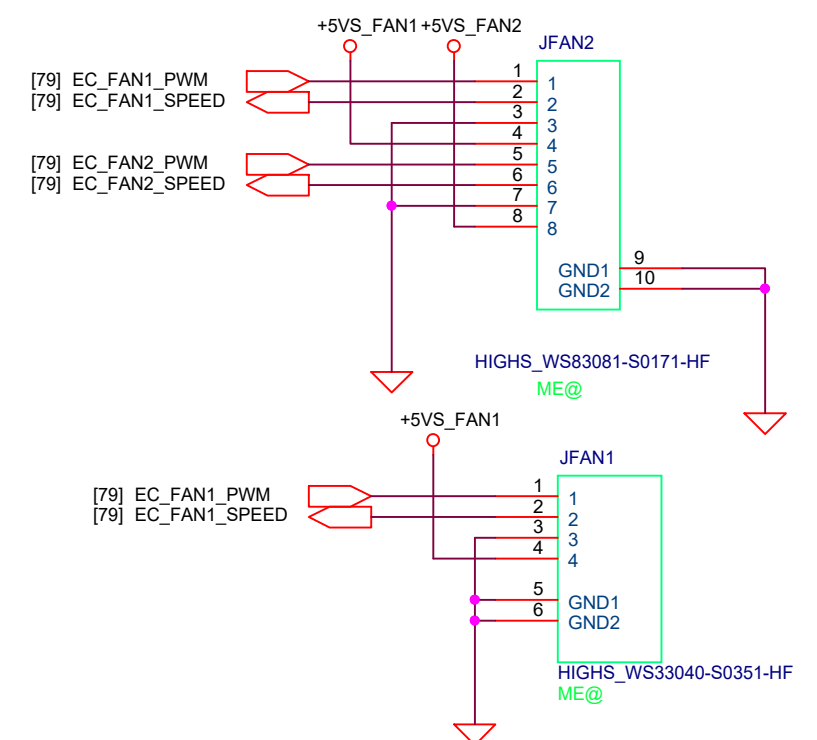
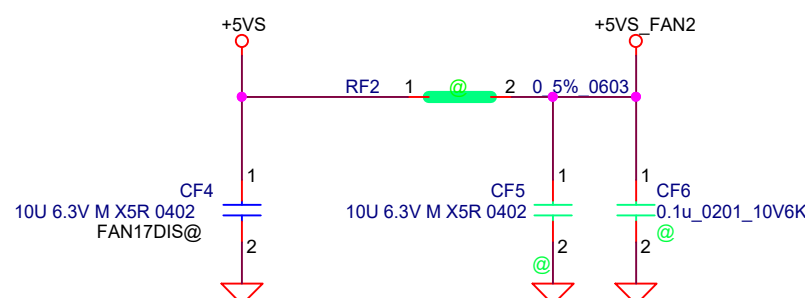
Address 1001_100xb

REMOTE+/-_R, REMOTE1+/-, REMOTE2+/-:
Trace width/space:10/10 mil
Trace length:<8"

FAN



FAN

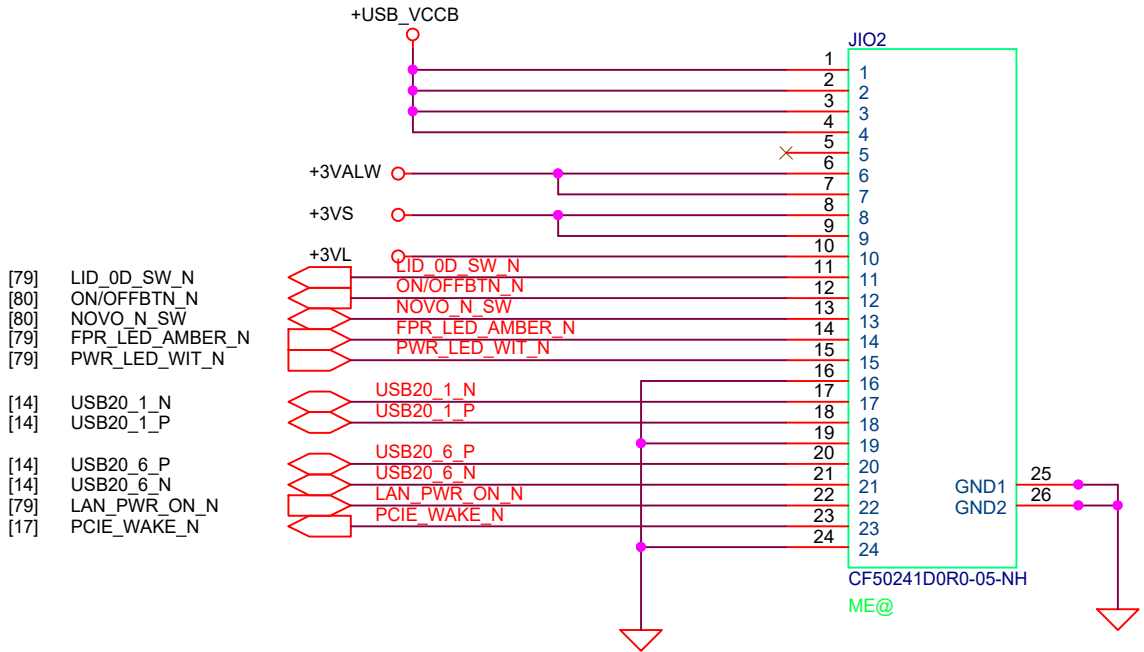
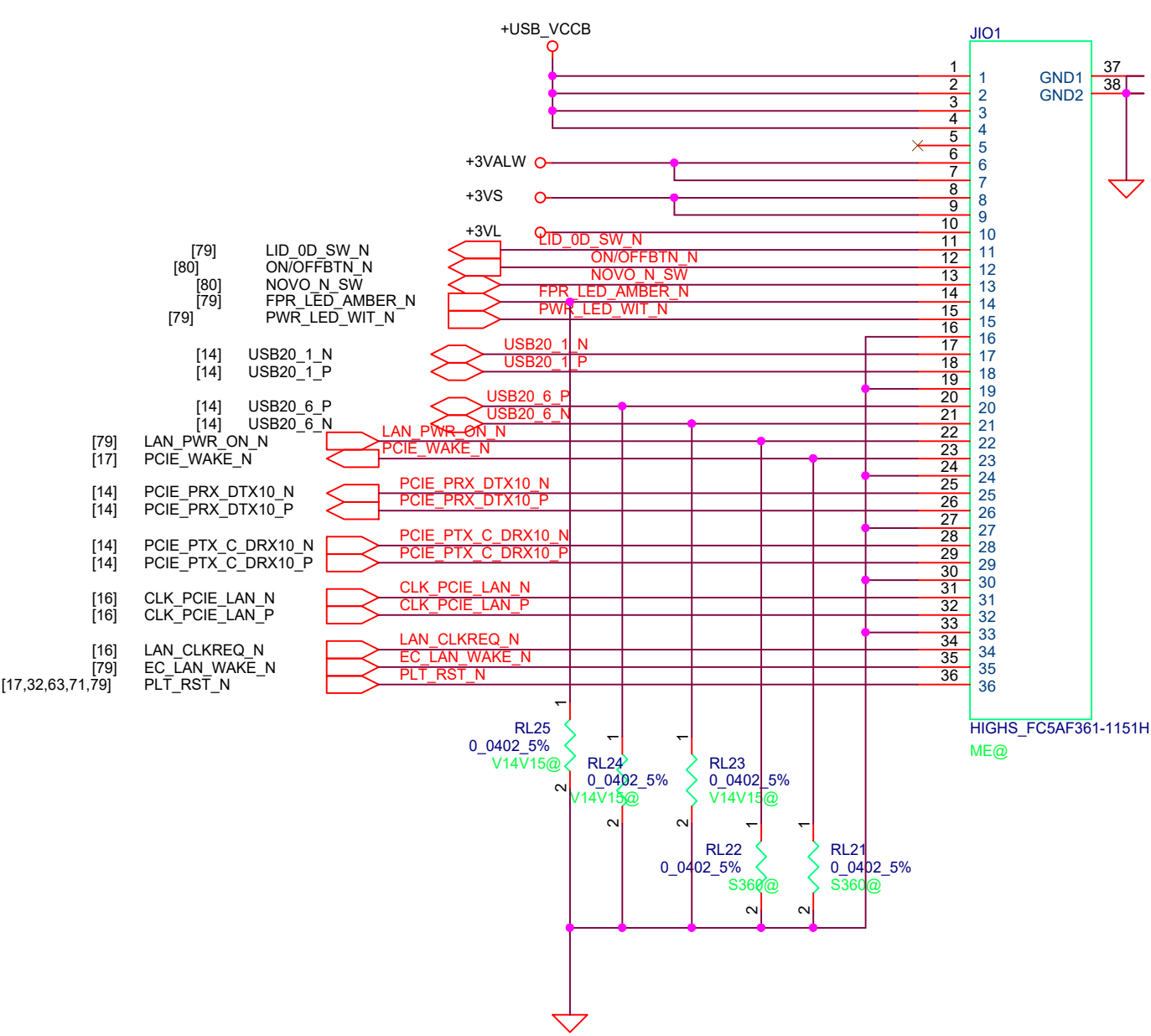


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				Date:	Wednesday, November 11, 2020	Sheet	77	of 110

FPC+cable+Wire

USB2.0_1 For USB2.0 Connector;
USB2.0_6 For S360 Card Reader

3VALW for LED&V360 Lanchip Power;
3VS for S360 Card Reader, V360 ISOLATE_N



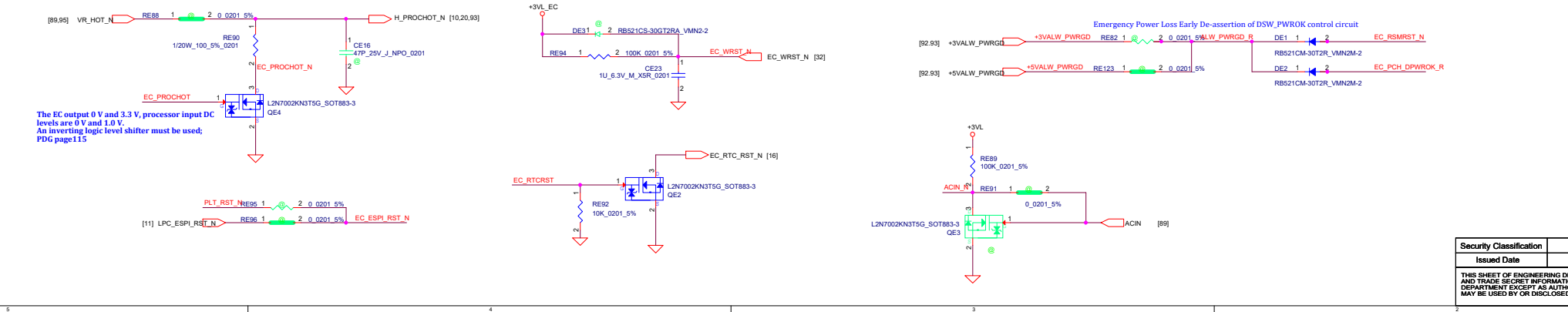
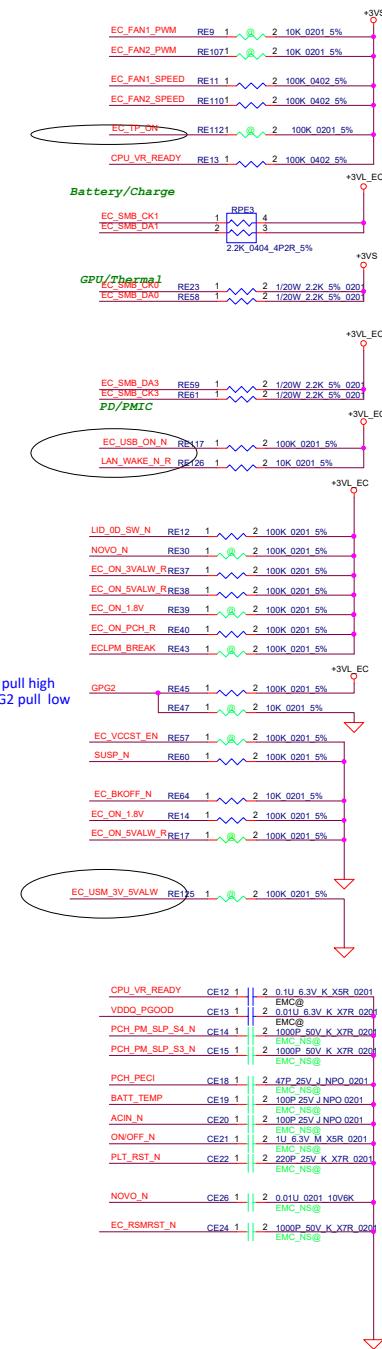
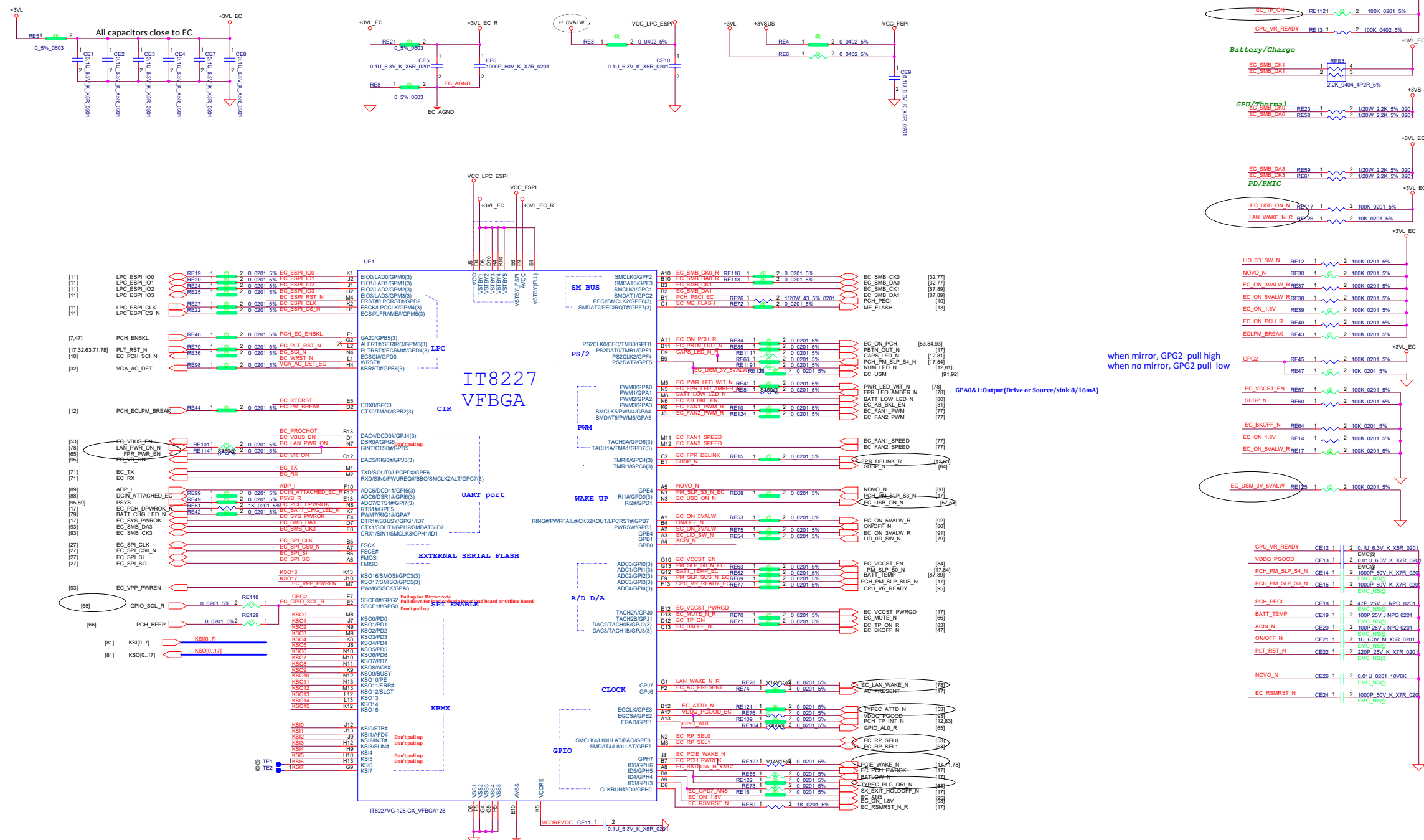
Security Classification		LCFC Highly Confidential Information				Title S360-TGL				
Issued Date	2012/07/01	Deciphered Date	2014/07/01							
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						Date:	Tuesday, November 10, 2020		Sheet 78 of 110	1

S360-TGL

LCFC

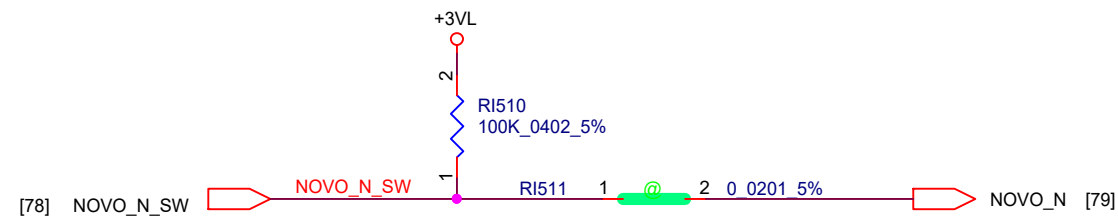
Sub Board CONN

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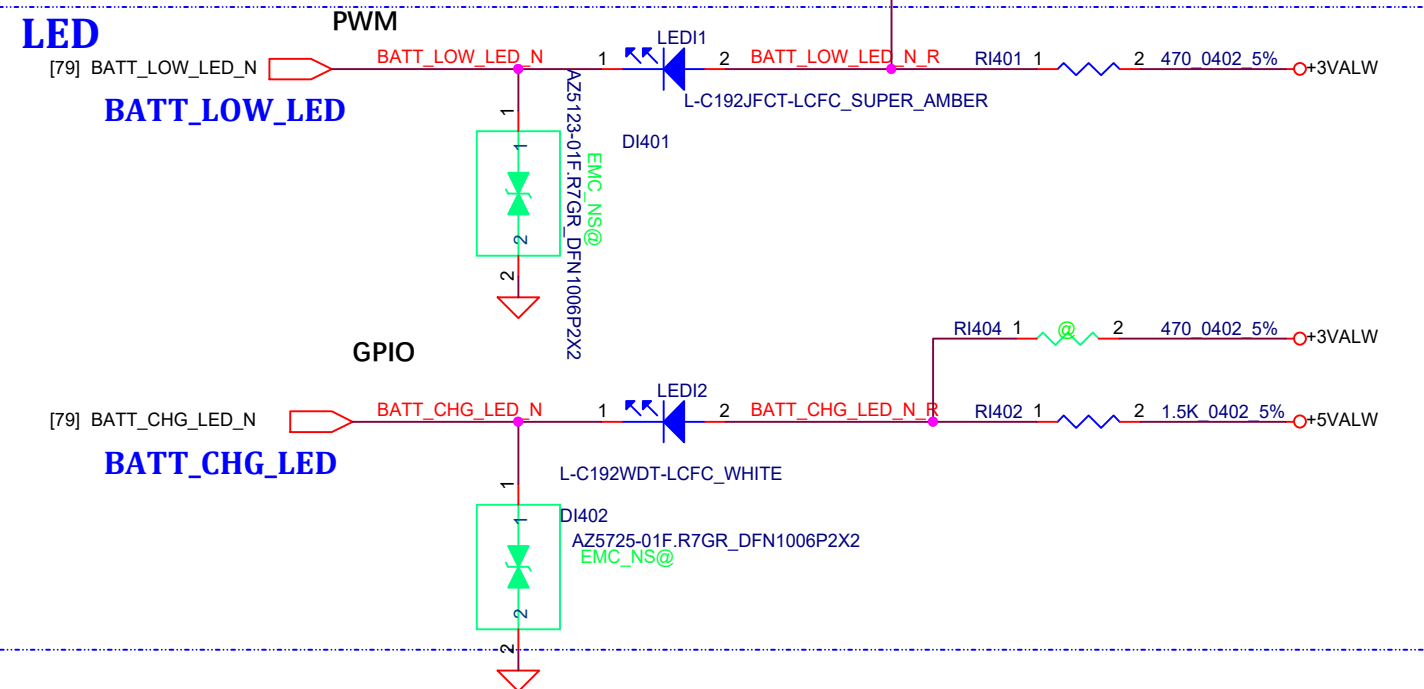


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Document Number		S360-TGL		EC		LCFC	
Date		Tuesday, November 17, 2020		Sheet		79 of 119	

NOVO switch

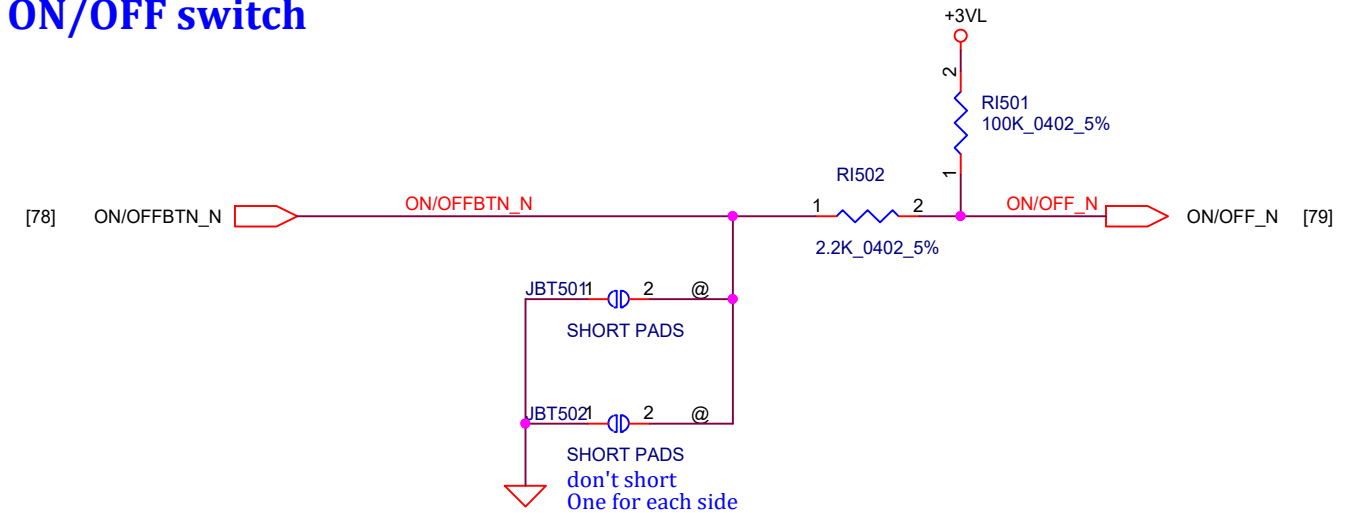


LED



The Power LED move to SUB board LENS

ON/OFF switch

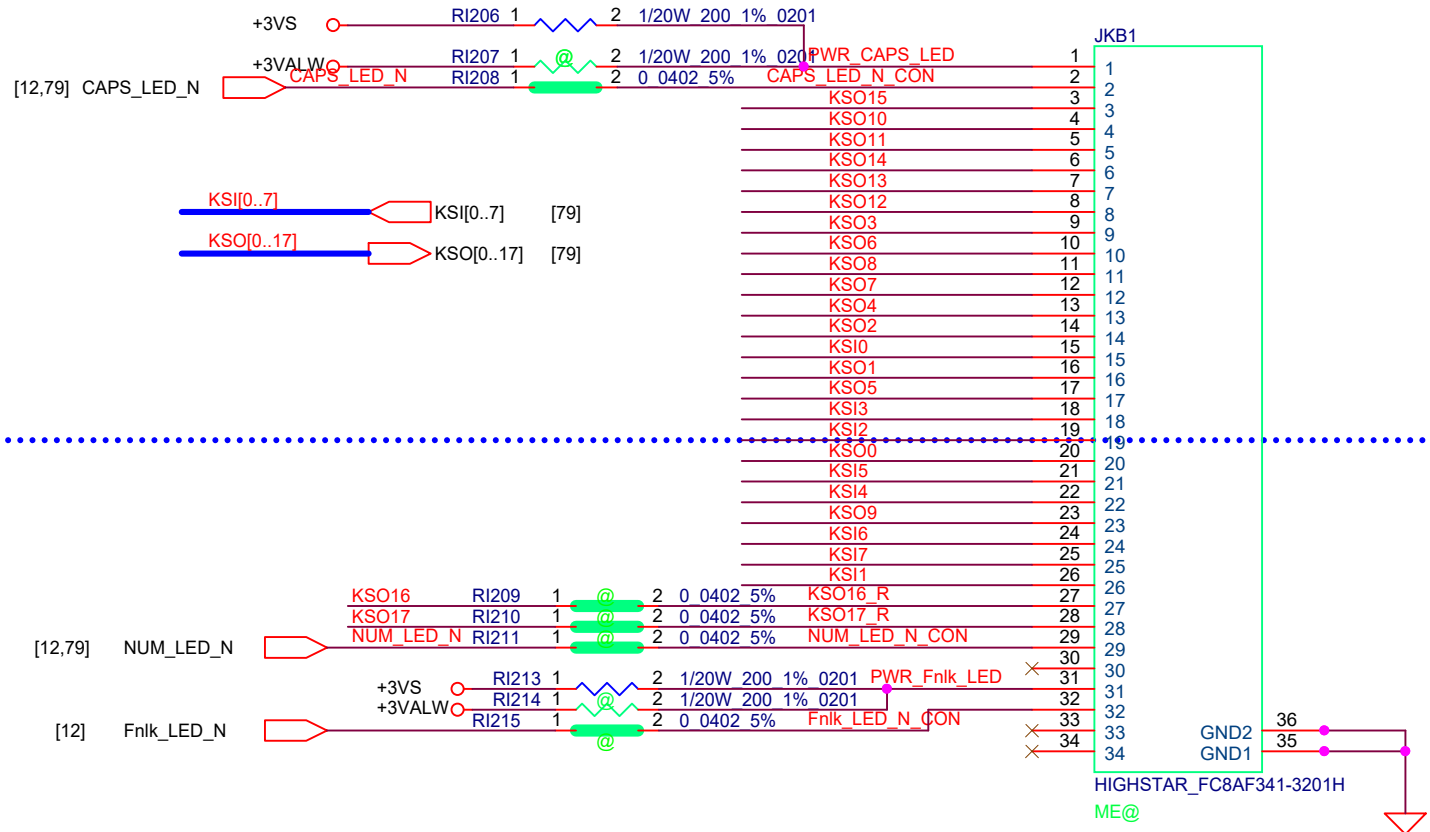


the diffrent with module:
1.module mos control LED,S360&V14V15 NO NEED

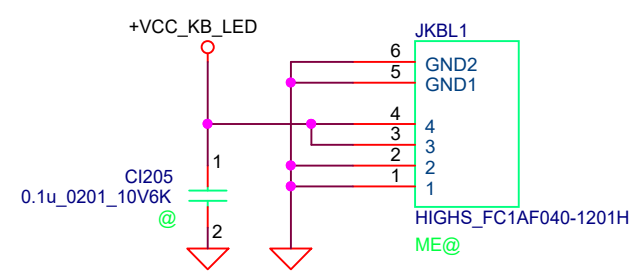
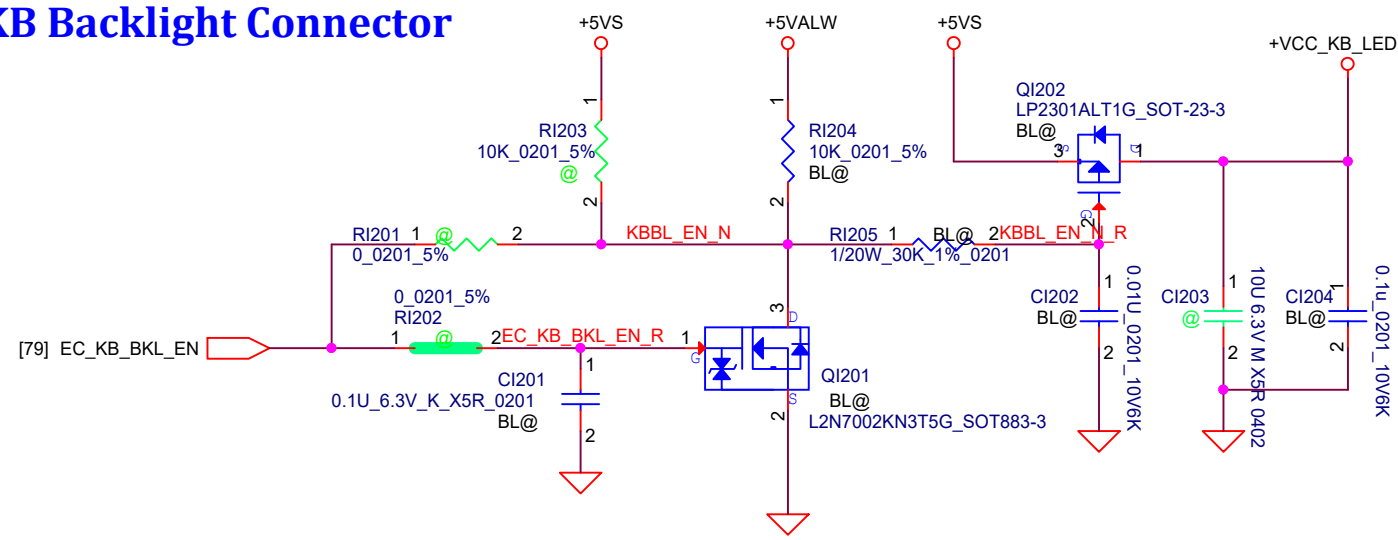
LED	Stute	LED Behavior
Power Button	System on	White_on (battery: 21%~100%)
		Amber_on (battery: 0%~20%)
	Standby LID closed	White_on (battery: 21%~100%)
		Amber_Blink_3S (battery: 0%~20%)
Charging	System off	OFF
	Battery only	OFF
	Charging	Amber_on (battery: 1%~90%) White_on (battery: 91%~100%)

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Size B		Document Number								Rev 0.1	
		LED/Button									
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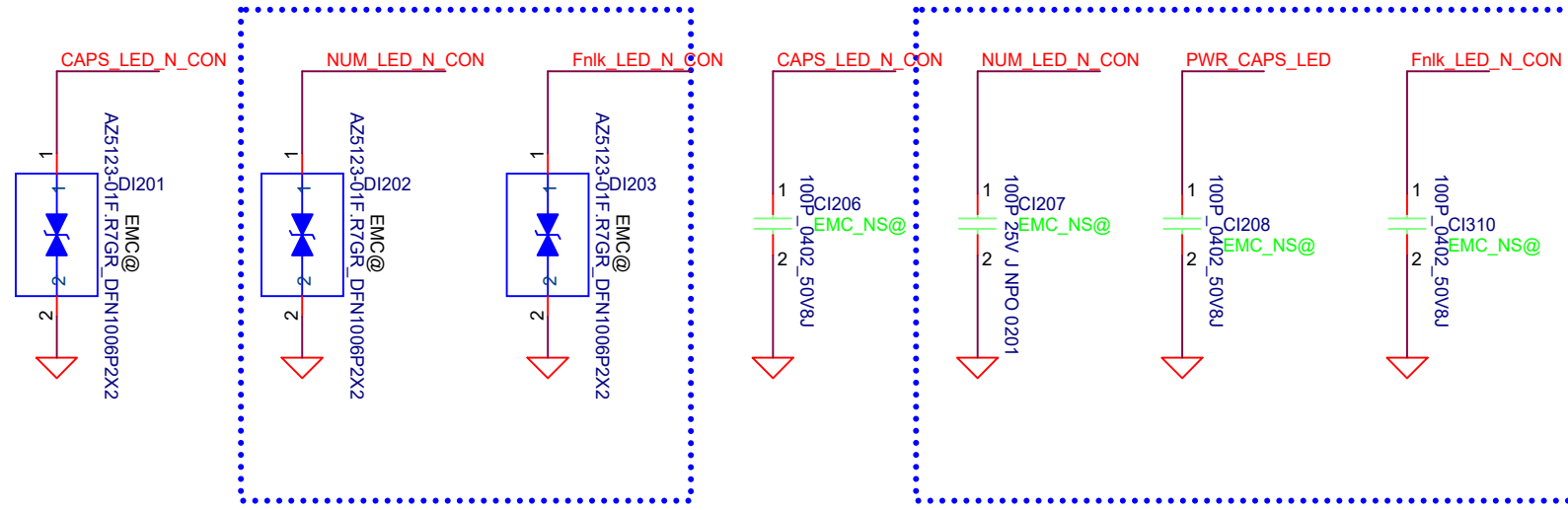
K/B Connector



KB Backlight Connector



EMC close to Conn



the diffrent with module:
1.modulel No 15 KB,S360&V14V15 have

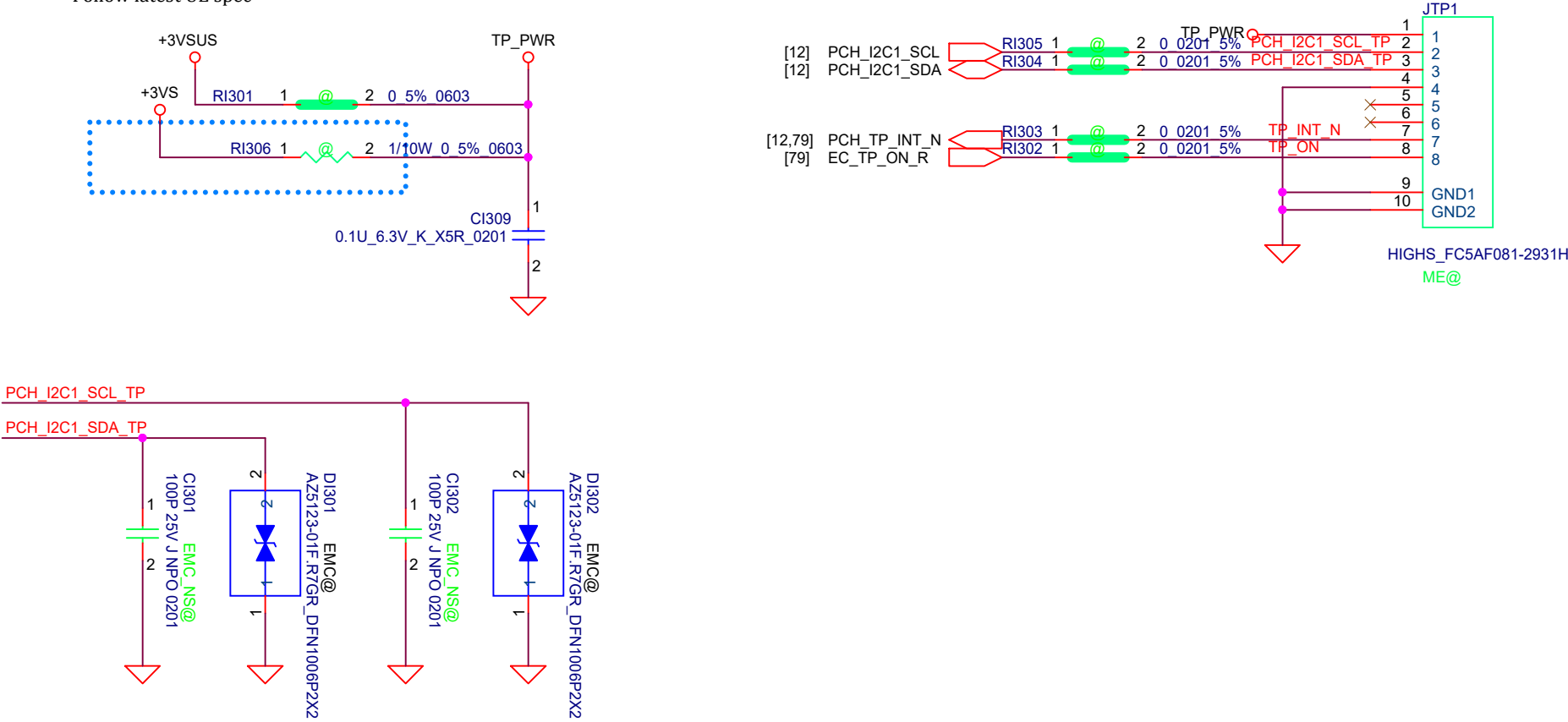
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				Date:	Tuesday, November 17, 2020	Sheet 81	of 110		
		3			2			1	

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						Anti Ghost KB					
Date:		Tuesday, November 10, 2020				Sheet	82	of	110		

the diffrent with Base MODULE:
1.Module no RI306,S360&V14&V15 Rerserve

Touch Pad

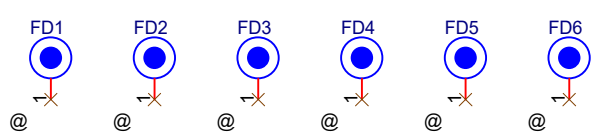
Need support wake up by touch pad
Follow latest UE spec



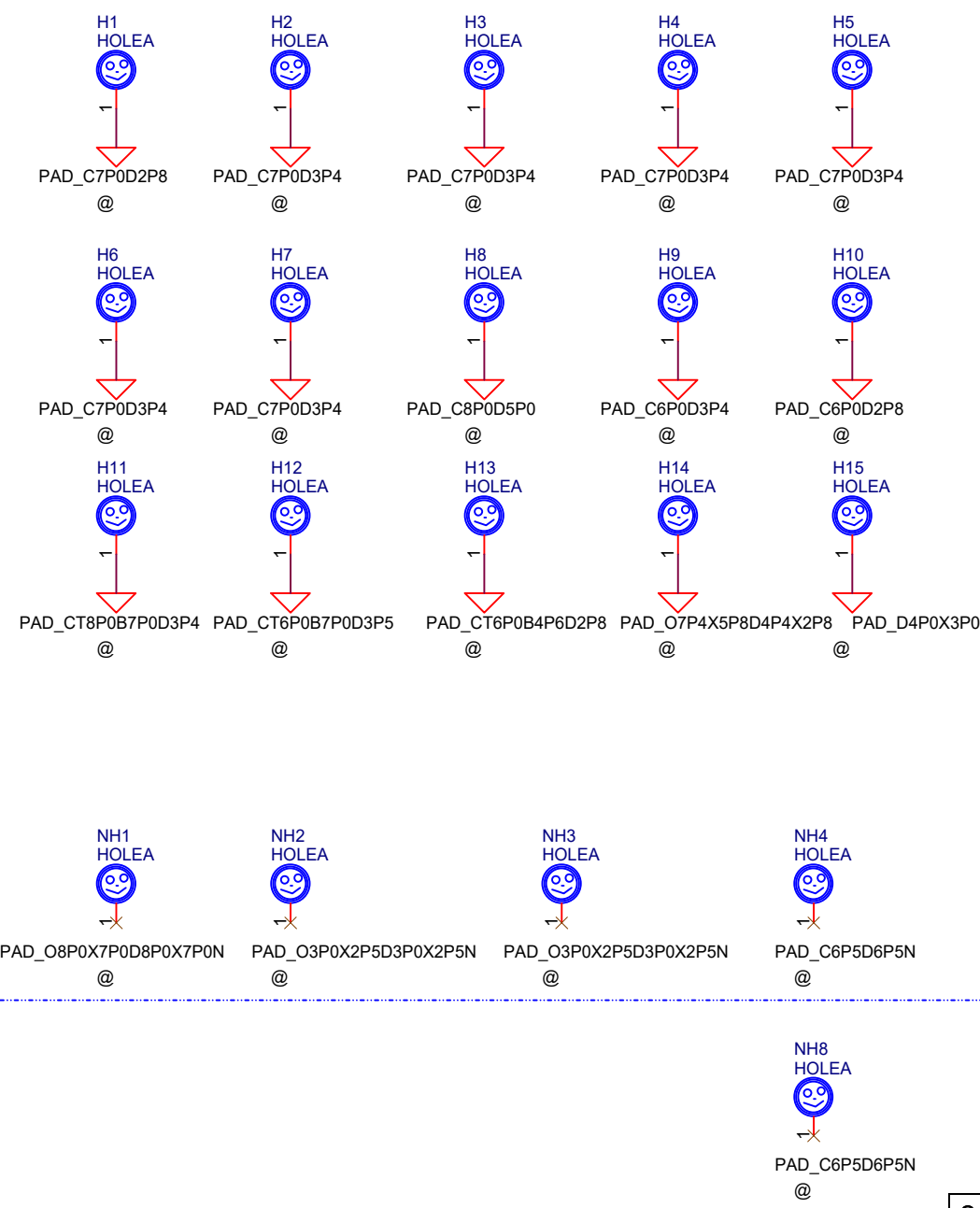
Security Classification		LCFC Highly Confidential Information		Title	
Issued Date	2012/07/01	Deciphered Date	2014/07/01	S360-TGL	
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PCB Federal Mark PAD

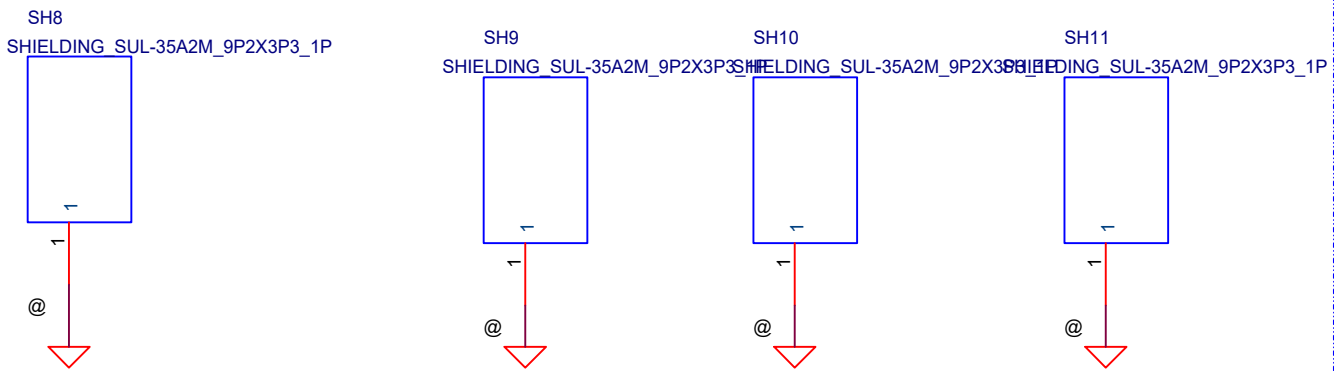


Hole

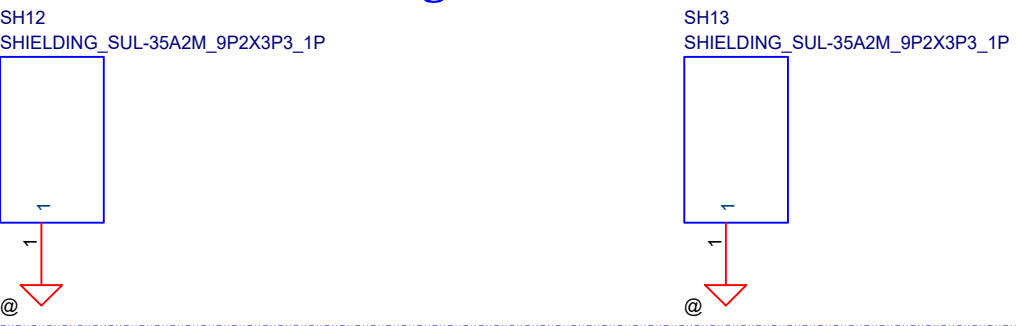


MD Shielding

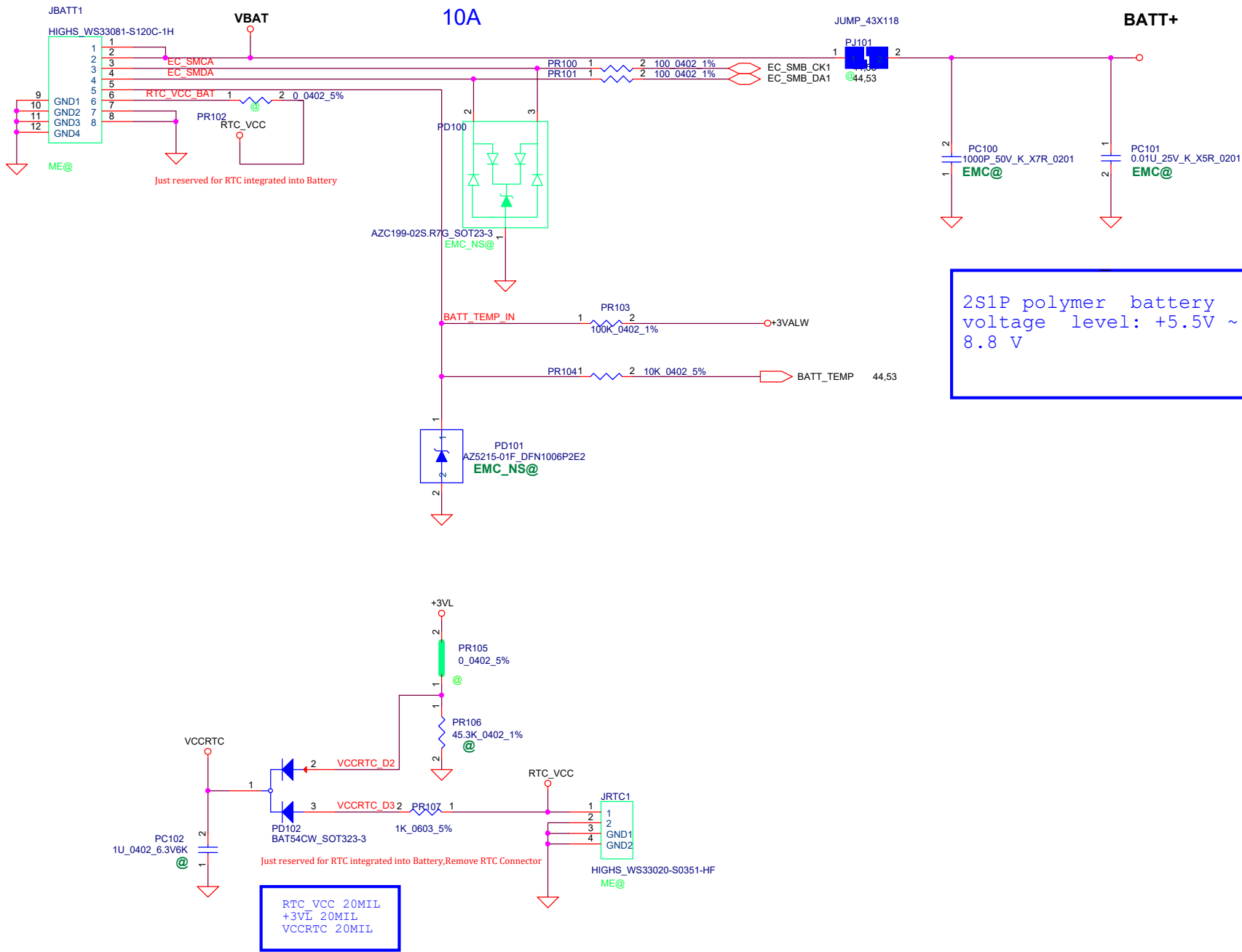
SO-DIMM Shielding

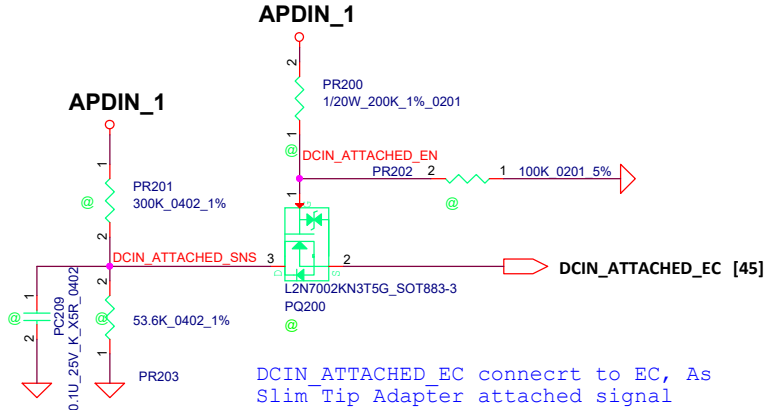
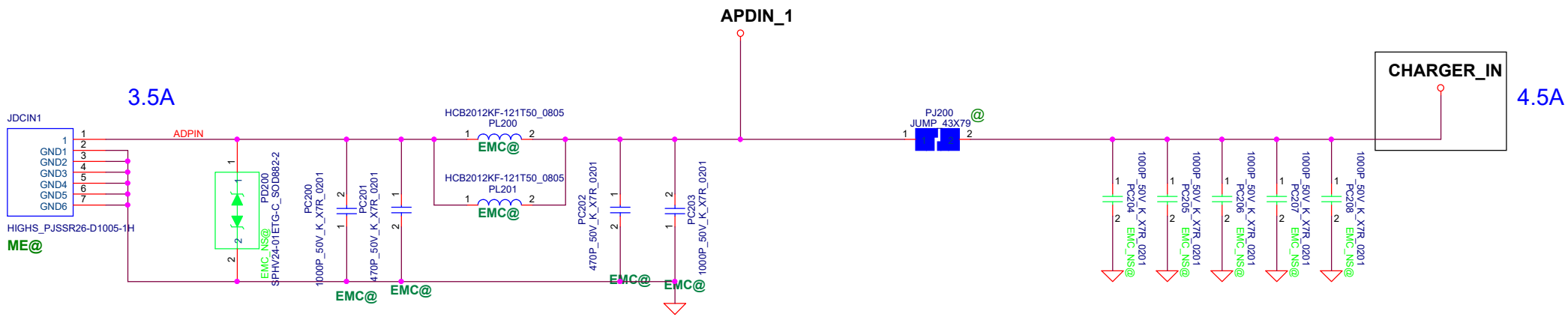


GPU or VRAM Shielding

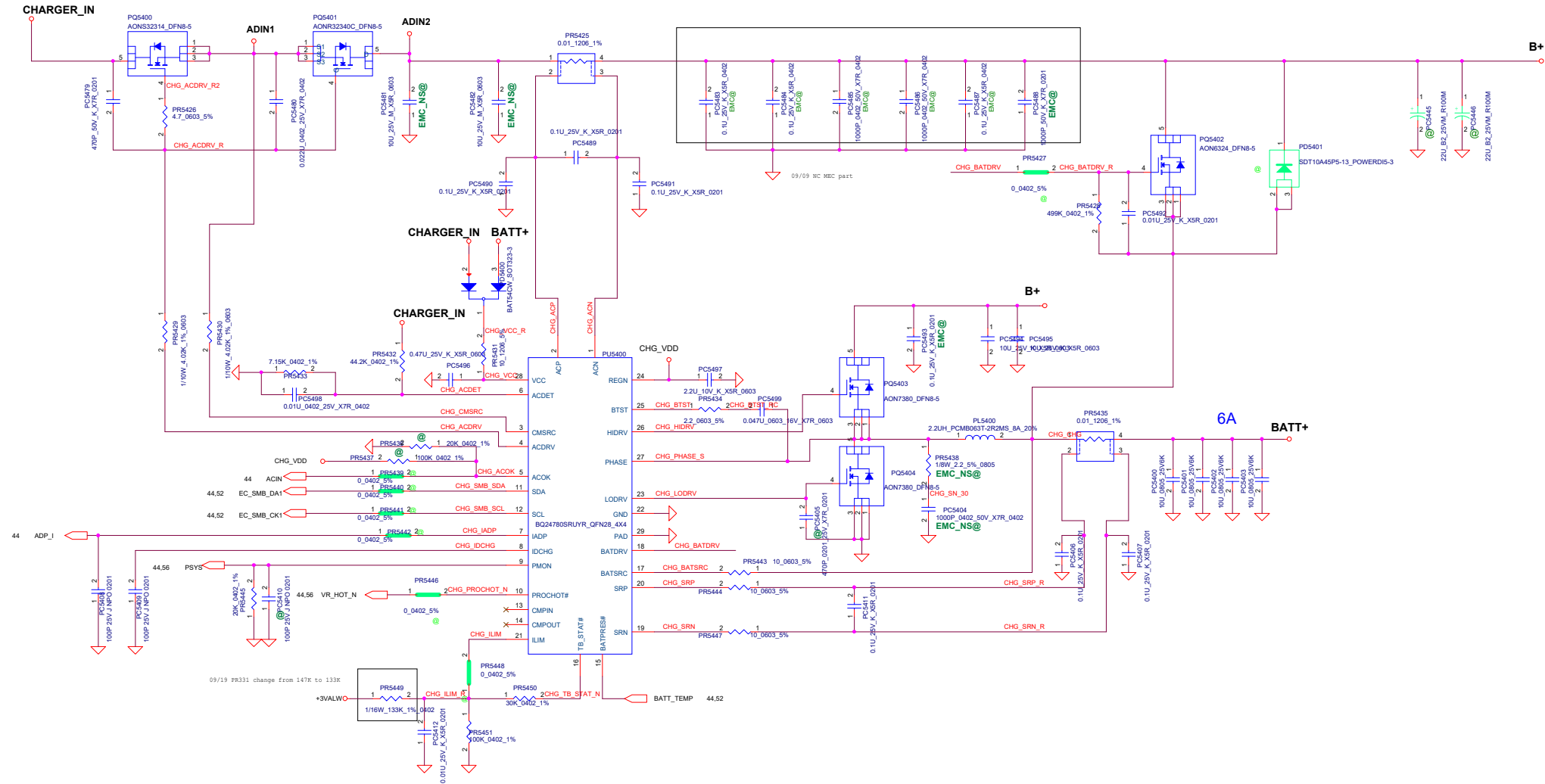


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								Date:		Tuesday, November 10, 2020			





DCIN ATTACHED EC connect to EC, As Slim Tip Adapter attached signal



ACDECT setting 17.2V
Charge current limit HW=7A
DC discharge limit =26A
Discharge current limit HW=9A during Turbo boost

IDPM	V(ILIM)	PR5320
500mA	1.2V	402K
1.0A	1.4V	332K
1.5A	1.6V	280K
2.0A	1.8V	237K
3.0A	2.2V	174K
3.25A	2.3V	162K
4.0A	2.6V	162K

← LOGIC

IND IN USE	PR5333
1uH	93K
2.2uH	137K
3.3uH	169K

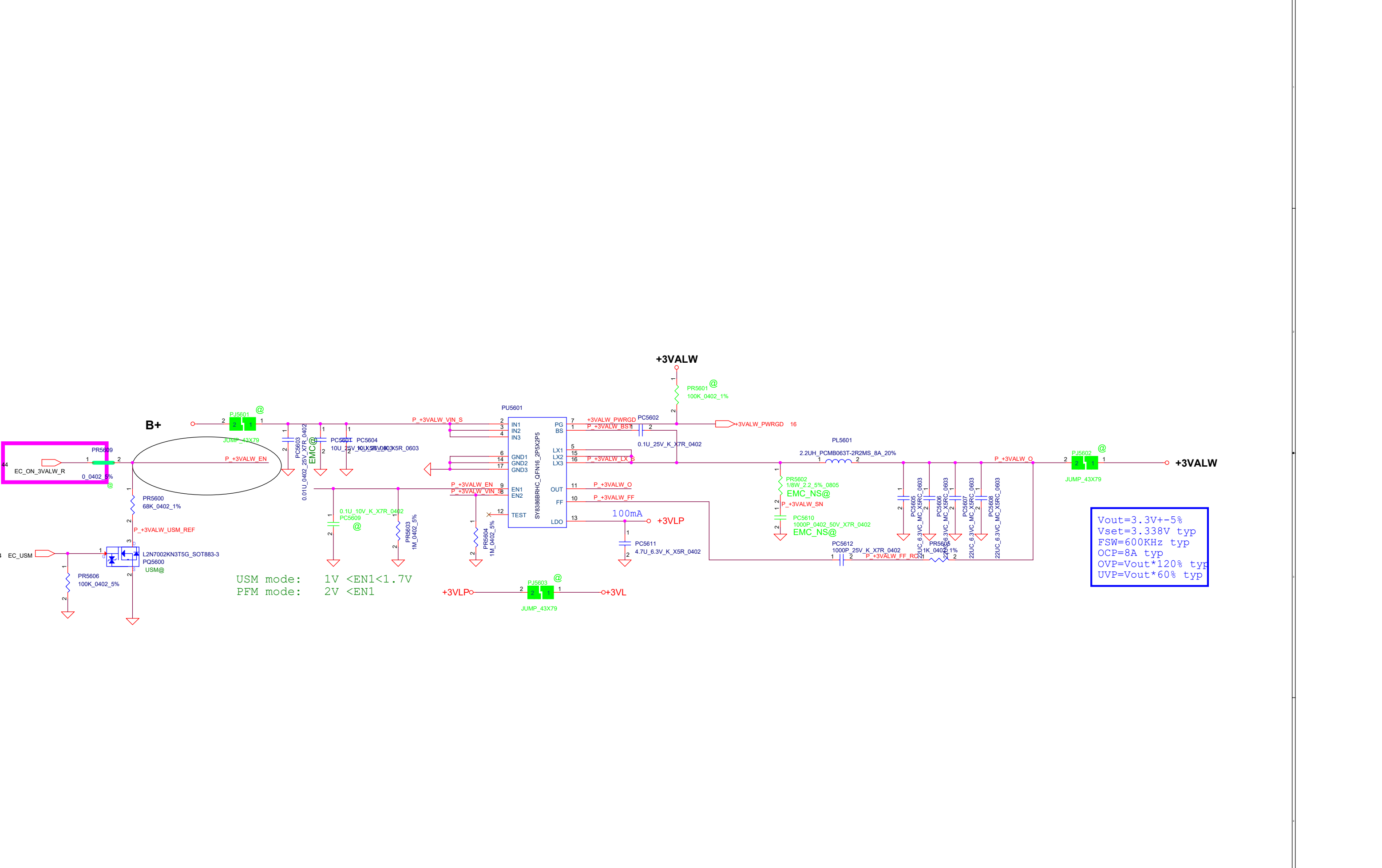
- LOGIC

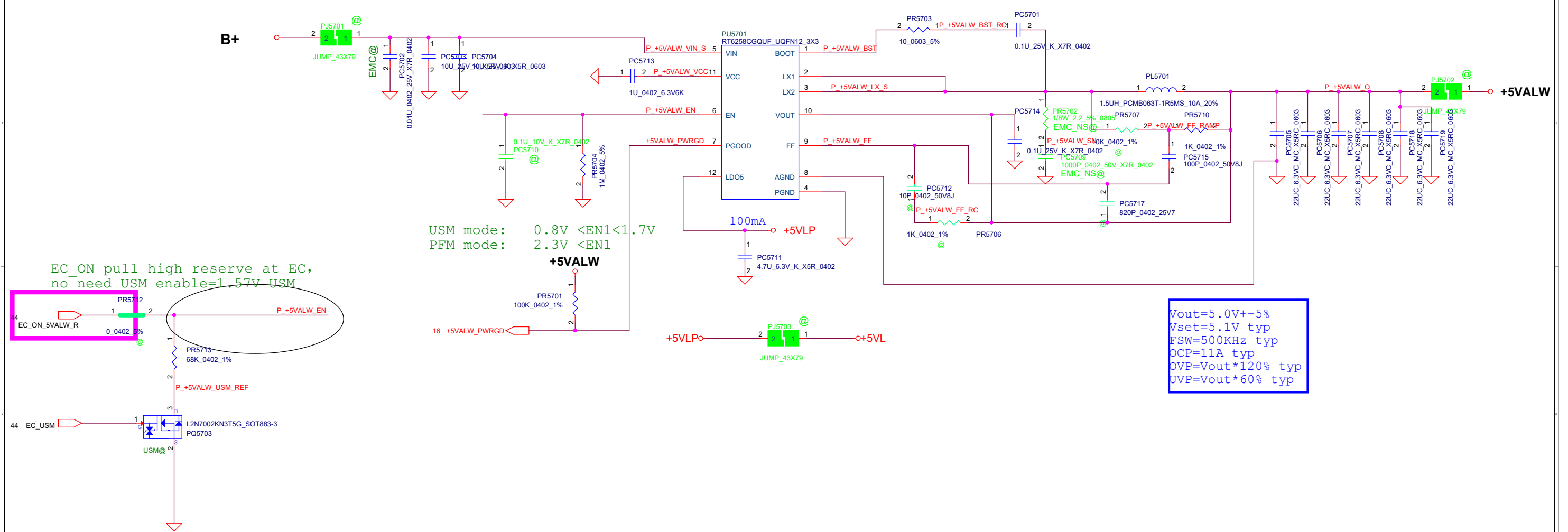
# of CELL	VCELL_PRES	PR5322
1-CELL	1.5V	301K
2-CELL	2.4V	150K
3-CELL	3.3V	82K
4-CELL	4.5V	33.2K

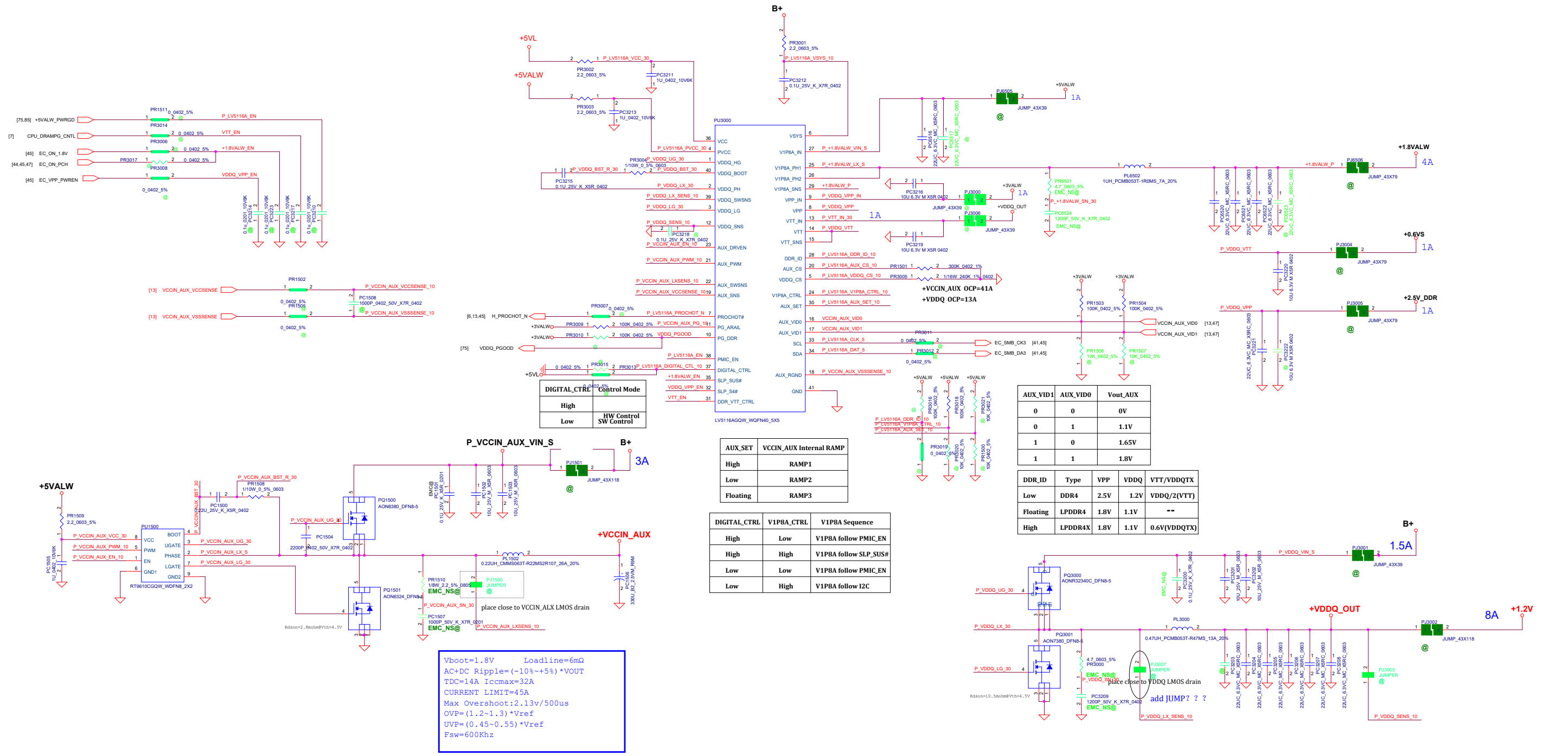
- LOGIC

$$V_{ILIM} = 1V + 40 \times (V_{ACP} - V_{ACN}) = 1 + 40 \times I_{DPM} \times R_{AC}$$

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				Date:	Tuesday, November 10, 2020	Sheet	90 of 110



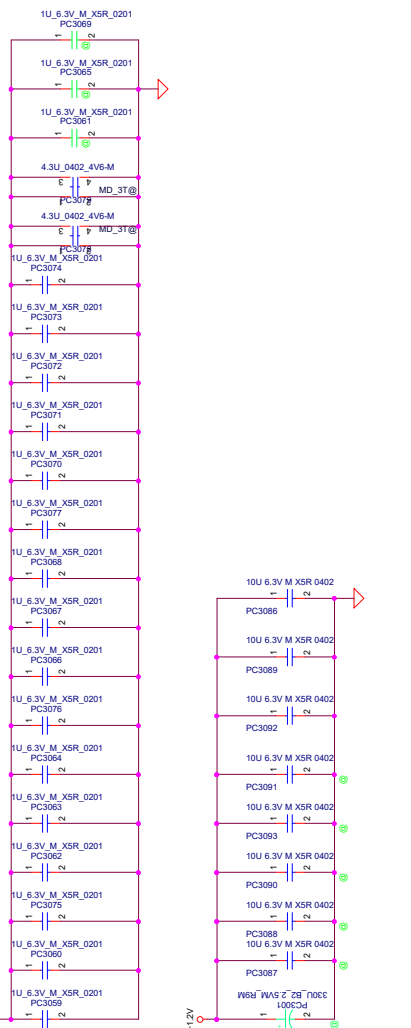




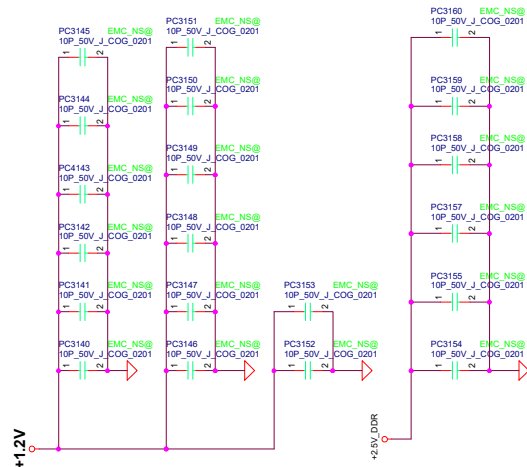
VTTC
PDG 2 Channel SODIMM: 2*10UF(0603,1 placeholder)+4*1UF(0402)
S360: channel A:1*10UF(0402)+ 2*1UF(0201)



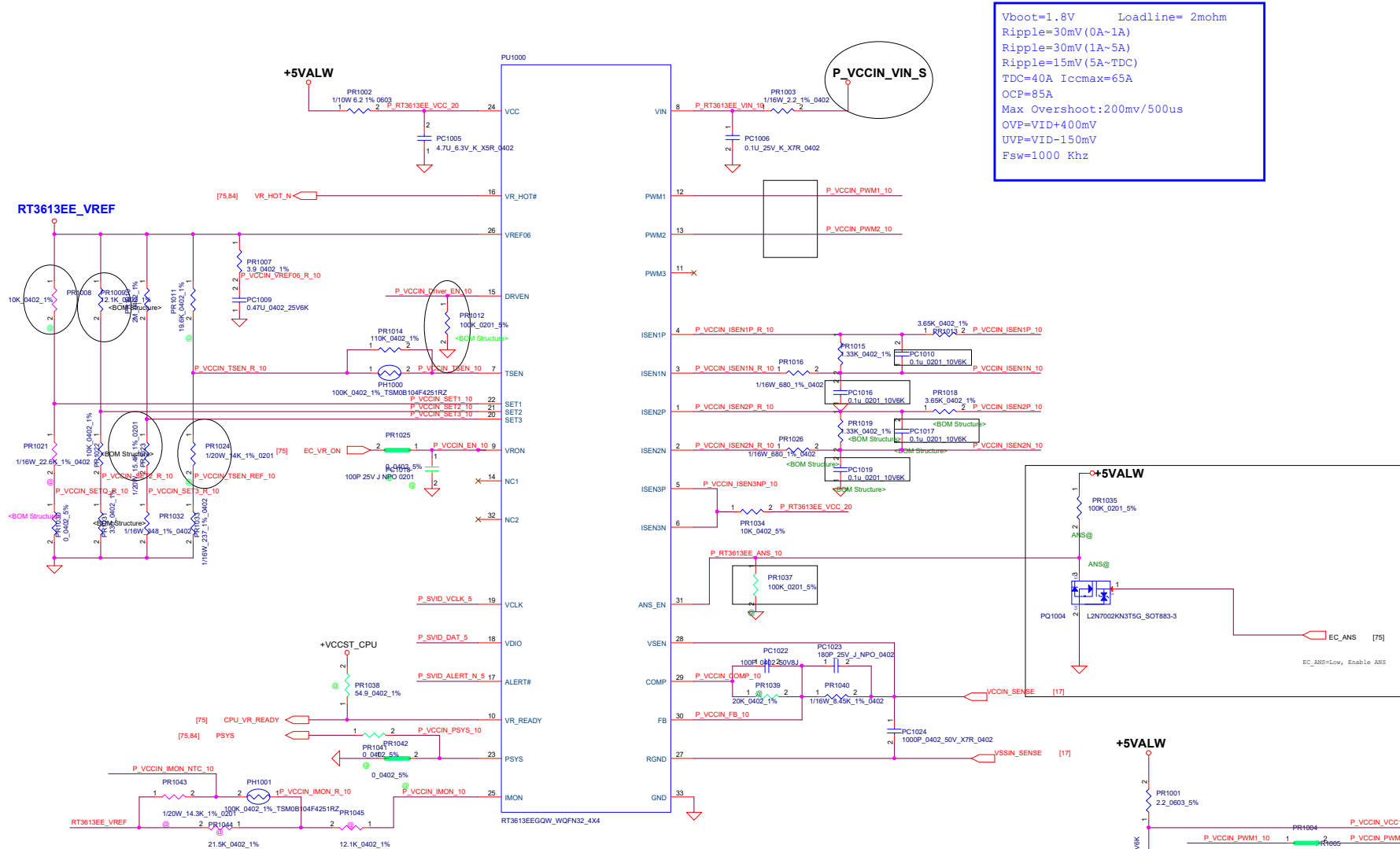
VT
PDG 2 Channel MD: 4*10UF(0603)+16*1UF(0402)
S360: channel B:2*10UF(0402)+ 2*1UF(0201)+2*4.3UF3T(P24)



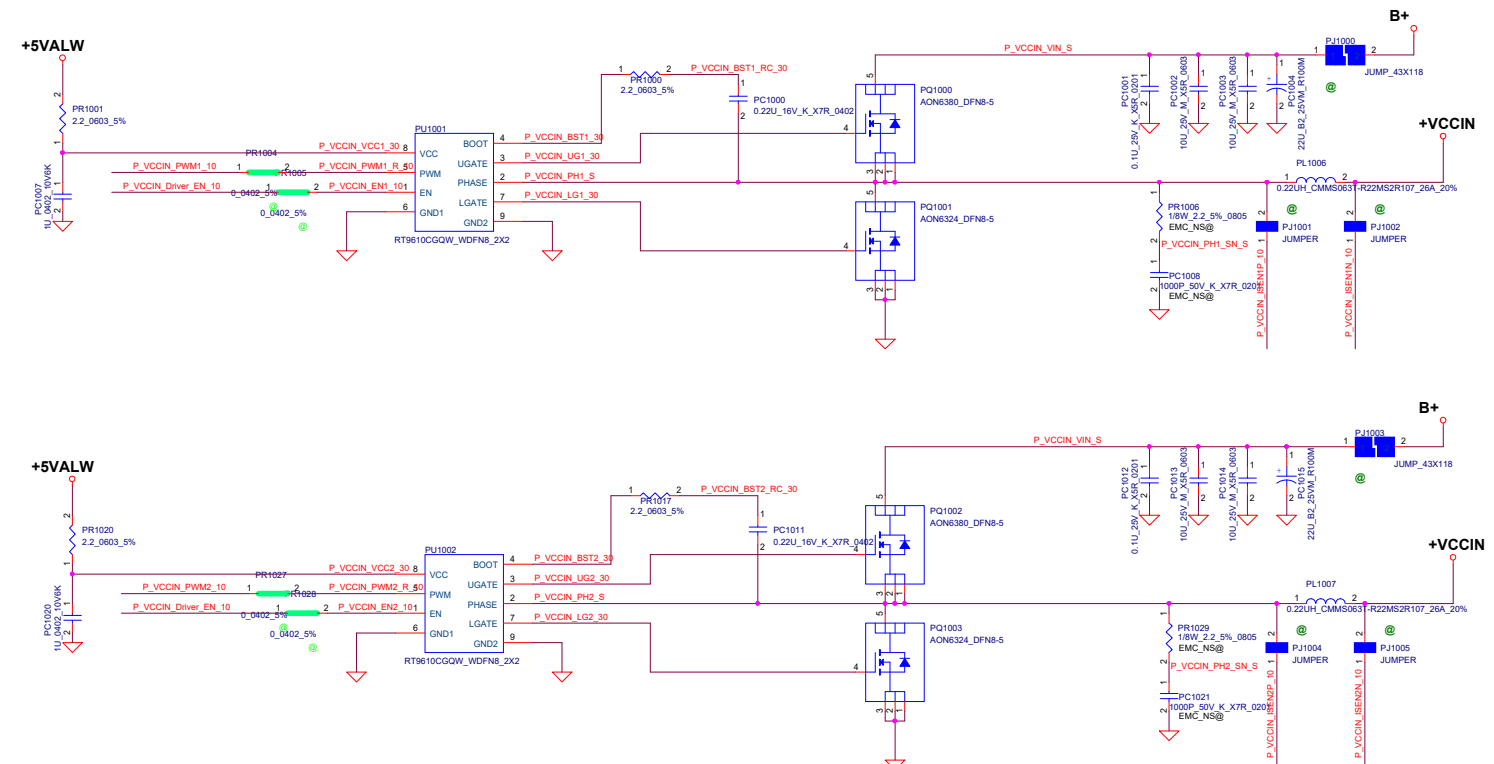
+1.2V PDG: 2*47UF(0603)+ 8*10UF(0402)+8*1UF(0402)
S360: 4*22UF(0603 P93)+ 6*10UF+9*1UF



Value	Qty	Note
10u.0402	8	Place on the primary side, as close as possible to the case that connects to the external power supply
47u.0603	2	Primary side 2 x optoisolated feedback
1u.0402	8	Place on the primary side, as close as possible to the case that connects to the external power supply. Do not use on the secondary side as the feedback signal is not isolated from the power supply. Do not use on the secondary side as the feedback signal is not isolated from the power supply. Do not use on the secondary side as the feedback signal is not isolated from the power supply.

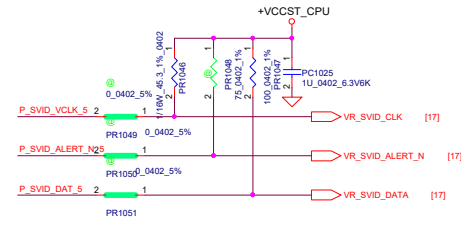


```
Vboot=1.8V      Loadline= 2mohm
Ripple=30mV(0A~1A)
Ripple=30mV(1A~5A)
Ripple=15mV(5A~TDC)
TDC=40A  Iccmax=65A
OCP=85A
Max Overshoot:200mv/500us
OVP=VID+400mV
UVP=VID-150mV
Fsw=1000 KHz
```



Virtual Symbol

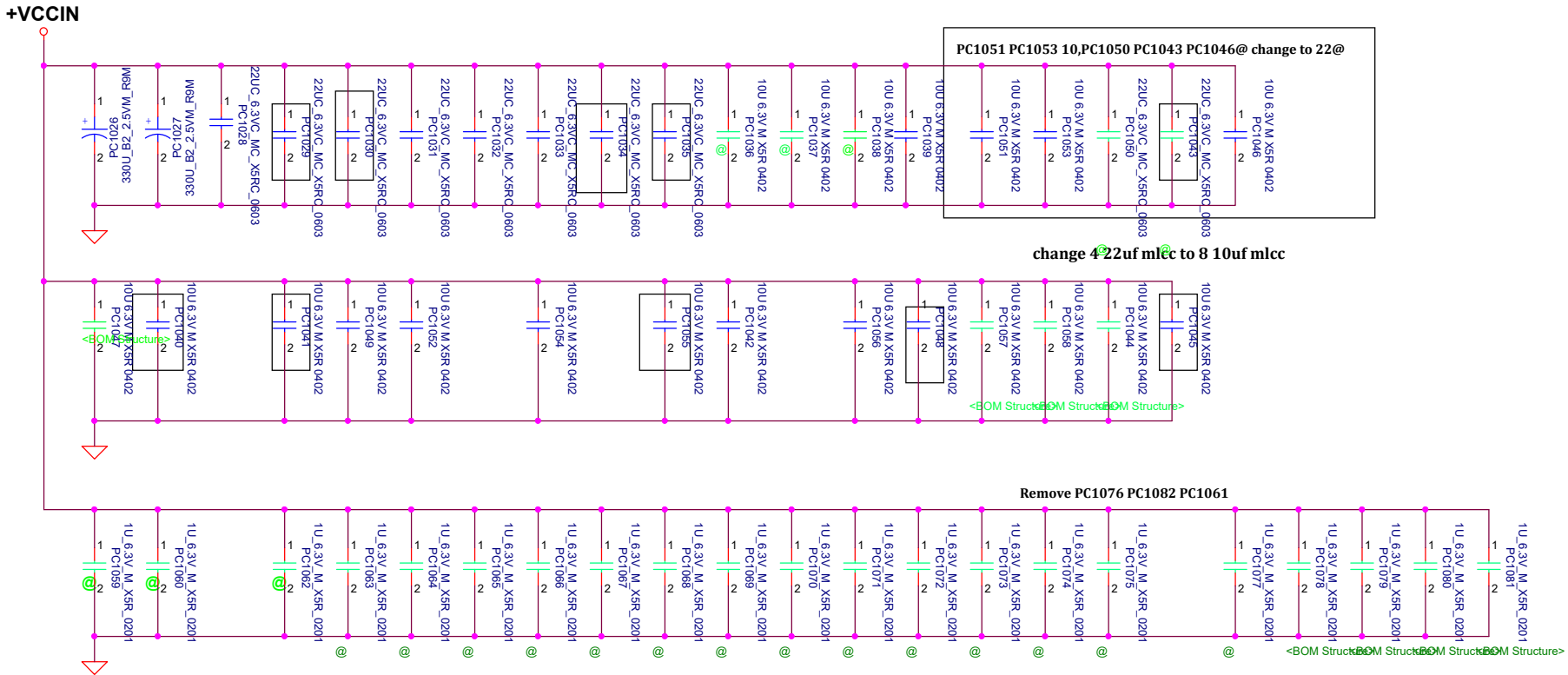
	15W	25W
PR1045	16.8K SD03416928Z	12.1K SD03412128Z
PR1044	24.3K SD000011300	21.5K SD03421528Z
PR1043	11.8K SD00001950T	14.3K SD00001D800
PR1039	42.2K SD03442228Z	20K SD0420028Z
PR1008	200K SD03420038Z	93.1K SD01110028Z
PR1021	19.6K SD00000358Z	22.6K SD000019W00
	2.37K SD03423718Z	3.65K SD000000D00
	1.8K SD00000R58Z	1.33K SD000000R0T
PR1011	178K SD03417838Z	19.6K SD00000358Z
PR1024	8.45K SD00001J38T	14K SD000001V00



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CHANGE VCCIN MLCC 0625
VCCIN
PDG: 2*220UF(7343)+ 8*22UF(0603)+12*10UF(0402)
S360: 2*330UF(B2)+ 10*22UF(0603)+10*10UF(0402)+3*1UF(0201)



PDG: VCCIN

Value	Qty	Note
10u_0402	12	Secondary Side Put them as directly below the BGAs as possible.
220u_7343	2	Primary Side MLCC cap recommendation based on VR solution with 150kHz VR bandwidth or 600kHz switching frequency, assuming BW=1/4 of Fsw. Place board caps as close as possible to package in this order where 0402 to be closest to package pins: VR->7343->0603->0402->Package
220u_7343 (Place holder)	2	
22u_0603	8	

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							PWR PCH PWR1		0.1			
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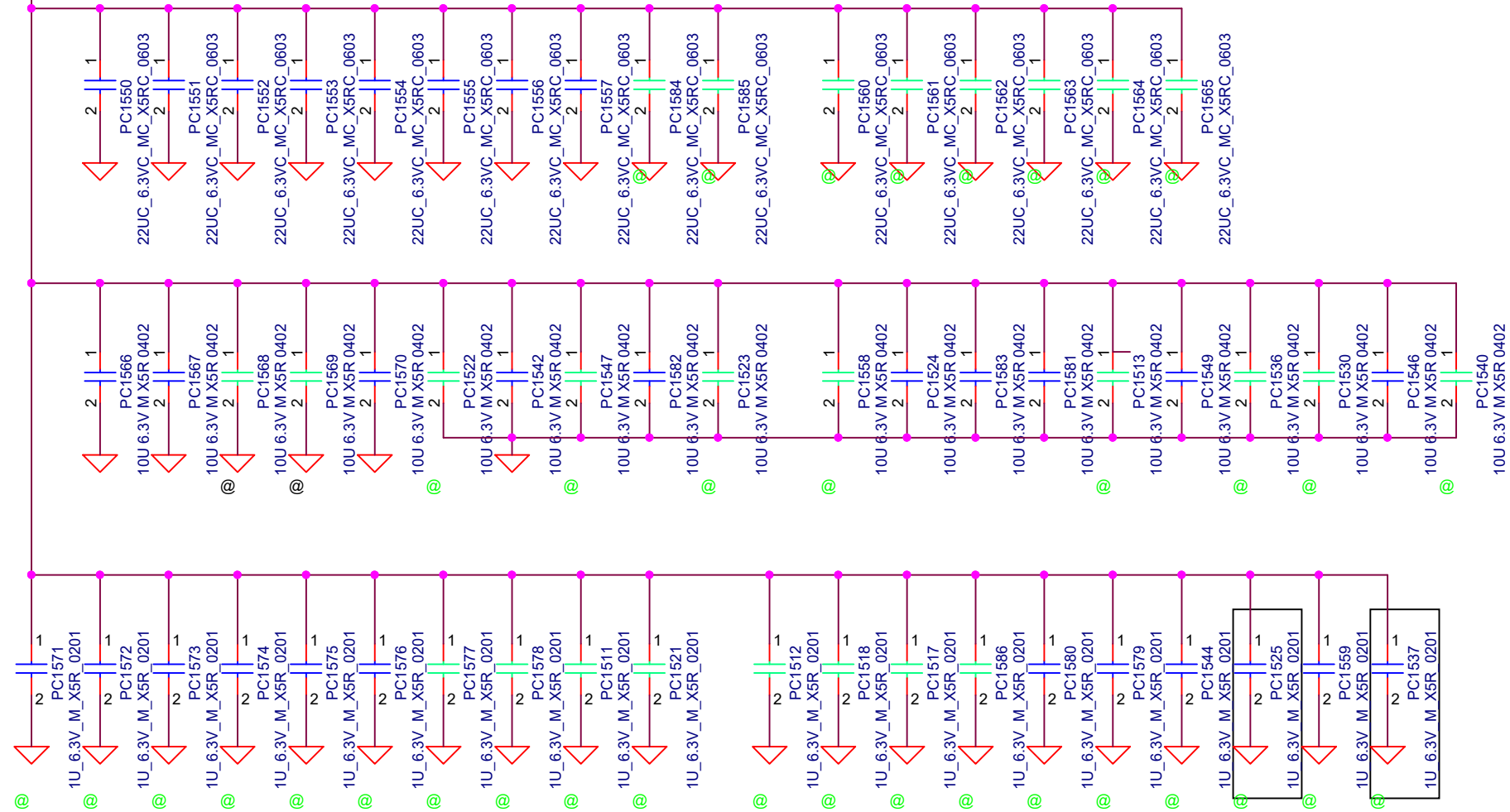
+VCCIN_AUX

VCCIN_AUX

PDG: 1*220UF+ 3*47UF+12*22UF+25*10UF

CHANGE VCCIN_AUX 1UF(6---12) MLCC 0625

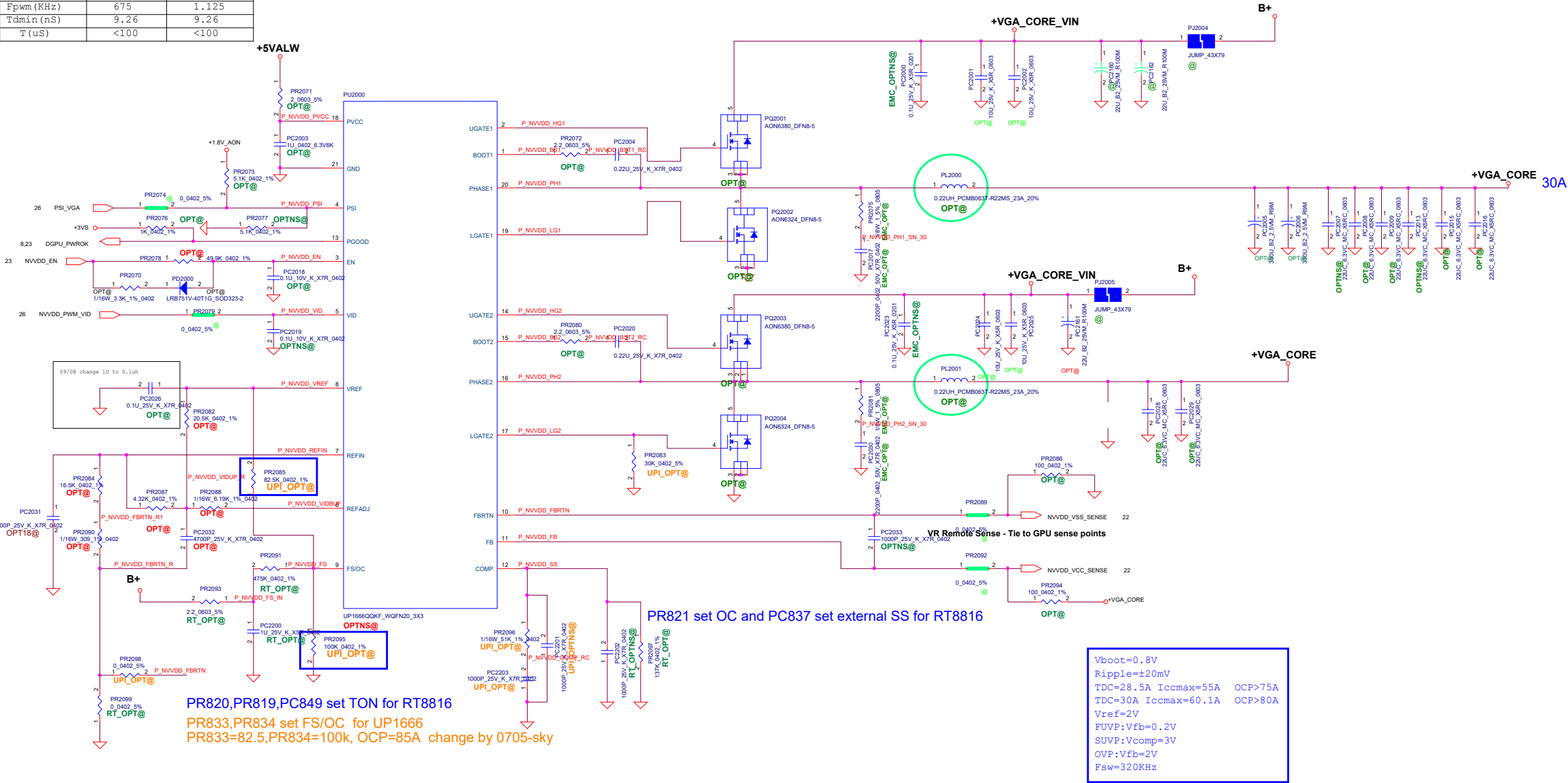
S360: 1*330UF(P93)+ 8*22UF+10*10UF*0*1UF



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				Document Number	
				PWR PCH PWR Decoupling	
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PWM-VID Specification		
	N17 Config	N16 Config B
Vmin (V)	0.3	0.6
Vmax (V)	1.3	1.2
Vboot (V)	0.8	0.9
Vstep (mV)	6.25	6.25
N(level)	160	96
Fpwm (KHz)	675	1.125
Tdmin (nS)	9.26	9.26
T(uS)	<100	<100

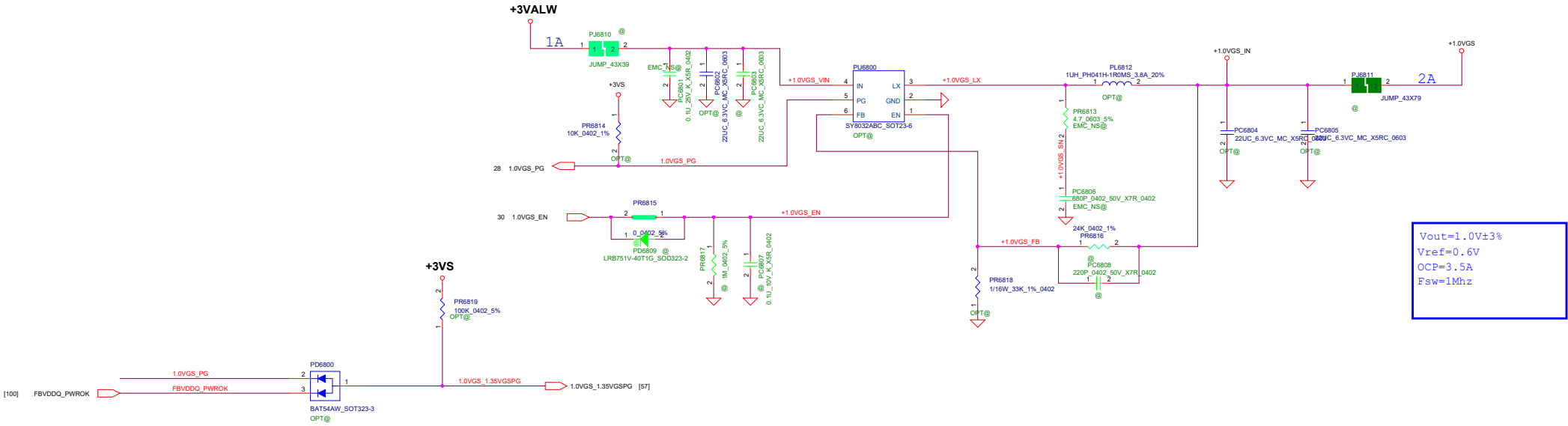
RT8816 PSI	UP1666 PSI	Phase Configuration
1.6V~5.5V	1.6~5.5V	2Phase CCM
1.08~1.35V	1~1.4V	2Phase DEM
0.7~0.88V	0.4V~0.8V	1Phase CCM
0~0.4V	0~0.2V	1Phase DEM



UPI_OPT@ : for UP1666
RT_OPT@ : for RT8816A

Vboot=0.8V
Ripple=±20mV
TDC=28.5A Iccmax=55A OCP>75A
TDC=30A Iccmax=60.1A OCP>80A
Vref=2V
FVUP:Vfb=0.2V
SUVP:Vcomp=3V
OVP:Vfb=2V
Fsw=320KHz

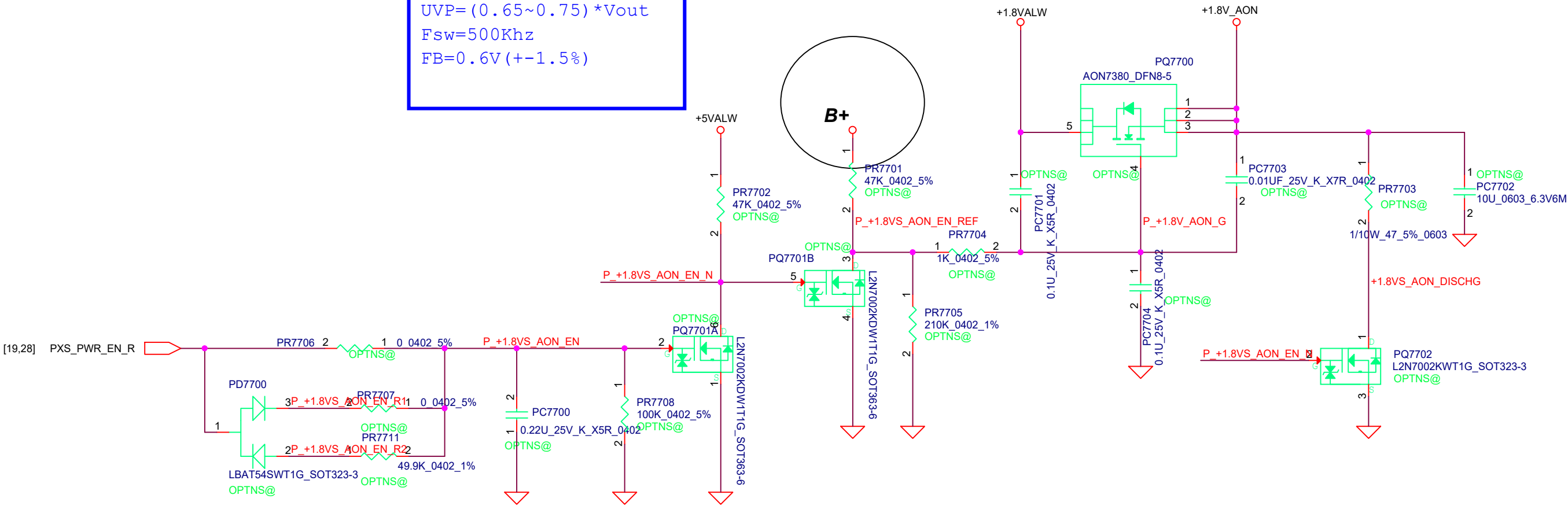
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Vout:
P5=1.2V±3%
P0=1.25V±3%
TDC=11A

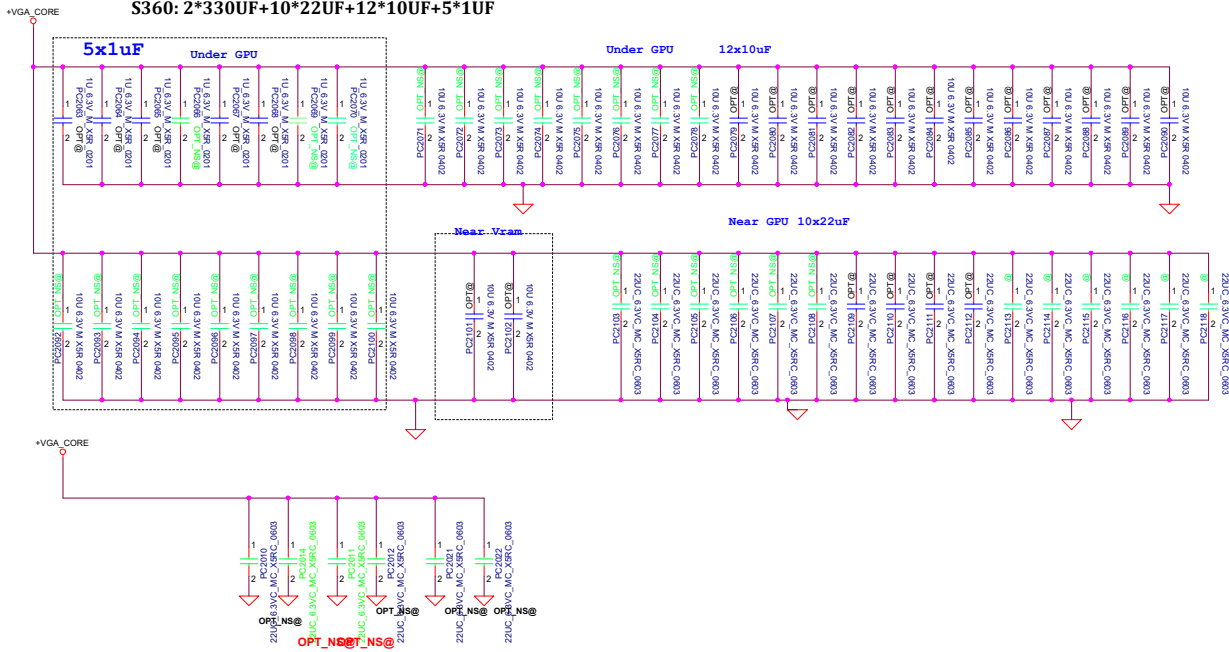
PK-PK :55mV
EDP-c=11A EDP-p=20A
OCP=26A
OVP=(1.2~1.3) *Vout
UVP=(0.65~0.75) *Vout
Fsw=500Khz
FB=0.6V (+-1.5%)



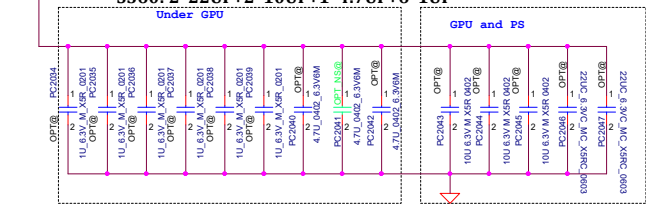
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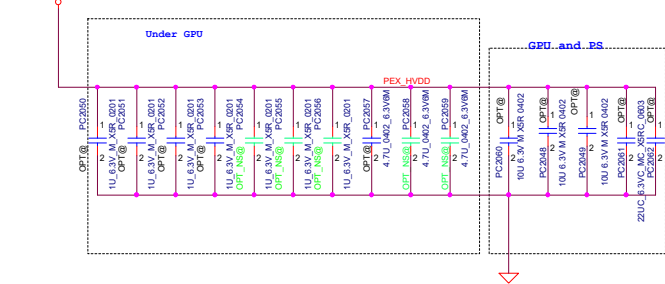
NVVD
PDG: 2*330UF+10*22UF+12*10UF+5*1UF
S360: 2*330UF+10*22UF+12*10UF+5*1UF



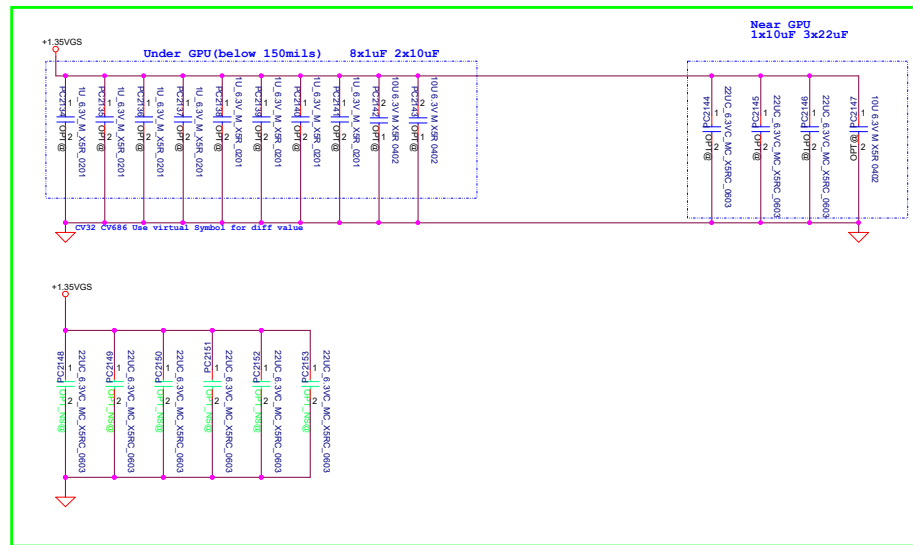
PEX_DVDD
PDG: 2*22UF+2*10UF+1*4.7UF+6*1UF
S360: 2*22UF+2*10UF+1*4.7UF+6*1UF



PEX_HVDD
PDG: 2*22UF+2*10UF+1*4.7UF+4*1UF
S360: 2*22UF+3*10UF+1*4.7UF+4*1UF



FBVDD
PDG: 3*22UF+3*10UF+8*1UF
S360: 3*22UF+3*10UF+8*1UF



XS_FLLVDD(N18:XSN_FLLVDD) Decoupling				
MLCC	N18/GB2E-64		N17S	location
0.1uF	X7R	0402	0	1
0.47uF	X6S	0201	1	0
Bead 30ohm@100Mhz (ESR=0.5ohm)			0	0
		0402	0	0

SP_FLLVDD&VID_FLLVDD Decoupling				
MLCC	N18/GB2E-64		N17S	location
0.1uF	X7R	0402	0	2
0.47uF	X6S	0201	2	0
Bead 30ohm@100Mhz (ESR=0.2ohm)			0	0
		0603	0	0

GPCPLL_AVDD Decoupling				
MLCC	N18/GB2E-64		N17S	location
0.1uF	X7R	0402	0	1
0.47uF	X6S	0201	1	0
4.7uF	X6S	0603	1	1
22uF	X5R	0805	0	1
Bead 30ohm@100Mhz (ESR=0.2ohm)			1	1

VDD_AON&VDD_MAIN Decoupling
N17_VDD18(N18_NC)

MLCC	N18/GB2E-64		N17S	location
0.1uF	X7R	0402	NA	2
1.0uF	X6S	0603	NA	1
4.7uF	X6S	0603	NA	1

MLCC	N18/GB2E-64		N17S	location
0.1uF	X7R	0402	0	2
0.47uF	X6S	0201W	4	0
1uF	X6S	0201/0402	0	1
0.47uF	X6S	0201W	6	0
4.7uF	X6S	0603	3	1

NVVD Decoupling				
MLCC	N18/GB2E-64		N17s	location
10uF	X6S	0603	12	12
4.7uF	X6S	0603	0	0
1.0uF	X6S	0201/0402	5	5
0.47uF	X6S	0201/0402	0	0
4.7uF	X6S	0805	0	4
10uF	X6S	0805	0	11
22uF	X6S	0805	10	4
330uF	POS	7343	2	2

FBVDD Decoupling				
MLCC	N18/GB2E-64		N17S	location
0.47uF	X6S	0201	0	0
1.0uF	X6S	0201/0402	8	8
10uF	X6S	0603	2	2
10uF	X6S	0603	1	1
22uF	X6S	0603	3	3

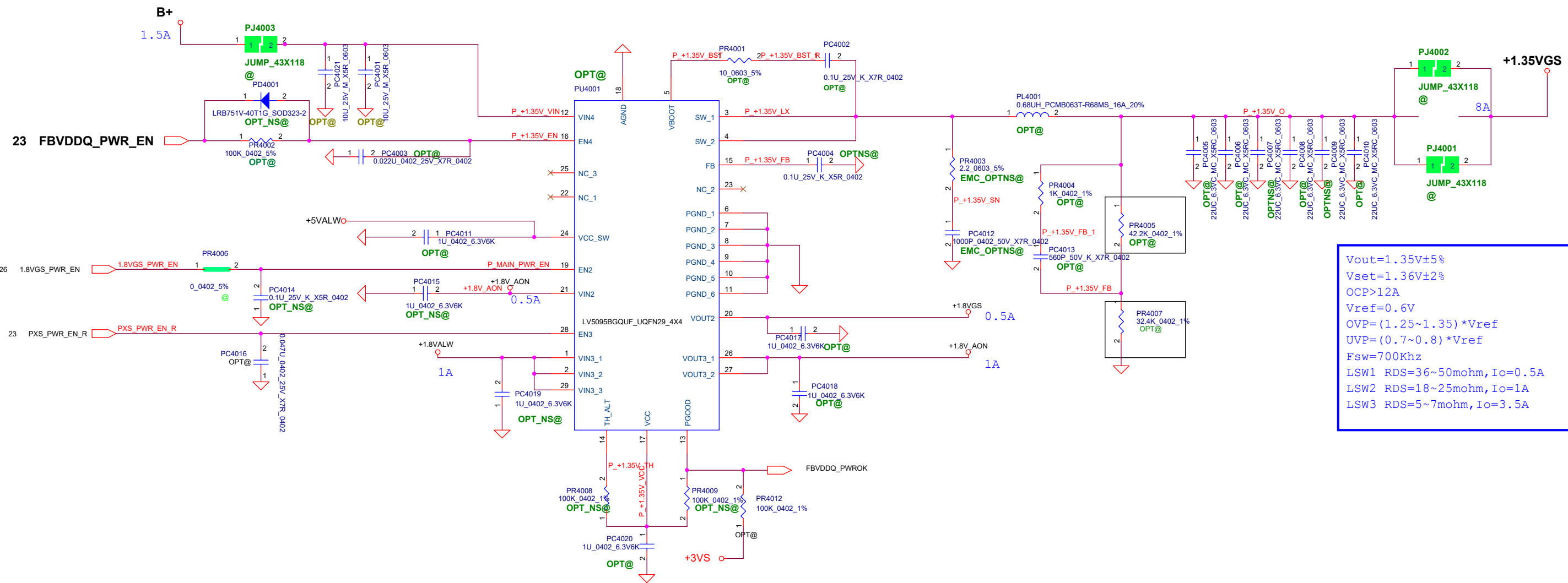
FBPLL Decoupling				
MLCC	N18/GB2E-64		N17S	location
0.47uF	X6S	0201	3	0
0.1uF	X7R	0402	0	4
4.7uF	X6S	0603	2	0
22uF	X5R	0603	0	1
22uF	X6S	0805	1	0
Bead 30ohm (ESR=0.01ohm)			0603	1

PEX_DVDD Decoupling				
MLCC	N18/GB2E-64		N17S	location
1.0uF	X6S	0201/0402	6	1
4.7uF	X6S	0603	1	1
4.7uF	X6S	0603	0	2
10uF	X6S	0805	2	2
22uF	X6S	0805	2	1

PEX_HVDD Decoupling				
MLCC	N18/GB2E-64		N17S	location
1.0uF	X6S	0201/0402	4	4
4.7uF	X6S	0603	1	0
4.7uF	X6S	0603	0	2
10uF	X6S	0805	2	2
22uF	X6S	0805	2	1

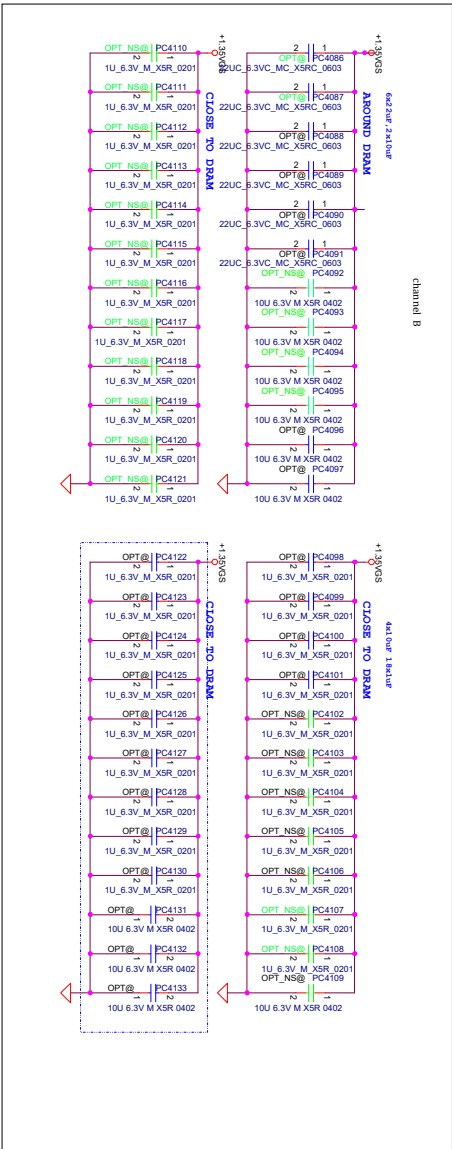
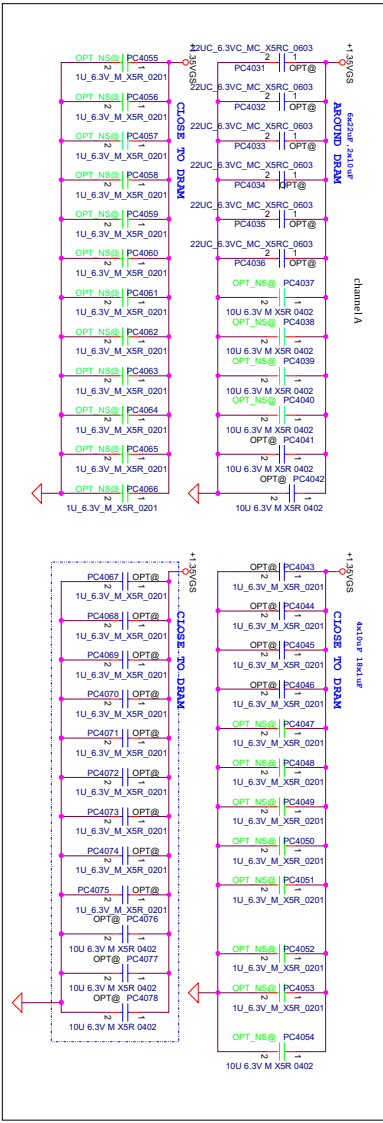
PEX_PLL HVDD Decoupling				
MLCC	N18/GB2E-64		N17S	location
0.1uF	X7R	0402	0	1
0.47uF	X6S	0201	1	0

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VRAM side
FBVDD:
PDG: 3*220UF+3*10UF+8*1UF
S350: 3*220UF+3*10UF+8*1UF

FBVDDQ Decoupling			
MLCC	N18/GB2E-64	N17	Location
0.47uF	X5S	0201	0
1.0uF	X5S	0201/0402	8
10uF	X5S	0603	2
10uF	X5S	0603	1
22uF	X5S	0603	3
			Near



2020/05/22
1.P30:RC2807---RC2806
2.P33:Add RG3102;
3.P34:RG3204 40.2OHM---60.4ohm
4.P35:DG3303 OPT_NS@---OPT@;
 DG3304 OPT_NS@---OPT@;
 DG3301 OPT@---OPT_NS@;
 CG3316:OPT@---OPT_NS@;
 RG3317:OPT_NS@---OPT@;
 RG3316:OPT@---OPT_NS@
5.P53:Remove RU34,RU35,RU36,RU37;
6.P08:DDRA_CLK1_N&P change ball BT41,BT42
7.P63:Remove RK207,QK201,CK216,CK215 change;
8.P25:DDR-DQ&DQS follow S350
9.Page66:SPKR_MUTE_N change to EC_MUTE_N
10.P86:add SH12~15 for GPU clip

2020/06/09
1.Compare the Connector list with 0601 version
change SSD &KB &HDD CONN, remove JTP

2020/06/11
1.page2 modify blockDiagram EE
2.page7 add RC863,remove Page47 RV6,
3.page11&79 ESPI_Xxxx net name change LPC_ESP_Xxxx
4.page14 RC863 change RC60

2020/06/16
1.EC_USB_ON_N swap pin&override

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		Change List PWR							
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